Quantum extensions of bi-conformal scalar-tensor gravity: particles as spacetime resonances, the Bernoulli mechanism for gravity, and falsifiable predictions

George Lotishvili[1,2,*]

[1] *East European University, Tbilisi, Georgia*

[2] *Independent research*

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We develop the quantum extension of bi-conformal scalar-tensor gravity (ISPG), a Horndeski-class theory defined by the action $S = (16\pi G)^{-1}\int d^4x\sqrt{-g}\,[R+\frac{1}{2}(\partial\varphi)^2]+S_{\rm m}$ with the bi-conformal metric $ds^2 = -e^{\varphi}c^2dt^2 + e^{-\varphi}(dx^2 + dy^2 + dz^2)$. In the classical regime, ISPG reproduces all solar-system tests ($\gamma = \beta = 1$), resolves the Schwarzschild singularity, derives MOND phenomenology ($a_0 = cH_0/2\pi$), and predicts falsifiable deviations at 2PN order [1]. The central aim of the present paper is to identify the *quantum mechanism of gravity*: how localized resonant excitations of the scalar-metric medium generate the static pressure deficit that becomes the gravitational field. Here we extend the theory to the quantum domain by interpreting elementary particles as localized resonant excitations of the scalar-metric medium. We derive: (i) the spectrum of scalar-field excitations on the bi-conformal background, establishing the effective mass formula $m_{\rm eff} = m_0e^{\varphi/2}$ at the quantum level; (ii) a Bernoulli-type mechanism in which the same scalar profile that defines the metric also carries a static pressure deficit in the scalar-metric medium, with exterior pressure $P_{\rm static} = -e^{\varphi}\varphi^4/(32\pi G\, r_{\rm s}^2)$ whose universal shape function is $\hat{f}(\varphi) = e^{\varphi}\varphi^4$; (iii) the cosmological negative-feedback loop that dynamically regulates the accumulated echo-background energy and, on the perturbative tail-power branch, gives an order-of-magnitude scale $\rho_\Lambda \sim H_0^2M_{\rm Pl}^2$, while the full zero-point decoupling and non-perturbative normalization remain open; (iv) the dark-energy equation of state $w(z) = -1 - \delta w_0\,(1 + z)^3/E(z)$ with $\delta w_0 \sim 10^{-2}$–10^{-1} (source-driven phantom, no ghost instability), whose magnitude remains conditional on the tail-power normalization and whose DESI DR2 CPL comparison is mixed (same sign of w_a, opposite sign of $w_0 + 1$); (v) a coherent gravitational phase/dephasing rate for entangled particles, $\Gamma_\varphi = 2m_0|\Delta\Phi_N|/h$, which reproduces the GR/COW redshift baseline at leading post-Newtonian order, while the irreversible decoherence channel associated with resonant tails and scalar fluctuations remains conditional on a dedicated correlator derivation; the formal eV-photon coefficient at 10 m is ~ 0.5 Hz but is not a clean flight-geometry test; (vi) a candidate topological route to the fermion spin projection and matter–antimatter winding labels $n = \pm1$; (vii) gravitational birefringence from the gradient-squared scalar channel, with $\Delta c/c \propto (\nabla\varphi)^2/m_e^2$ distinct from the GR+QED Drummond–Hathrell curvature channel; the numerical coefficient and detectability forecast are future one-loop calculation targets; (viii) a candidate profile-dependent WKB tunneling effect in inhomogeneous $\varphi(x)$ backgrounds, with any nonreciprocal $T_\downarrow/T_\uparrow$ channel tied to a future resonant-tail/anchoring boundary-condition calculation and any macroscopic or cumulative astrophysical relevance left to dedicated modelling; (ix) a schematic cosmological variation of the fine-structure constant $|\Delta\alpha/\alpha| \sim 10^{-7}$–$10^{-6}$ at $z \sim 1$–3, subject to a fixed-scale/renormalization prescription and testable by ESPRESSO and ELT-ANDES; (x) numerical support for the Koide relation $Q = 2/3$ from the MASS/Mathieu lepton ladder and the E1D/rank-1 cavity analysis, with Q reproduced to 0.024% in N-space and to 0.009% in the E1D check, and with the 3D-to-1D cavity bridge and the self-consistent $x = 33 + 24\sqrt{2}$ selection identified as the remaining numerical task;

(xi) a perturbative localized-resonance derivation of the inertial response $\mathbf{F} = -C_{\text{in}} m_{\text{eff}} \mathbf{a}$ from the retarded potential of an accelerated oscillon, with $C_{\text{in}} = 1$ at leading order in $a/(c\omega_0)$, exhibiting m_{inertial} and $m_{\text{gravitational}}$ as the same scalar-profile/Bernoulli-deficit bookkeeping within the weak-field, slow-acceleration, spherical equilibrium regime, and connecting inertia to medium hysteresis via a Klein–Gordon effective field whose microscopic tracking stiffness is motivated by the oscillon's quasi-normal mode structure; (xii) a structural argument that the scalar sensitivity $s = 1/2$ and the suppression of scalar dipole radiation are protected at perturbative/EFT precision by a shift-symmetry Ward identity (exact for vacuum compact objects at all loop orders, leading-order for matter-filled bodies; the exact non-perturbative matter-coupled cancellation remains an open task per MAIN/16), supporting a perturbatively controlled strong equivalence statement in the scalar sector; (xiii) a homogeneous-background no-inflation scenario in which the first excitations nucleate quantum-mechanically from the quiescent medium, addressing the horizon, flatness, and topological-defect issues at the background-consistency level, while the primordial perturbation spectrum and superhorizon correlations remain open numerical work. The central result—closed on the static spherically symmetric exterior branch and at leading zero-frequency for time-periodic oscillons—is the oscillon $\to$ Bernoulli pressure-deficit $\to$ source $\to$ $1/r$ field chain for gravity, together with the reduced-gap/coercive closure showing that, after projection onto the source manifold, non-neutral deformations remain perturbatively controlled while the exterior field follows the neutral collective coordinate. The remaining sectors show how the same medium picture extends into cosmology, particle structure, and quantum phenomena; where a sector still requires a dedicated numerical or structural closure, that status is stated explicitly in the text and does not affect the core gravity derivation.

* g_lotishvili@eeu.edu.ge

CONTENTS

I. INTRODUCTION

A. Motivation

The reconciliation of gravity with quantum mechanics remains the central unsolved problem in theoretical physics. General relativity (GR) describes spacetime as a dynamical geometric entity, while quantum field theory (QFT) treats spacetime as a fixed background arena populated by quantized fields. Neither framework, taken alone, provides a satisfactory account of the other's domain: GR predicts singularities where quantum effects should dominate [4, 5], while QFT on curved spacetime leads to the 10^{120} cosmological constant catastrophe [6, 7] and leaves the origin of gravitational decoherence unexplained [8, 9].

Attempts to merge these frameworks have produced string theory [10], loop quantum gravity [11], and asymptotic safety [12], each introducing substantial theoretical apparatus (extra dimensions, spin foams, or UV fixed points) whose empirical signatures remain elusive.

In the classical domain, bi-conformal scalar-tensor gravity (ISPG) [1] has demonstrated that a minimally coupled phantom scalar field φ with the bi-conformal metric

$$ds^2 = -e^{\varphi}\, c^2\, dt^2 + e^{-\varphi}\big(dx^2 + dy^2 + dz^2\big) \tag{1}$$

and the action

$$S = \frac{1}{16\pi G}\int d^4x\,\sqrt{-g}\left[R + \frac{1}{2}\,g^{\mu\nu}\partial_\mu\varphi\,\partial_\nu\varphi\right] + S_{\rm m}, \tag{2}$$

reproduces the classical 1PN solar-system tests ($\gamma = \beta = 1$), predicts a falsifiable ISPG–GR 2PN Shapiro-delay difference $\Delta B = \pi/4$ (closed regularization-independent observable; standalone absolute coefficients are finite-endpoint convention-dependent), resolves the Schwarzschild singularity, is compatible with the CMB angular power spectrum at linear order, and derives MOND phenomenology with $a_0 = cH_0/(2\pi)$ and the baryonic Tully–Fisher relation; the 15% offset between the instantaneous formula and the observed value is resolved by a vortex-lag mechanism that predicts galaxy-age-dependent a_0 [2]. The same classical compact-object sector predicts a test-particle ISCO at $r_{\rm ISCO} = \frac{3+\sqrt{5}}{2}\, r_{\rm s}$, an ISCO orbital frequency 6.9% lower than in GR for the same total mass, a photon sphere at $r = r_{\rm s}$ with a 4.6% larger shadow, and suppressed scalar dipole radiation in the controlled sensitivity regimes (vacuum exact, matter leading order) where $s = 1/2$ [1]. The physical interpretation of φ as the pressure state of an information medium, whose scalar profile both defines the physical metric and carries pressure-deficit and transport information, provides a mechanical picture of gravity: the effective mass of a particle at position φ is

$$m_{\rm eff}(\varphi) = m_0\, e^{\varphi/2}, \tag{3}$$

which is locally undetectable because all masses, clocks, and rulers participate in the same bi-conformal rescaling.

The classical theory, however, does not address the fundamental nature of matter itself. It treats particles as point sources in the field equation $\Box\varphi = -(8\pi G/c^4)\,T$, without specifying what constitutes these sources at the microscopic level.

B. Central thesis

The present work fills this gap by proposing and formalizing a single, unifying hypothesis:

> *Elementary particles are not objects placed in spacetime. They are localized resonant excitations of the scalar-metric medium itself.*

This identification has far-reaching consequences. If matter *is* spacetime vibration, then:

- Mass is the amplitude of resonance (energy stored in the vibrational mode), naturally explaining Eq. (3).
- Gravity arises because resonant oscillations redistribute kinetic energy density $\frac{1}{2}(\nabla\varphi)^2$, lowering the static pressure of the medium via an analog of Bernoulli's principle.
- Entropy increase is the irreversible spreading of resonant tails into the surrounding medium.
- Vacuum energy is regulated by a negative-feedback loop between resonance production and pressure depletion.
- Wave-particle duality, the uncertainty principle, and quantum entanglement follow as natural properties of spatially extended resonant structures.

C. Relation to existing approaches

The proposal that particles are excitations of a medium has precursors. Kelvin's vortex theory of atoms [13] proposed that atoms are knots in the luminiferous ether. Superfluid vacuum theory (SVT) models spacetime as a Bose–Einstein condensate [14, 15], and emergent gravity approaches derive gravitational dynamics from thermodynamic or entropic considerations [16, 17].

ISPG shares the philosophical starting point of SVT and emergent gravity but differs in three critical respects:

1. **No extra dimensions.** Unlike string theory, ISPG operates in standard $(3+1)$-dimensional spacetime. The diversity of particle species arises from the internal degrees of freedom of a single medium (analogous to longitudinal, transverse, and vortical modes of a fluid), not from compactified extra dimensions.
2. **Exact Lorentz invariance.** SVT generically breaks Lorentz symmetry at high energies through the preferred rest frame of the condensate. The ISPG bi-conformal structure preserves local Lorentz invariance by construction: the speed of light is locally invariant ($c_{\text{local}} = c$) despite position-dependent coordinate speed [1].

3. **A concrete mechanical origin of gravity.** Entropic gravity [16] derives gravitational force from information-theoretic considerations but does not specify a physical mechanism. ISPG provides one: the Bernoulli-type kinetic-energy transfer from resonant oscillations to the static pressure of the medium.

D. Structure of the paper

The logical center of the paper is the derivation of the quantum gravitational mechanism itself: localized oscillon sources, the Bernoulli pressure deficit, the static $1/r$ field, Newtonian recovery, and the source-following collective mode. The remaining sections develop broader consequences of the same scalar-metric picture for inertia, cosmology, particle structure, and quantum phenomenology. These sectors do not all play the same logical role: the gravity chain is the primary closed result of the paper, while several wider extensions are carried to the level needed here and are marked in the text where a separate full closure remains future work.

The paper is organized as follows.

Section II derives the canonical quantization of the coupled scalar-metric system using the ADM formalism, obtains the reduced Lagrangian and Hamiltonian constraint, analyses the constraint algebra and the relationship between reduced and Dirac quantization, derives the Wheeler–DeWitt equation, verifies the semi-classical limit via WKB expansion, and discusses the ultralocal kinetic structure and its implications.

Section III derives the spectrum of scalar-field excitations on the bi-conformal background, establishes the identification of particles as resonant modes, and argues for the all-orders quantum stability of the phantom sector within the pure-gravity constrained sector via kinematic ghost freedom, the structural constraint analysis of Appendix 11 (full constraint-algebra closure left to a separate technical proof), and non-perturbative boundedness on the constrained background. The matter-coupled extension is controlled through the EFT completion described below. It derives the structure of the quantum effective action, demonstrating shift-symmetry protection, controlled one-loop divergence structure with negligible physical corrections, and systematic EFT power counting at all loop orders. It then presents the structural argument for the quantum persistence of the scalar sensitivity ($s = 1/2$ at all loop orders for vacuum compact objects; leading-order in $|\varphi|$ for matter-filled bodies, with the exact non-perturbative matter-coupled cancellation marked open per MAIN/16), supporting suppressed scalar dipole radiation and a perturbatively controlled strong equivalence statement in the scalar sector.

Section IV derives the Bernoulli mechanism: the kinetic energy of particle resonances generates the static pressure deficit that manifests as the gravitational potential $\varphi = -r_{\mathrm{s}}/r$.

Section V obtains the inertial response $\mathbf{F} = -C_{\mathrm{in}}\, m_{\mathrm{eff}}\, \mathbf{a}$ ($C_{\mathrm{in}} = 1$ at leading order) from the retarded potential of an accelerated oscillon, exhibits m_{inertial} and $m_{\mathrm{gravitational}}$ as the same scalar-profile/Bernoulli-deficit bookkeeping within the weak-field, slow-acceleration, spherical equilibrium regime, connects inertia to medium hysteresis via a Klein–Gordon field, and provides a microscopic motivation for the effective hysteresis action template and its microscopic tracking stiffness from the quasi-normal mode structure of the oscillon.

Section VI analyzes entropy production through irreversible resonance spreading, derives the cosmological negative-feedback loop, shows that the echo-background energy density (observable dark energy) is dynamically regulated to a value compatible with observation on the perturbative branch, obtains the dark-energy equation of state $w(z) < -1$ (source-driven phantom) as a falsifiable test target rather than a current DESI confirmation, and addresses the earliest-epoch dynamics: quantum nucleation of the first oscillons from the quiescent medium, the onset cascade, singularity avoidance, and a homogeneous-background no-inflation scenario in which the horizon, flatness, and topological-defect issues are addressed at the background-consistency level while the primordial perturbation spectrum and superhorizon correlations remain open numerical work.

Section VII analyzes the spectrum of three-dimensional resonant modes, formulates the self-consistent nonlinear eigenvalue problem in dimensionless form, shows that the confining potential is a geometric property of the nonlinear wave equation independent of G, establishes the formal direct-cavity amplitude–frequency ansatz with gravitational self-energy corrections of order $(m/M_{\rm Pl})^2 \sim 10^{-45}$, presents the lattice formulation suitable for numerical solution, and connects the structural cavity sector to the MASS/Mathieu lepton ladder and the Koide consistency condition.

Section VIII derives the bi-conformal scaling of an effective electroweak VEV, $v(\varphi) = v_0\, e^{\varphi/2}$, and the broken-phase mass matrix $m_W(\varphi) = g\, v(\varphi)/2$, $m_Z(\varphi) = \sqrt{g^2 + g'^2}\, v(\varphi)/2$ once the SM electroweak sector (gauge algebra, couplings, and Higgs doublet structure) is supplied. The absolute value $v_0 = 246$ GeV and the first-principles map $\Phi_0 \to v_0$ remain open. This demonstrates that the standard tree-level electroweak mass ratios and local dispersion relations are preserved locally in the conditional embedding, with photon masslessness protected by the imported exact $U(1)_{\rm EM}$ Ward identity.

Section IX formulates quantum entanglement as correlated resonance structure and derives the gravitational decoherence rate.

Section X develops a proposed topological organization of particle species: fermion spin projection as vortex winding, antimatter as opposite-winding vortex label, and Pauli exclusion through vortex topology, with the field-equation and spin-statistics closures stated explicitly there.

Section XI isolates the gravitational-birefringence channel and its leading $(\nabla\varphi)^2/m_e^2$ scaling, with the numerical coefficient deferred to a dedicated one-loop calculation, and presents the cosmological variation estimate of the fine-structure constant α.

Section XII summarizes all quantitative predictions in a unified table and identifies the most immediately testable signatures.

Section XIV concludes.

E. Notation and conventions

We use the $(-,+,+,+)$ metric signature and natural units $\hbar = c = 1$ unless explicitly restored. The reduced Planck mass is $M_{\rm Pl} = (8\pi G)^{-1/2}$. Greek indices $\mu, \nu, \ldots$ run over $0, 1, 2, 3$; Latin indices $i, j, \ldots$ over spatial coordinates $1, 2, 3$. The d'Alembertian is $\Box \equiv g^{\mu\nu}\nabla_\mu\nabla_\nu$. The scalar field φ is dimensionless, with the static weak-field solution $\varphi = -r_{\rm s}/r$ where $r_{\rm s} = 2GM/c^2$ is the Schwarzschild radius. Equation numbers from the classical paper [1] are cited as (I.n).

II. CANONICAL QUANTIZATION OF THE SCALAR-METRIC SYSTEM

The subsequent sections treat φ as a quantum field on a classical background. Here we derive the canonical quantization of the *coupled* scalar-metric system, exploiting the fact that the bi-conformal ansatz reduces the full gravitational *configuration* space to a single scalar degree of freedom (the corresponding phase space is two-dimensional, (φ, π_φ)).

A. ADM decomposition of the bi-conformal metric

Following Arnowitt, Deser, and Misner [40], we foliate spacetime into spatial hypersurfaces Σ_t and write

$$ds^2 = -N^2\,dt^2 + h_{ij}(dx^i + N^i\,dt)(dx^j + N^j\,dt), \tag{4}$$

where N is the lapse function, N^i the shift vector, and h_{ij} the induced metric on Σ_t. The bi-conformal metric (1) fixes the spatial geometry and shift,

$$h_{ij} = e^{-\varphi}\,\delta_{ij}, \qquad N^i = 0, \tag{5}$$

with $\sqrt{h} = e^{-3\varphi/2}$, while the lapse N is retained as a free function for the canonical analysis; its on-shell value $N = e^{\varphi/2}$ follows from the classical field equations. The extrinsic curvature of Σ_t is

$$K_{ij} = \frac{1}{2N}\,\dot{h}_{ij} = -\frac{\dot\varphi\,e^{-\varphi}}{2N}\,\delta_{ij}, \tag{6}$$

giving

$$K \equiv h^{ij}K_{ij} = -\frac{3\dot\varphi}{2N}, \qquad K_{ij}K^{ij} = \frac{3\dot\varphi^2}{4N^2}, \tag{7}$$

and the kinetic combination

$$K_{ij}K^{ij} - K^2 = -\frac{3\dot\varphi^2}{2N^2}. \tag{8}$$

The spatial Ricci scalar of the conformally flat metric $h_{ij} = e^{2\omega}\delta_{ij}$ with $\omega = -\varphi/2$ follows from the standard n-dimensional conformal transformation $\bar{R} = e^{-2\omega}[-2(n-1)\nabla_0^2\omega - (n-1)(n-2)(\nabla_0\omega)^2]$ evaluated at $n = 3$ and $R_0 = 0$:

$${}^{(3)}\!R = e^{\varphi}\Big(2\,\nabla^2\varphi - \tfrac{1}{2}(\nabla\varphi)^2\Big), \tag{9}$$

where ∇^2 and ∇ denote the flat-space Laplacian and gradient.

B. Reduced action

Substituting the bi-conformal ADM quantities (5)–(9) into the full action (2), the ADM Lagrangian density $\mathcal{L} = (N\sqrt{h}/16\pi G)({}^{(3)}R + K_{ij}K^{ij} - K^2 + \frac{1}{2}\,g^{\mu\nu}\partial_\mu\varphi\,\partial_\nu\varphi)$ separates into four terms (using $g^{00} = -N^{-2}$, $g^{ij} = h^{ij} = e^{\varphi}\delta^{ij}$, and $N^i = 0$):

(i) Einstein–Hilbert kinetic: $\frac{N\sqrt{h}}{16\pi G}(K_{ij}K^{ij} - K^2) = -\frac{3\,e^{-3\varphi/2}\,\dot{\varphi}^2}{32\pi GN}$;

(ii) Einstein–Hilbert potential: $\frac{N\sqrt{h}}{16\pi G}\,{}^{(3)}R = \frac{N\,e^{-\varphi/2}}{16\pi G}(2\nabla^2\varphi - \frac{1}{2}(\nabla\varphi)^2)$;

(iii) Scalar kinetic: $-\frac{\sqrt{h}\,\dot{\varphi}^2}{32\pi GN} = -\frac{e^{-3\varphi/2}\,\dot{\varphi}^2}{32\pi GN}$;

(iv) Scalar gradient: $\frac{N\,e^{-\varphi/2}}{32\pi G}(\nabla\varphi)^2$.

A remarkable cancellation occurs: the $(\nabla\varphi)^2$ contribution from (ii) is $-\frac{N\,e^{-\varphi/2}}{32\pi G}(\nabla\varphi)^2$, which exactly cancels (iv). The four time-derivative contributions from (i) and (iii) sum to a single kinetic term, and the surviving spatial term gives the reduced Lagrangian density

$$\mathcal{L} = -\frac{e^{-3\varphi/2}}{8\pi GN}\,\dot{\varphi}^2 + \frac{N\,e^{-\varphi/2}}{8\pi G}\,\nabla^2\varphi. \tag{10}$$

The total kinetic coefficient is negative-definite, reflecting the combined effect of the gravitational trace constraint and the phantom kinetic sign in the action (2).

C. Hamiltonian formulation

The momentum conjugate to φ is

$$\pi_\varphi \equiv \frac{\partial\mathcal{L}}{\partial\dot{\varphi}} = -\frac{e^{-3\varphi/2}}{4\pi GN}\,\dot{\varphi}. \tag{11}$$

The overall negative sign is inherited from the negative total kinetic term. Inverting gives $\dot{\varphi} = -4\pi GN\,e^{3\varphi/2}\,\pi_\varphi$.

Performing the Legendre transform $\mathcal{H} = \pi_\varphi\dot{\varphi} - \mathcal{L}$ and expressing the result in terms of phase-space variables:

$$H = \int d^3x\; N(\mathbf{x})\,\mathcal{H}_0(\mathbf{x}), \tag{12}$$

where the Hamiltonian constraint density is

$$\mathcal{H}_0 = -2\pi G\,e^{3\varphi/2}\,\pi_\varphi^2 - \frac{e^{-\varphi/2}}{8\pi G}\,\nabla^2\varphi \approx 0. \tag{13}$$

Here $\approx$ denotes weak equality on the constraint surface; the lapse N enters (12) as a Lagrange multiplier enforcing $\mathcal{H}_0 = 0$ at every spatial point.

Consistency check.—For a static configuration ($\pi_\varphi = 0$), the constraint reduces to $\nabla^2\varphi = 0$, the vacuum field equation $\Box\varphi = 0$ evaluated on the static bi-conformal background. The weak-field solution $\varphi = -r_s/r$ satisfies $\nabla^2\varphi = 0$ for $r > 0$, confirming consistency with the classical theory [1].

Phase-space restriction.—For dynamical configurations ($\pi_\varphi \neq 0$), the constraint requires $\nabla^2\varphi \leq 0$: the kinetic term $-2\pi G\, e^{3\varphi/2}\pi_\varphi^2$ is non-positive, so the potential term must be non-negative. This condition is automatically preserved by the Hamilton equations and is satisfied by all physically relevant configurations—static backgrounds ($\nabla^2\varphi = 0$), gravitationally bound perturbations (subharmonic near sources), and cosmological modes (damped oscillations on FLRW). The constraint thus acts as a self-consistency selection principle on the space of allowed initial data, analogous to the positive-energy theorem in general relativity.

D. Constraint structure

The bi-conformal ansatz $h_{ij} = e^{-\varphi}\delta_{ij}$ with $N^i = 0$ exhausts the spatial diffeomorphism gauge freedom: the spatial metric carries no transverse-traceless modes, and the momentum constraint $\mathcal{H}_i \approx 0$ is satisfied identically. The sole remaining constraint is the Hamiltonian constraint (13), which generates time reparametrizations via $\dot{F} = \{F,\, H[N]\}$ for any phase-space functional $F[\varphi, \pi_\varphi]$.

The algebra of smeared Hamiltonian constraints,

$$\{H[N],\, H[M]\} = H_{\text{diff}}[\vec{V}], \qquad V^i = h^{ij}(N\,\partial_j M - M\,\partial_j N), \tag{14}$$

reproduces the standard Dirac hypersurface-deformation algebra [41]. Since the momentum constraint vanishes identically in the reduced theory, $H_{\text{diff}}[\vec{V}] = 0$ and the constraint algebra is Abelian strongly (not merely on the constraint surface), confirming that the bi-conformal reduction introduces no anomalous constraints.

A methodological remark is in order. The quantization adopted here is of the “reduce first, then quantize” type: the bi-conformal ansatz is imposed classically, and the resulting single-field system is quantized. The alternative Dirac approach—quantize the full metric and scalar system, then impose $h_{ij} = e^{-\varphi}\delta_{ij}$ as a quantum constraint—is known to yield inequivalent results in general [42]. Two considerations favor the reduced approach for ISPG. First, the bi-conformal structure is not a gauge choice but a *symmetry reduction* that defines the theory; it is analogous to the minisuperspace truncation in quantum cosmology, where reduced quantization is standard practice [43]. Second, the Abelian constraint algebra demonstrated above guarantees that no ordering ambiguities arise in the constraint imposition, so the two approaches coincide at the semi-classical level [44].

E. Wheeler–DeWitt equation

Following DeWitt [41], canonical quantization promotes the Poisson bracket to the commutator

$$[\hat{\varphi}(\mathbf{x}),\, \hat{\pi}_\varphi(\mathbf{y})] = i\hbar\, \delta^{(3)}(\mathbf{x} - \mathbf{y}). \tag{15}$$

In the Schrödinger representation $\hat{\pi}_\varphi = -i\hbar\,\delta/\delta\varphi$, imposing the quantum Hamiltonian constraint $\hat{\mathcal{H}}_0\,\Psi = 0$ yields the Wheeler–DeWitt equation for the bi-conformal theory:

$$\left[2\pi G\hbar^2\, e^{3\varphi/2}\,\frac{\delta^2}{\delta\varphi(\mathbf{x})^2} - \frac{e^{-\varphi/2}}{8\pi G}\,\nabla^2\varphi(\mathbf{x})\right]\Psi[\varphi] = 0. \tag{16}$$

The factor $e^{3\varphi/2}$ does not commute with $\hat{\pi}_\varphi$, giving rise to an operator-ordering ambiguity. Equation (16) adopts the ordering $e^{3\varphi/2}\hat{\pi}_\varphi^2$; the Laplace–Beltrami ordering on the DeWitt supermetric $\mathcal{G}_{\varphi\varphi} = -(4\pi G)^{-1}e^{-3\varphi/2}$ would introduce an additional first-derivative term $\propto \hbar^2 e^{3\varphi/2}\,\delta/\delta\varphi$. The two orderings agree at the semi-classical level; the distinction is relevant only in the deep quantum regime and does not affect the perturbative results of subsequent sections.

F. Semi-classical limit

Write $\Psi[\varphi] = \mathcal{A}[\varphi]\,\exp\bigl(iS[\varphi]/\hbar\bigr)$ with real $\mathcal{A}$ and S. Substituting into the Wheeler–DeWitt equation (16) and noting that $\nabla^2\varphi$ acts multiplicatively on Ψ while $\delta^2/\delta\varphi^2$ produces

$$\frac{\delta^2\Psi}{\delta\varphi^2} = \left[\frac{\delta^2\mathcal{A}}{\delta\varphi^2} + \frac{2i}{\hbar}\frac{\delta\mathcal{A}}{\delta\varphi}\frac{\delta S}{\delta\varphi} + \frac{i}{\hbar}\mathcal{A}\frac{\delta^2 S}{\delta\varphi^2} - \frac{\mathcal{A}}{\hbar^2}\left(\frac{\delta S}{\delta\varphi}\right)^2\right] e^{iS/\hbar}, \tag{17}$$

we collect powers of $\hbar$.

Order $\hbar^0$ (Hamilton–Jacobi).—The leading term gives

$$-2\pi G\, e^{3\varphi/2}\left(\frac{\delta S}{\delta\varphi}\right)^2 - \frac{e^{-\varphi/2}}{8\pi G}\,\nabla^2\varphi = 0. \tag{18}$$

Identifying $\pi_\varphi = \delta S/\delta\varphi$, this is exactly the classical Hamiltonian constraint (13). The corresponding classical equations of motion, derived from the action (2) in Sec. III A, yield the wave equation

$$e^{-\varphi_0}\,\delta\ddot{\varphi} = e^{\varphi_0}\,\nabla^2\delta\varphi \tag{19}$$

for perturbations around a static background φ_0 with $\nabla^2\varphi_0 = 0$.

Order $\hbar^1$ (continuity).—The next-to-leading order yields

$$\frac{\delta}{\delta\varphi}\left(\mathcal{A}^2\,\frac{\delta S}{\delta\varphi}\right) = 0, \tag{20}$$

which is the continuity equation for the probability density $|\Psi|^2 = \mathcal{A}^2$ in configuration space, ensuring conservation of probability in the semi-classical regime.

The semi-classical quantization of Sec. III is therefore the correct leading approximation to the full quantum theory.

G. Physical interpretation

The canonical analysis reveals three structural features of the bi-conformal quantum theory:

1. *Degree-of-freedom reduction.* The bi-conformal ansatz collapses the six components of h_{ij} into a single scalar φ. The gravitational *configuration* space is one-dimensional per spatial point (phase space two-dimensional, (φ, π_φ)), dramatically simplifying the constraint structure relative to full quantum gravity.

2. *Phantom kinetic sign.* The negative total kinetic term in (10) reflects the phantom nature of φ. In the Wheeler–DeWitt equation (16), this converts the kinetic operator from elliptic to hyperbolic type, which is the feature that allows the constraint $\hat{\mathcal{H}}_0\Psi = 0$ to admit non-trivial, normalizable solutions with oscillatory behaviour in the scalar-field direction, enabling a well-posed initial-value formulation in superspace. Moreover, the hyperbolic signature provides a natural candidate for an internal time variable in superspace—a structure absent in standard (elliptic) quantum gravity, where the "problem of time" admits no straightforward resolution.

3. *Consistency with the classical theory.* The Hamiltonian constraint reproduces the vacuum field equation in the static limit, the semi-classical expansion recovers the wave equation (25), and the hypersurface-deformation algebra matches that of general relativity. The all-orders quantum stability of the phantom sector within the pure-gravity constrained sector is argued for in Sec. III G (full bracket-closure proof deferred; matter-coupled extension via the EFT completion).

4. *Ultralocality of the kinetic sector.* The exact cancellation of the $(\nabla\varphi)^2$ terms in the reduced action (10) renders the kinetic energy ultralocal: different spatial points are dynamically independent at the kinetic level. Spatial coupling re-enters exclusively through the $\nabla^2\varphi$ term in the Hamiltonian constraint (13). This property implies a natural lattice regularization in which each spatial site carries an independent copy of the one-dimensional quantum mechanics of φ, coupled to its neighbours only through the discrete Laplacian in the constraint—a structure well suited to the numerical eigenvalue problem for the mass spectrum (Sec. VII).

III. SCALAR-FIELD EXCITATIONS AS PARTICLES

The central thesis of this work is that elementary particles are localized resonant excitations of the scalar-metric medium. In this section we make this identification precise by quantizing the scalar field on the bi-conformal background, deriving the excitation spectrum, and establishing the position-dependent effective mass formula at the quantum level.

A. Wave equation on the bi-conformal background

Consider perturbations $\delta\varphi$ of the scalar field around a static background $\varphi_0(\mathbf{x})$:

$$\varphi = \varphi_0(\mathbf{x}) + \delta\varphi(t, \mathbf{x}). \tag{21}$$

The background satisfies the vacuum field equation $\Box\varphi_0 = 0$, and the perturbation satisfies the linearized equation

$$\Box_{\varphi_0}\,\delta\varphi = 0, \tag{22}$$

where $\Box_{\varphi_0}$ is the d'Alembertian evaluated on the background metric $g_{\mu\nu}(\varphi_0) = \mathrm{diag}(-e^{\varphi_0},\, e^{-\varphi_0},\, e^{-\varphi_0},\, e^{-\varphi_0})$.

Computing the d'Alembertian explicitly (in Cartesian spatial coordinates with $c = 1$):

$$\sqrt{-g} = e^{-\varphi_0}, \qquad g^{00} = -e^{-\varphi_0}, \qquad g^{ij} = e^{\varphi_0}\,\delta^{ij}, \tag{23}$$

and noting that $\sqrt{-g}\,g^{ij} = e^{-\varphi_0} \cdot e^{\varphi_0}\,\delta^{ij} = \delta^{ij}$ (the spatial prefactors cancel exactly), we obtain

$$\Box_{\varphi_0}\,\delta\varphi = e^{\varphi_0}\left[-e^{-2\varphi_0}\,\ddot{\delta\varphi} + \nabla^2\delta\varphi\right] = 0, \tag{24}$$

where $\ddot{f} \equiv \partial^2 f/\partial t^2$ and ∇^2 is the flat-space Laplacian. The cancellation of spatial prefactors is a remarkable property of the bi-conformal structure: it ensures that the spatial part of the wave operator is the *ordinary* Laplacian, with the entire effect of gravity concentrated in the temporal prefactor. Rearranging:

$$\boxed{e^{-\varphi_0}\,\ddot{\delta\varphi} = e^{\varphi_0}\,\nabla^2\delta\varphi.} \tag{25}$$

This gives the local propagation speed

$$c^2_{\mathrm{coord}} = e^{2\varphi_0}, \tag{26}$$

which varies with position. However, the locally measured speed—using proper time $d\tau = e^{\varphi_0/2}\,dt$ and proper length $d\ell = e^{-\varphi_0/2}\,dr$—is

$$c_{\mathrm{local}} = \frac{d\ell}{d\tau} = \frac{e^{-\varphi_0/2}}{e^{\varphi_0/2}}\,\frac{dr}{dt} = e^{-\varphi_0} \cdot e^{\varphi_0} = 1 = c, \tag{27}$$

confirming that local Lorentz invariance is preserved for all perturbation modes, as established in [1].

B. Mode decomposition in the static field

For the spherically symmetric background $\varphi_0 = -r_{\mathrm{s}}/r$ generated by a mass M, we seek time-harmonic solutions $\delta\varphi = \psi(\mathbf{x})\,e^{-i\omega t}$. Substituting into Eq. (25):

$$\nabla^2\psi + \omega^2\,e^{-2\varphi_0}\,\psi = 0. \tag{28}$$

For $\varphi_0 = -r_s/r$, the exponential is $e^{-2\varphi_0} = e^{2r_s/r}$. Expanding in spherical harmonics, $\psi(r,\theta,\phi) = \sum_{lm} u_l(r)/r \cdot Y_{lm}(\theta,\phi)$, the radial function u_l satisfies the Schrödinger-like equation

$$\boxed{u_l'' + \left[\omega^2\, e^{2r_s/r} - \frac{l(l+1)}{r^2}\right] u_l = 0,} \tag{29}$$

where $u_l'' \equiv d^2u_l/dr^2$. The key feature is the exponential factor $e^{2r_s/r}$ multiplying ω^2, which acts as a position-dependent effective frequency: modes are gravitationally *blue-shifted* near the source ($r \to r_s$, $e^{2r_s/r} \gg 1$) and approach the asymptotic value ($e^{2r_s/r} \to 1$) far from the source ($r \to \infty$).

In the weak-field limit ($r_s/r \ll 1$), the exponential can be expanded:

$$e^{2r_s/r} \approx 1 + \frac{2r_s}{r} + \frac{2r_s^2}{r^2} + \cdots\,, \tag{30}$$

and the radial equation reduces to

$$u_l'' + \left[\omega^2 + \frac{2\omega^2 r_s}{r} - \frac{l(l+1)}{r^2} + \cdots\right] u_l = 0. \tag{31}$$

This is formally identical to the radial Schrödinger equation for a particle of energy ω^2 in an *attractive* Coulomb-like potential $-2\omega^2 r_s/r$. The analogy is *structurally* exact: the ISPG radial equation shares the same form as the non-relativistic Schrödinger equation for a particle in a Coulomb potential, though the coupling constants here relate ω (scalar-wave energy) to GM rather than a particle mass m to GM. Modes of the scalar field in the gravitational potential therefore experience an effective attraction toward the source.

For $l = 0$ (s-wave), the solutions in the asymptotic region ($r \gg r_s$) are spherical waves:

$$u_0(r) \sim \sin(\omega r + \delta_0), \tag{32}$$

where δ_0 is the gravitationally induced phase shift. Near the source, the amplitude grows as $e^{r_s/r}$ – an exponential *amplitude* enhancement of the mode near the source; the corresponding mode energy as seen from infinity is given by Eq. (35) and is in fact *reduced* (gravitational redshift), not blue-shifted.

C. Energy of excitations and the effective mass

We now establish the central result connecting mode energy to the effective mass formula.

Coordinate and proper frequencies.—A mode oscillating as $\delta\varphi \propto e^{-i\omega t}$ has coordinate frequency ω. A local observer at position φ_0 measures proper time $\tau = e^{\varphi_0/2}\, t$, so the locally measured frequency is

$$\omega_{\text{local}} = \frac{d(\omega t)}{d\tau} = \omega\, e^{-\varphi_0/2}. \tag{33}$$

Local rest energy.—The rest energy of a localized excitation, as measured in its own frame, is

$$E_{\text{local}} = \hbar\,\omega_{\text{local}} = \hbar\omega\, e^{-\varphi_0/2} \equiv m_0 c^2, \tag{34}$$

which defines the bare mass m_0 as the locally measured rest mass. This is the mass that appears in all local experiments: atomic spectra, Compton scattering, and particle decays. It is independent of position because the definition is purely local.

Energy at infinity.—The energy of the same excitation as measured by an asymptotic observer ($\varphi \to 0$) is simply

$$E_\infty = \hbar\omega = m_0 c^2\, e^{\varphi_0/2}, \tag{35}$$

which yields the effective mass

$$\boxed{m_{\rm eff}(\varphi_0) = \frac{E_\infty}{c^2} = m_0\, e^{\varphi_0/2}.} \tag{36}$$

This reproduces Eq. (I.meff) of the classical theory [1], now derived from the quantum-mechanical energy of a scalar-field excitation rather than from the geodesic worldline interval. In the measurable-existence identification, the same result may be written as

$$\frac{\mathcal{N}_{\rm act}(\varphi_0)}{\mathcal{N}_{\rm act}(0)} = \frac{m_{\rm eff}(\varphi_0)}{m_0} = e^{\varphi_0/2}. \tag{37}$$

This does not introduce a new conserved charge or a microscopic particle count. It states that $m_{\rm eff}$ is the asymptotic mass/energy projection of the operational active measurable content carried by the localized excitation. The identification does not derive the discrete mass ratios of the charged leptons.

Physical interpretation.—In a region of lower pressure ($\varphi_0 < 0$), the resonant mode carries less energy as seen from infinity. The mode "vibrates on a looser string": the reduced background tension (pressure) diminishes the amplitude and energy of the excitation. This is the quantum-mechanical realization of the guitar-string analogy: a string under less tension produces a softer note (less energy), even though the local pitch (identity of the note, i.e., the particle species) remains the same.

D. Local undetectability: the quantum formulation

The classical theory established that the ratio of any two masses is position-independent, $m^A_{\rm eff}/m^B_{\rm eff} = m^A_0/m^B_0$ [1]. We now show that this extends to *all* quantum observables. Operationally, the cancellation occurs because the measured system and the measuring apparatus are built from the same interaction-active measurable content and therefore co-scale under a uniform background shift. A local experiment can compare only ratios internal to that co-scaled standard.

Transition frequencies.—Consider an atomic transition between two energy levels E_1 and E_2. The emitted photon frequency as measured locally is

$$\nu_{\rm local} = \frac{E_{2,\rm local} - E_{1,\rm local}}{h} = \frac{(E_2 - E_1)_0}{h}, \tag{38}$$

since both levels scale by the same factor $e^{\varphi_0/2}$, which cancels in the difference measured by local instruments. A local spectroscopist obtains the same spectrum regardless of the value of φ_0.

Compton wavelength.—In the local tetrad, the locally measured rest mass is m_0 and the locally measured speed of light is c (Eq. 27). The Compton wavelength measured against local rulers is therefore

$$\lambda_{C,\mathrm{local}} = \frac{h}{m_0\, c}, \tag{39}$$

which is the standard position-independent value. The coordinate representation follows from $\ell_{\mathrm{coord}} = e^{\varphi_0/2} \ell_{\mathrm{local}}$:

$$\lambda_{C,\mathrm{coord}} = e^{\varphi_0/2}\, \lambda_{C,\mathrm{local}} = \frac{h}{m_0 c}\, e^{\varphi_0/2} = \frac{h}{m_{\mathrm{eff}}\, c}. \tag{40}$$

The bi-conformal scaling moves λ_C in coordinate units; the local Compton wavelength is unchanged.

De Broglie wavelength.—In the local tetrad, a particle with locally measured momentum p_{local} has de Broglie wavelength

$$\lambda_{\mathrm{dB,local}} = \frac{h}{p_{\mathrm{local}}}, \tag{41}$$

the standard relation in local units, with no position dependence. The coordinate representation follows by converting both momentum and length: $p_{\mathrm{local}} = e^{\varphi_0/2} p_{\mathrm{coord}}$ and $\lambda_{\mathrm{coord}} = e^{\varphi_0/2} \lambda_{\mathrm{local}}$, so

$$\lambda_{\mathrm{dB,coord}} = e^{\varphi_0/2}\, \lambda_{\mathrm{dB,local}} = \frac{h}{p_{\mathrm{local}}\, e^{-\varphi_0/2}} = \frac{h}{p_{\mathrm{coord}}}. \tag{42}$$

Thus the de Broglie relation $\lambda = h/p$ holds separately in the local tetrad and in the coordinate chart.

General theorem.—All dimensionless quantum observables (branching ratios, coupling constants, cross-section ratios) are invariant under the bi-conformal rescaling $\varphi_0 \to \varphi_0 + \delta$. This follows from the fact that the bi-conformal transformation is a simultaneous rescaling of $\{m, E, \omega\} \to e^{\delta/2}\{m, E, \omega\}$ and $\{\ell, t\} \to e^{-\delta/2}\{\ell, t\}$, which preserves all dimensionless combinations.

This is the **quantum local undetectability principle**:

> *No quantum experiment confined to a local laboratory can detect the value of the background scalar field φ_0. The bi-conformal rescaling of masses, frequencies, and length scales is simultaneously absorbed by the rescaling of all measuring apparatus.*

E. Self-consistent resonances and particle stability

The linearized treatment of Sec. III B describes freely propagating scalar excitations. *Particles*, by contrast, are localized, self-sustaining resonant structures. Their existence requires the nonlinear self-interaction inherent in the bi-conformal coupling between φ and $g_{\mu\nu}$.

A localized excitation $\delta\varphi$ modifies the metric through Eq. (1), and the modified metric in turn affects the propagation of $\delta\varphi$ through Eq. (25). A self-consistent solution satisfies both simultaneously. Specifically, we seek time-periodic, localized solutions of the form

$$\varphi(t,r) = \Phi(r)\cos(\omega t) + \mathcal{O}\big(\Phi^2\big)\,, \tag{43}$$

where $\Phi(r) \to 0$ as $r \to \infty$.

To make the metric backreaction explicit, we expand the full field equation $\Box\varphi = 0$ to quadratic order in Φ, including the φ-dependent metric coefficients. The d'Alembertian on the bi-conformal background (Eq. 24) yields, for $\varphi = \varphi_0 + \Phi\cos(\omega t)$:

$$\Box\varphi = e^{\varphi_0}\big[\nabla^2\Phi + \omega^2 e^{-2\varphi_0}\Phi\big]\cos(\omega t) + \mathcal{O}\big(\Phi^2\big) = 0. \tag{44}$$

At order Φ^2, the backreaction modifies the effective potential for Φ through the exponential factors. The self-consistency condition becomes the *nonlinear Helmholtz equation*

$$\nabla^2\Phi - U'(\Phi) + \omega^2 e^{-2\varphi_0}\Phi + \frac{1}{2}\omega^2 e^{-2\varphi_0}\Phi^2 - \varphi_0'\,\Phi' + \mathcal{O}\big(\Phi^3\big) = 0, \tag{45}$$

where $\varphi_0' \equiv d\varphi_0/dr$ is the background gradient and $U'(\Phi)$ represents the effective potential from the metric coupling. The term $-\varphi_0'\Phi'$ acts as a position-dependent damping that localizes the solution.

Effective potential from metric backreaction.— The potential term $U'(\Phi)$ arises from the expansion of the exponential metric factors $e^{-2\varphi}$ to quadratic order in Φ. Expanding $e^{-2\varphi} = e^{-2\varphi_0}(1 - 2\Phi\cos\omega t + \mathcal{O}(\Phi^2))$ in the time-periodic d'Alembertian and time-averaging over one oscillation period yields an effective static potential:

$$U'(\Phi) = -\frac{1}{2}\omega^2 e^{-2\varphi_0}\Phi + \mathcal{O}\big(\Phi^2\big)\,, \tag{46}$$

where the linear term represents the frequency shift due to the background pressure gradient. The full nonlinear form of $U(\Phi)$ requires solving the coupled scalar-metric equations self-consistently.

Epistemic note on the linear sector.—The leading term in Eq. (46) carries a negative sign, so $U'(\Phi) \propto -\Phi$ at leading order: the quadratic part of $U(\Phi)$ is *tachyonic* ($m_{\text{eff}}^2 < 0$). A purely linear analysis therefore admits no bound state – stabilization of a localized configuration relies essentially on the nonlinear backreaction $+\frac{1}{2}\omega^2 e^{-2\varphi_0}\Phi^2$ entering Eq. (45), not on a positive quadratic confining potential. This distinguishes the ISPG oscillon from a bound state in a static well and aligns the sector with the Gleiser-type nonlinear-Klein–Gordon oscillons referenced below.

Connection to the 1D oscillon model.—To demonstrate that localized solutions exist, we derive a simplified 1D model from the ISPG field structure. Near the oscillon center ($r \ll r_{\text{s}}$), the background $\varphi_0 \approx 0$ and the exponential factors $e^{-2\varphi_0} \approx 1$. Defining dimensionless variables

$$x = \frac{r}{R_{\text{core}}}, \qquad \Phi = \frac{\phi}{\phi_0}, \qquad \Omega^2 = \omega^2 R_{\text{core}}^2, \tag{47}$$

where $R_{\text{core}} = \hbar/(mc)$ is the Compton natural scale (a *post-hoc* identification, fixed once m is known from the continuation below rather than supplied as an independent input – see also Eq. (46) and the cyclicity remark in the following subsection), and ϕ_0 is a *dimensionless* amplitude convention used to canonicalize the rescaling, not the physical weak-field value (which satisfies $\varphi \ll 1$ for elementary sources of mass $m \ll M_{\text{Pl}}$). In these variables, Eq. (45) reduces in the 1D (planar) limit to

$$\frac{d^2\Phi}{dx^2} - \Phi + \Phi^2 + \Omega^2\Phi = 0. \tag{48}$$

The nonlinear term Φ^2 arises from the quadratic backreaction in Eq. (45); the effective potential $U(\Phi) = \frac{1}{2}(1-\Omega^2)\Phi^2 - \frac{1}{3}\Phi^3$ has a local minimum at $\Phi = 0$ (oscillon core) separated by a barrier from the runaway region at large Φ. This equation admits the localized solution

$$\Phi(x) = A\,\text{sech}^2(Bx), \tag{49}$$

with $A = 3(1-\Omega^2)/2$ and $B = \sqrt{1-\Omega^2}/2$, provided $\Omega^2 < 1$ (bound state condition). This is the standard oscillon profile [18–20].

Epistemic status of the reduction.—We emphasize that the passage from Eq. (45) to Eq. (48) is an *identification* of the ISPG nonlinear Helmholtz sector with the canonical Gleiser-type oscillon equation, obtained through the dimensionless rescaling $\{x, \phi_0, \Omega\}$ that absorbs the effective coefficients of the linear and quadratic terms in Eq. (45) into the canonical form $-\Phi+\Phi^2+\Omega^2\Phi$. It is *not* a sign-preserving derivation in the original variables: the negative linear coefficient $-\Phi$ of Eq. (48) reflects the canonical oscillon rescaling, not a direct reading of the coefficient of Φ in Eq. (45). The substantive content carried by this identification is that the nonlinear sector of ISPG falls into the same mathematical class as the well-studied Bogolyubsky–Gleiser–Copeland oscillons, for which localized sech^2-type cores and 3D spherical continuation are established; the gravitational construction below uses only this existence of localized finite-energy sources, not the specific canonical coefficients.

3D spherical generalization.—The full 3D equation adds the radial Laplacian $\frac{2}{r}\Phi'$ and the damping $-\varphi_0'\Phi'$ from the background gradient. Numerical boundary-value solutions of the spherical problem exhibit regular localized profiles, with the core structure well-approximated by the 1D solution and the tail matching the outgoing behavior discussed in Sec. VI A. The restricted existence claim needed below is therefore this: the nonlinear bi-conformal equations admit localized finite-energy source configurations, so the scalar-metric medium contains concrete bound sources rather than only freely propagating waves. For the gravitational construction developed below, this is the essential output of the nonlinear sector. The stronger identification of these solutions with the full Standard-Model spectrum is logically separate and is not used in the derivation of gravity.

More concretely, the spherical boundary-value problem admits regular profiles under continuation in the central amplitude Φ_0, and linearization about these backgrounds yields a discrete cavity spectrum rather than a continuum of delocalized plane-wave modes. This is the precise numerical content needed in the present section: not a complete nonlinear time-evolution theorem, but the existence of localized source backgrounds on which the gravitational dressing and its long-range field can be built.

The stationary source statement used below can therefore be written in theorem-like form. Consider the spherical coupled boundary-value problem with central amplitude $\Phi(0) = \Phi_c$, regularity conditions $\Phi'(0) = \varphi_0'(0) = 0$, and localization conditions $\Phi(r_{\max}) = \varphi_0(r_{\max}) = 0$ at large $r_{\max}$. Numerical continuation in Φ_c produces smooth branches $(\Phi(r;\Phi_c), \varphi_0(r;\Phi_c), \Omega(\Phi_c))$ for which

$$\Phi(r) = \Phi_c + \mathcal{O}\big(r^2\big)\,, \qquad \varphi_0(r) = \varphi_0(0) + \mathcal{O}\big(r^2\big) \quad (r \to 0), \tag{50}$$

so the source is regular at the origin. On the bound branch ($\Omega^2 < 1$),

$$4\pi \int_0^\infty \Big[\tfrac{1}{2}\Omega^2\Phi^2 + \tfrac{1}{2}(\Phi')^2\Big] r^2\, dr < \infty, \qquad \Phi(r) \sim A\, e^{-\kappa r}/r, \tag{51}$$

so the source energy is finite and localized. Linearization about each such background yields isolated cavity eigenvalues below threshold rather than a continuum of delocalized modes. This is the precise theorem-like numerical statement used in the gravity section: ISPG contains a stationary spherical source sector of regular finite-mass oscillon backgrounds with a discrete small-oscillation spectrum. The continuation in Φ_c and the associated discrete cavity spectrum are verified numerically by the companion scripts `QUANTUM/Python/test_3d_bvp.py`, `test_cavity_spectrum.py`, `test_ispg_oscillon.py`, and `test_phi3_cont.py`, which provide the explicit $(\Phi(r;\Phi_c), \Omega(\Phi_c))$ branches and small-oscillation eigenvalues referred to here. This stationary statement is the base point for the modulation-control argument developed below; a full $3D + 1$ nonlinear orbital-stability theorem would extend the same source control from the present metastable regime to fully global evolution.

At the level needed in this section, the localized solutions fall into two classes:

- A **bound oscillon** is a regular finite-energy configuration with an exponentially localized core.
- A **quasi-bound oscillon** is a localized core accompanied by an outgoing tail, so that energy slowly leaks into propagating modes.
- The **decay width** Γ of a quasi-bound configuration is the imaginary part of the quasi-normal frequency $\omega = \omega_R - i\Gamma/2$, with lifetime $\tau = \hbar/\Gamma$.

Quasi-normal mode analysis.—For a quasi-bound oscillon, the scalar field has the time dependence $\Phi(t, r) = \phi(r)e^{-i\omega t}$ with complex frequency $\omega = \omega_R - i\Gamma/2$. The spatial profile

$\phi(r)$ satisfies the radial equation with outgoing boundary condition at infinity:

$$\frac{d^2\phi}{dr^2} + \frac{2}{r}\frac{d\phi}{dr} + \left[\omega^2 e^{-2\varphi_0(r)} - V_{\text{eff}}(r)\right]\phi = 0, \tag{52}$$

where $V_{\text{eff}}(r)$ is the effective potential from the nonlinear self-interaction. The outgoing condition $\phi(r) \sim e^{i\omega r}/r$ as $r \to \infty$ makes the frequency complex.

For a shallow bound state (weak nonlinearity), the decay width can be estimated by the WKB tunneling formula:

$$\Gamma \sim \omega_R \exp\left[-2\int_{r_1}^{r_2} \sqrt{V_{\text{eff}}(r) - \omega_R^2 e^{-2\varphi_0(r)}}\, dr\right], \tag{53}$$

where r_1 and r_2 are the classical turning points. The width is exponentially suppressed for deep bound states ($V_{\max} \gg \omega_R^2$), corresponding to long-lived particles like the muon ($\tau \sim 2.2\,\mu$s). For shallow states or high-mass resonances, the barrier penetration is efficient and the lifetime short (e.g., W/Z bosons with $\tau \sim 10^{-25}$ s).

The stability of a resonant mode depends on the background pressure φ_0. At higher pressure (φ_0 closer to zero), more energetic modes can be sustained—heavier particles are stable. As the pressure decreases over cosmic time (Sec. VI), previously stable modes may become quasi-bound and eventually decay. This provides a natural explanation for the evolving particle content of the universe: in the early universe, where the pressure was far higher, particles that are unstable today (e.g., the quark-gluon plasma phase) were stable, compatibly with the experimental recreation of that phase in heavy-ion collisions at CERN's ALICE facility [21].

Metastable modulation regime.—For the gravity construction below, the dynamical statement required is weaker than a full orbital-stability theorem but stronger than the mere existence of stationary profiles. Equations (160) and (168) imply that for deep bound or quasi-bound states with $\Gamma \ll \omega_R$, there exists a parametrically large time window

$$\omega_R^{-1} \ll t \ll \Gamma^{-1}, \tag{54}$$

in which the evolution remains close to the finite-dimensional oscillon manifold generated by translations and slow motion along the bound branch:

$$\varphi(\mathbf{x}, t) \simeq \varphi_{\text{grav}}\big(|\mathbf{x} - \mathbf{R}(t)|; \Phi_c(t)\big) + \Phi_0\big(|\mathbf{x} - \mathbf{R}(t)|; \Phi_c(t)\big)\cos\Theta(t) + \eta(\mathbf{x}, t). \tag{55}$$

Here $\mathbf{R}(t)$ is the collective position, $\Theta(t)$ the fast internal phase, and η the radiative remainder carried by damped quasi-normal modes. The translational channel is the $\ell = 1$ Goldstone mode proven below, while leakage into non-bound channels is controlled by the exponentially small width Γ. Thus, on all timescales relevant to the Newtonian far-zone field, the oscillon behaves as a *metastable particle source*: its center and phase may modulate adiabatically, and its parameters drift only on the long leakage timescale Γ^{-1}, while the gravitational dressing follows the same collective coordinates. A full $3D + 1$ nonlinear orbital-stability theorem would strengthen this metastable modulation statement to a global theorem, but is not required for the source concept used in the gravity section.

Neutral versus damped channels.—The linearized dynamical channels relevant to the source concept therefore split into two classes. First, the translational $\ell = 1$ Goldstone mode is *neutral*: it shifts the oscillon center without changing the internal energy, so it moves the source along the manifold rather than away from it. Second, the non-Goldstone perturbations admit the quasi-normal expansion discussed below, with mode amplitudes satisfying

$$\ddot{a}_n + \Gamma_n \dot{a}_n + \omega_n^2 a_n = f_n(t), \tag{56}$$

where $\Gamma_n \geq 0$ measures leakage into radiative channels. For stable particles, $\Gamma_n \ll \omega_n$ and slow forcing couples dominantly to the lowest mode, so higher overtones are suppressed rather than amplified. Thus, modulo the neutral symmetry directions, the oscillon source sector is *perturbatively attractive* on the timescale relevant to the gravitational field: deformations damp or remain adiabatically slaved to the collective coordinates instead of generating runaway growth.

Gravity-use source-control statement.—The dynamical claim actually needed in Sec. IV can now be stated in modulation form. Let

$$\mathcal{M}_{\rm osc} \equiv \Big\{ \varphi_{\rm grav}\big(|\mathbf{x}-\mathbf{R}|;\Phi_c\big) + \Phi_0\big(|\mathbf{x}-\mathbf{R}|;\Phi_c\big)\cos\Theta \Big\} \tag{57}$$

denote the finite-dimensional oscillon manifold generated by translations, motion along the bound branch, and phase shifts of the leading harmonic. For a field configuration that remains near this manifold in the metastable regime, write

$$\varphi(\mathbf{x},t) = \varphi_{\rm grav}\big(|\mathbf{x}-\mathbf{R}(t)|;\Phi_c(t)\big) + \Phi_0\big(|\mathbf{x}-\mathbf{R}(t)|;\Phi_c(t)\big)\cos\Theta(t) + \eta(\mathbf{x},t). \tag{58}$$

Choose $\mathbf{R}(t)$, $\Phi_c(t)$, and $\Theta(t)$ so that the remainder is orthogonal to the tangent directions of $\mathcal{M}_{\rm osc}$:

$$\langle \eta, T_i \rangle = 0, \qquad \langle \eta, T_c \rangle = 0, \qquad \langle \eta, T_\Theta \rangle = 0, \tag{59}$$

where

$$T_i \equiv \partial_{R_i}\Big[\varphi_{\rm grav}\big(|\mathbf{x}-\mathbf{R}|;\Phi_c\big) + \Phi_0\big(|\mathbf{x}-\mathbf{R}|;\Phi_c\big)\cos\Theta\Big], \tag{60}$$

$$T_c \equiv \partial_{\Phi_c}\Big[\varphi_{\rm grav}\big(|\mathbf{x}-\mathbf{R}|;\Phi_c\big) + \Phi_0\big(|\mathbf{x}-\mathbf{R}|;\Phi_c\big)\cos\Theta\Big], \tag{61}$$

$$T_\Theta \equiv \partial_\Theta\Big[\Phi_0\big(|\mathbf{x}-\mathbf{R}|;\Phi_c\big)\cos\Theta\Big], \tag{62}$$

and $\langle f, g\rangle \equiv \int_{\mathbb{R}^3} f(\mathbf{x}) g(\mathbf{x})\, d^3x$. At leading harmonic order, T_i and T_c reduce to the familiar translation and branch directions $\partial_i\Phi_0$ and $\partial_{\Phi_c}\Phi_0$, while T_Θ is the phase tangent. Thus the neutral collective coordinates are absorbed into $\mathbf{R}(t)$, $\Phi_c(t)$, and $\Theta(t)$ rather than double-counted inside η.

Projecting the linearized dynamics onto the complement of these tangent directions gives a discrete quasi-normal expansion

$$\eta(\mathbf{x},t) = \sum_{n \notin \rm neutral} a_n(t)\, \phi_n(\mathbf{x}), \tag{63}$$

with amplitudes satisfying Eq. (168). Define the quadratic linearized source energy

$$E_{\rm lin}[\eta] \equiv \frac{1}{2} \sum_{n\notin\text{neutral}} \left(\dot a_n^2 + \omega_n^2 a_n^2\right). \tag{64}$$

Differentiating and using Eq. (168) yields the exact identity

$$\frac{d}{dt} E_{\rm lin}[\eta] = -\sum_{n\notin\text{neutral}} \Gamma_n\, \dot a_n^2 + \sum_{n\notin\text{neutral}} \dot a_n\, f_n(t). \tag{65}$$

Introducing the projected forcing norm

$$F_{\rm ad}(t) \equiv \left(\sum_{n\notin\text{neutral}} |f_n(t)|^2\right)^{1/2} = \mathcal{O}\Big(|\dot{\mathbf{R}}(t)|^2 + |\dot\Phi_c(t)|^2 + E_{\rm lin}[\eta]^{1/2}\Big), \tag{66}$$

which expresses that, after removal of the tangent directions, non-neutral forcing starts only at higher order in the adiabatic drift and in the remainder itself. By Cauchy–Schwarz,

$$\frac{d}{dt} E_{\rm lin}[\eta] \le -\sum_{n\notin\text{neutral}} \Gamma_n\, \dot a_n^2 + E_{\rm lin}[\eta]^{1/2}\, F_{\rm ad}(t). \tag{67}$$

Coercive reduced stability estimate.—Let $\eta_\perp$ denote the remainder after projection away from the neutral tangent directions. The self-adjoint part of the linearized operator on this reduced subspace is diagonal in the mode basis:

$$\mathcal{L}_\perp \phi_n = \omega_n^2\, \phi_n, \qquad n \notin \text{neutral}. \tag{68}$$

Because the neutral symmetry directions have been removed and the remaining discrete branch is the stable source branch used below, there is a positive reduced spectral gap

$$\lambda_{\rm gap} \equiv \inf_{n\notin\text{neutral}} \omega_n^2 > 0. \tag{69}$$

Define the coercive source energy by

$$E_{\rm coer}[\eta_\perp] \equiv \frac{1}{2}\, \|\partial_t \eta_\perp\|_{L^2}^2 + \frac{1}{2}\, \langle \eta_\perp, \mathcal{L}_\perp \eta_\perp\rangle = \frac{1}{2} \sum_{n\notin\text{neutral}} \left(\dot a_n^2 + \omega_n^2 a_n^2\right). \tag{70}$$

Therefore

$$E_{\rm coer}[\eta_\perp] \ge \frac{1}{2}\, \|\partial_t \eta_\perp\|_{L^2}^2 + \frac{\lambda_{\rm gap}}{2}\, \|\eta_\perp\|_{L^2}^2, \tag{71}$$

so every non-neutral deformation costs strictly positive quadratic energy. Combining Eq. (65) with Cauchy–Schwarz now gives

$$\frac{d}{dt} E_{\rm coer}[\eta_\perp] \le \sqrt{2E_{\rm coer}[\eta_\perp]}\, F_{\rm ad}(t), \tag{72}$$

and hence

$$E_{\rm coer}[\eta_\perp](t)^{1/2} \le E_{\rm coer}[\eta_\perp](0)^{1/2} + \frac{1}{\sqrt{2}} \int_0^t F_{\rm ad}(s)\, ds. \tag{73}$$

This is the coercive stability statement needed by the gravity mechanism: after removing translation, branch drift, and phase, the remainder cannot wander at zero cost. It is separated from the source manifold by the positive gap $\lambda_{\rm gap}$ and stays small under adiabatic forcing if it is initially small. Hence, for sufficiently small initial remainder and adiabatic motion along $\mathcal{M}_{\rm osc}$, Grönwall's inequality implies that $E_{\rm lin}[\eta]$ remains small throughout the same window

$$\omega_R^{-1} \ll t \ll \Gamma^{-1}, \tag{74}$$

with Γ the leakage scale of the least-damped non-neutral channel. This is the precise source-control statement used by the gravity section: the only neutral drift relevant to the static source is translation of the localized core and slow motion along the bound branch, while the fast phase is averaged out and non-neutral deformations are damped or remain perturbatively slaved to those collective coordinates. The far-zone gravitational field therefore follows the same $\mathbf{R}(t)$ that moves the core.

Projected collective-coordinate dynamics.—The orthogonality conditions (59) not only define the modulation parameters; they also determine their dynamics. Let

$$q_A(t) \equiv \big(R_1, R_2, R_3, \Phi_c, \Theta\big). \tag{75}$$

Projecting the full field equation onto the tangent basis $\{T_A\}$ gives the finite-dimensional modulation system

$$\sum_B \mathcal{M}_{AB}(q)\,\ddot{q}_B + \sum_B \mathcal{C}_{AB}(q,\dot{q})\,\dot{q}_B = \mathcal{F}_A^{\rm ext}(t) + \mathcal{R}_A[\eta], \qquad \mathcal{M}_{AB}(q) \equiv \langle T_A, T_B\rangle, \tag{76}$$

where $\mathcal{C}_{AB}$ comes from differentiating the tangent bundle along $\mathcal{M}_{\rm osc}$, $\mathcal{F}_A^{\rm ext}$ is the adiabatic forcing projected on T_A, and $\mathcal{R}_A[\eta] = \mathcal{O}\big(E_{\rm lin}[\eta]^{1/2}\big)$ is the residual backreaction of the non-neutral remainder. Because the Gram matrix $\mathcal{M}_{AB}$ is positive definite on the reduced tangent space, the collective drift is controlled by

$$\ddot{\mathbf{R}} = \mathcal{O}\big(F_{\rm ad} + E_{\rm lin}[\eta]^{1/2}\big)\,, \qquad \dot{\Phi}_c = \mathcal{O}\big(\Gamma + F_{\rm ad} + E_{\rm lin}[\eta]^{1/2}\big)\,, \qquad \dot{\Theta} = \Omega(\Phi_c) + \mathcal{O}\big(F_{\rm ad} + E_{\rm lin}[\eta]^{1/2}\big)\,. \tag{77}$$

Thus translations remain the neutral channel, branch drift occurs only on the long leakage timescale, and the phase follows the instantaneous bound frequency up to adiabatic corrections.

Numerical reduced-gap check.—For the reference background used in the cavity scan below ($\Phi_0 = 2.35$), the translation mode is the only neutral direction, while the first non-neutral bound mode occurs at

$$\omega_{\rm min,nn} = 0.663853, \qquad \lambda_{\rm gap} \approx \omega_{\rm min,nn}^2 = 0.440701. \tag{78}$$

Thus the reduced source sector is not merely qualitatively but also numerically separated from zero by a finite gap: after projecting out the translational mode, no additional soft deformation competes with the source manifold.

Far-zone closure for a moving source.—At distances $r \gg \kappa^{-1}$, the oscillatory core is exponentially small and the remainder contributes only through the controlled quantity $E_{\rm lin}[\eta]$. Therefore the static part of the field generated by the modulated source has the universal form

$$\varphi_{\rm grav}(\mathbf{x}, t) = -\frac{2G\, M_{\rm grav}(\Phi_c(t))}{c^2\, |\mathbf{x} - \mathbf{R}(t)|} + \mathcal{O}\big(E_{\rm lin}[\eta]^{1/2}\big) + \mathcal{O}\big(e^{-\kappa|\mathbf{x}-\mathbf{R}(t)|}\big)\,, \tag{79}$$

and hence

$$\partial_t \varphi_{\rm grav} = -\dot{\mathbf{R}}(t)\cdot\nabla\varphi_{\rm grav} + \dot{\Phi}_c(t)\, \partial_{\Phi_c}\varphi_{\rm grav} + \mathcal{O}\big(E_{\rm lin}[\eta]^{1/2}\big)\,. \tag{80}$$

On timescales $t \ll \Gamma^{-1}$ the mass drift $M_{\rm grav}(\Phi_c(t))$ is adiabatic, so the leading effect is rigid translation of the universal $1/r$ field. This closes the single-source dynamical statement needed by the gravity mechanism: the moving oscillon core, its source term, and its far-zone field are carried by one and the same collective coordinate $\mathbf{R}(t)$.

F. Internal degrees of freedom and particle diversity

A single scalar field admits only spin-0 excitations. The full particle spectrum—with spins 0, $\frac{1}{2}$, 1, and 2—requires the additional degrees of freedom present in the coupled scalar-metric system. We emphasize at the outset that the *scalar–vector–tensor (SVT) decomposition* developed in this section supplies the bosonic sectors only ($s = 0, 1, 2$); the half-integer (fermion) sector of the spectrum is *not* read off from the SVT sectors but is treated below through a candidate vortex-topology route in the oscillon envelope, using Komar angular momentum and Onsager–Feynman circulation quantization in Sec. X A. The decomposition below should therefore be read as the bosonic half of the spectrum argument, complemented by the vortex construction of Sec. X.

The bi-conformal metric (1) is determined by one scalar function φ on the *background*, but general perturbations decompose into three independent sectors:

1. **Scalar perturbations** ($\delta\varphi$): longitudinal, compressional modes of the medium. These are the breathing modes analyzed above.

2. **Tensor perturbations** (h_{ij}^{TT}): transverse-traceless deformations of the spatial metric. These are the two TT polarizations h_+ and $h_\times$, which propagate at c as established in [1].

3. **Vector perturbations** (δg_{0i}): rotational, vortical modes of the medium. These are frame-dragging degrees of freedom, analogous to vortices in a fluid.

This decomposition—longitudinal (scalar), transverse (tensor), and rotational (vector)—parallels exactly the three independent modes of a continuous medium (sound waves, shear waves, and vortices). The analogy is not superficial: it reflects the mathematical structure of the SVT (scalar-vector-tensor) decomposition theorem for perturbations of a Lorentzian manifold [22, 23].

The diversity of particle species arises from combinations of these bosonic excitation types together with the vortex topology of the oscillon sector introduced in Sec. III E. The latter supplies two ingredients that the SVT sectors alone do not: (i) an integer vortex winding number $n \in \mathbb{Z}$, which supports a conditional Komar-normalization route to the half-integer axial spin-projection quantum $J = n\hbar/2$ for $n = \pm 1$ once the scalar-phase $\to$ gravitomagnetic-slot bridge is supplied (the fermion sector, Sec. X A), and (ii) a Berry-exchange phase for two such vortices, which is proposed as the topological route to antisymmetric exchange statistics and the Pauli exclusion principle (Sec. X C). The full topological classification—spin, matter versus antimatter (vortex sign), and exchange statistics—is developed in Sec. X.

Remark.—This framework requires no extra dimensions: the excitation types above (plus the vortex sector of Sec. X) are all supported by a three-dimensional medium, just as a fluid supports diverse wave phenomena (acoustic, transverse, vortical) without additional spatial dimensions. The *epistemic status* of the internal quantum numbers is layered, however. The construction above derives (i) the bosonic sectors $s = 0, 1, 2$ from SVT and (ii) integer vortex winding on the complex oscillon envelope. It then gives a conditional route from that winding to the fermionic axial spin-projection quantum $J = \hbar/2$, pending the field-equation derivation of the scalar-phase $\to$ gravitomagnetic-slot bridge noted in Sec. X A. Matter versus antimatter from the sign of the winding (Sec. X B) and Pauli exclusion from the Berry-exchange phase (Sec. X C) are structural proposals whose closures are treated as separate open steps below. The remaining Standard-Model internal structure—electric charge, color $SU(3)_c$, and flavor—is *not* derived here: the expectation that these likewise arise from topological invariants and coupled mode structure in the 3D scalar-metric medium is a *working hypothesis* in the spirit of the vortex analogy, and a full topological/representation-theoretic derivation of the SM gauge content $SU(3)_c \times SU(2)_L \times U(1)_Y$ from ISPG is left for future work.

G. Quantum stability of the phantom sector

The ISPG scalar field has phantom kinetic sign ($\mathcal{K} > 0$ in Eq. 99), which classically leads to a stable, singularity-free theory due to the bi-conformal coupling [1]. In a generic phantom theory, quantum corrections destabilize the vacuum through ghost pair production at a rate set by the UV cutoff [45, 46]. Below we give the argument that this instability is absent in ISPG within the pure-gravity constrained sector to all orders in the loop expansion (matter-coupled extension is controlled by the EFT completion of Appendix 11; full bracket-closure/anomaly proof deferred per A.5). This question belongs to the consistency of the full quantum extension and is logically downstream of the oscillon-to-gravity construction developed above, which requires only localized finite-energy sources and their exterior field.

Kinematic ghost freedom.—On the bi-conformal background, the constraint reduces the gravitational phase space from three propagating degrees of freedom (two tensor, one scalar) to a single scalar mode φ, as established by the Dirac–Bergmann analysis of [1] and confirmed by the canonical reduction in Sec. II B. (General perturbations that break the bi-conformal symmetry—gravitational waves—propagate in the unconstrained 3-DOF sector; their classical stability follows from the strong hyperbolicity of the principal symbol proven in [1].) Ghost pair production—the process $|0\rangle \to |\text{ghost}\rangle + |\text{tensor}\rangle$—requires at least two independent propagating fields: one carrying negative energy (the ghost) and one carrying positive energy (the tensor mode), so that their sum matches the vacuum energy. On the constraint surface, no such partner exists: the tensor modes are fully determined by φ through the bi-conformal conditions, and the momentum constraint vanishes identically (Sec. II D). This elimination is *kinematic*, not dynamical: it is a property of the constraint surface and holds independently of the loop order.

Anomaly freedom of the constraint.—The kinematic argument above is valid at all orders only if the constraint is not broken by quantum corrections (a constraint anomaly). In the present theory, the smeared Hamiltonian constraint algebra is Abelian (Sec. II D, Eq. 14): the structure *constants* on the right-hand side vanish identically, rather than being field-dependent structure *functions*. Quantum anomalies in constraint algebras arise from the regularization of products of structure functions at coincident points [41]; with vanishing structure functions, no such regularization ambiguity arises in the kinematic argument. At the structural level (Appendix 11), the Abelian algebra is therefore expected to remain anomaly-free at the quantum level, with the single-DOF reduction persisting at all orders; a fully rigorous bracket-closure/anomaly proof is deferred to a separate technical appendix.

Unitarity of the physical S-matrix.—The optical theorem requires

$$2\,\mathrm{Im}\,\mathcal{M}(i \to i) = \sum_{f \in \mathcal{H}_{\text{phys}}} |\mathcal{M}(i \to f)|^2, \tag{81}$$

where the sum runs over the physical Hilbert space $\mathcal{H}_{\text{phys}}$. In the constrained theory, the pure-gravity sector of $\mathcal{H}_{\text{phys}}$ contains only the φ excitations—no independent ghost or tensor gravitational state contributes to the intermediate-state sum. Matter sectors of $\mathcal{H}_{\text{phys}}$ (fermions, gauge fields minimally coupled to $g_{\mu\nu}[\varphi]$) enter the sum as ordinary positive-norm contributions, so their total inclusion preserves the non-negativity of the right-hand side of (81) provided the φ–matter coupling is gravitational (Planck-suppressed). The right-hand side is therefore manifestly non-negative at every loop order in the Planck-suppressed expansion, supporting perturbative unitarity of the S-matrix order-by-order within the EFT regime; the full bookkeeping of mixed matter-ghost configurations belongs to the EFT completion of Appendix 11 and is not asserted here as an exact non-perturbative unitarity theorem. This is the same mechanism by which cuscuton gravity [47] and mimetic gravity [48] achieve ghost freedom: kinematic removal by constraint, not dynamical stabilization. The ISPG case is distinguished by the Abelian constraint algebra, which further excludes the ordering ambiguities that could reintroduce ghosts in non-Abelian constrained systems.

Perturbative suppression of loop corrections.—Beyond the structural all-orders argument, the *magnitude* of quantum corrections is controlled by the EFT expansion parameter. The scalar perturbation on the bi-conformal background has dispersion relation $\omega^2 = k^2$

(Sec. III A), so the propagator

$$\langle \delta\varphi(x)\, \delta\varphi(y)\rangle = -\int \frac{d^4k}{(2\pi)^4}\, \frac{i}{k^2 + i\epsilon} \tag{82}$$

has the same pole structure as a canonical massless scalar but with a negative residue—the hallmark of the phantom kinetic sign. Within the pure-gravity sector (where φ is the only propagating mode), the negative residue does not lead to a unitarity deficit: the physical inner product on the φ-subsector of $\mathcal{H}_{\rm phys}$ can be redefined by absorbing the sign (equivalently, the field redefinition $\tilde{\varphi} = i\varphi$ maps the pure-gravity sector to a canonical scalar with modified vertex signs). This redefinition does not simultaneously rotate the positive-norm matter sector, so the combined theory retains mixed-signature states whose consistency is controlled by the gravitational suppression of φ–matter couplings (see "Kinematic ghost freedom" above). At n-loop order, the corrections to any observable scale as $(E/M_{\rm Pl})^{2n}$, where E is the characteristic energy of the process. On the classical background $\varphi_0 = -r_{\rm s}/r$, the relevant energy scale is set by the local curvature, $E \sim |\varphi_0'| = r_{\rm s}/r^2$. Near a neutron star surface ($r \sim 10$ km), this gives $E \sim 10^{-10}$ eV, so the loop expansion parameter is

$$\left(\frac{E}{M_{\rm Pl}}\right)^2 \sim 10^{-76}, \tag{83}$$

placing the theory deep within the controlled EFT regime at every astrophysically relevant scale.

Non-perturbative stability.—The Hamiltonian constraint $\mathcal{H}_0 \approx 0$ (Eq. 13) restricts all physical configurations of the pure-gravity sector to the zero-energy surface; no runaway to arbitrarily negative energy is possible *within this sector.* With matter included, the constraint reads $\mathcal{H}_\varphi + \mathcal{H}_m \approx 0$, so a bookkeeping configuration with $\mathcal{H}_\varphi \to -\infty$ balanced by $\mathcal{H}_m \to +\infty$ is in principle allowed; such a runaway is kinematically controlled by the Planck-suppressed gravitational φ–matter coupling strength rather than by the constraint itself, and is deferred to the EFT-completion analysis above. Additionally, the energy density $\rho_\varphi = e^{\varphi_0}|\nabla\varphi_0|^2/(32\pi G) \geq 0$ (Lemma A 2) is non-negative, and $\rho_\varphi \to 0$ as $\varphi \to -\infty$ due to the e^φ prefactor, so the effective potential for φ is bounded from above with no runaway direction. In the path-integral formulation, the functional integral is restricted to field histories satisfying the constraint; on this restricted configuration space, the classical solutions are local minima of the action and the quadratic fluctuation operator has no negative eigenvalues that would signal a tunneling instability. Tunneling from the ISPG vacuum to any non-perturbative configuration would require traversal of a field-space barrier with action $\Delta S \gg \hbar$, exponentially suppressing such transitions. The global structure of the constrained configuration space therefore admits no non-perturbative vacuum instability.

Summary.—The argument for all-orders quantum stability of the phantom sector within the pure-gravity constrained sector rests on three independent pillars:

(i) *Kinematic ghost freedom*: the single-DOF constraint eliminates the pair-production channel at all loop orders;

(ii) *Structurally Abelian constraint algebra*: the structural argument of Appendix 11 (full bracket-closure proof deferred) suggests that the constraint preserves the DOF reduction at the quantum level;

(iii) *Non-perturbative boundedness on the constrained background*: the Hamiltonian constraint confines all physical configurations to the zero-energy surface, and the bounded effective potential excludes tunneling to runaway configurations within the pure-gravity constrained sector.

Together, these arguments suggest that the phantom kinetic sign in ISPG does not compromise quantum consistency within the pure-gravity constrained sector; matter-coupled extension is controlled by the EFT completion. On the contrary, as discussed in Sec. II G, the phantom sign is the very feature that renders the Wheeler–DeWitt equation hyperbolic, enabling non-trivial quantum states and providing a natural candidate for an internal time variable in superspace.

H. Structure of the quantum effective action

Having outlined the structural argument for all-orders stability of the theory within the pure-gravity constrained sector, we now derive the explicit structure of the quantum effective action $\Gamma[\varphi_0]$, which governs the leading quantum corrections to all classical ISPG predictions.

Background field expansion.—Split the scalar field into a classical background and a quantum fluctuation,

$$\varphi(x) = \varphi_0(x) + \delta\varphi(x), \tag{84}$$

and expand the action (2) to quadratic order in $\delta\varphi$. On the bi-conformal background the fluctuation operator takes the form

$$\mathcal{D}(\varphi_0) = -\Box + \mathcal{V}(\varphi_0), \tag{85}$$

where $\Box = g^{\mu\nu}\nabla_\mu\nabla_\nu$ is the covariant d'Alembertian on the background metric $g_{\mu\nu}(\varphi_0)$ and $\mathcal{V}$ collects the curvature-dependent potential terms arising from the expansion of $\sqrt{-g}\,R$.

The effective action through one loop is then

$$\Gamma[\varphi_0] = S[\varphi_0] + \frac{\hbar}{2}\,\mathrm{Tr}\ln\mathcal{D}(\varphi_0) + \mathcal{O}(\hbar^2), \tag{86}$$

where the functional trace includes integration over spacetime and summation over all fluctuation modes (scalar and tensor); we use the standard loop-expansion convention $\Gamma = S + \sum_{n\geq 1}\hbar^n\,\Gamma^{(n)}$, so the one-loop term proper is $\Gamma^{(1)} = \frac{1}{2}\mathrm{Tr}\ln\mathcal{D}$.

Shift symmetry and radiative protection.—At the unreduced action level, the ISPG action (2) possesses a continuous shift symmetry $\varphi \to \varphi + \text{const}$: the scalar field enters only through its derivatives $\partial_\mu\varphi$. This symmetry forbids the perturbative radiative generation of a mass term $m^2\varphi^2$ or any non-derivative potential $V(\varphi)$ at any loop order. The shift-symmetric counterterms include not only the pure-kinetic powers $(\partial\varphi)^{2n}$, but also curvature-derivative operators such as $R\,(\partial\varphi)^2$, $R_{\mu\nu}\partial^\mu\varphi\,\partial^\nu\varphi$, $(\Box\varphi)^2$, and combinations thereof, each suppressed by additional powers of $M_{\rm Pl}^{-2}$:

$$\Delta\mathcal{L}_{\rm ct} = \sum_{n=2}^{\infty} \frac{c_n}{M_{\rm Pl}^{2(n-1)}}\,(\partial_\mu\varphi\,\partial^\mu\varphi)^n + (\text{curvature-derivative shift-invariants}). \tag{87}$$

This radiative stability is the same mechanism that protects the inflaton mass in shift-symmetric inflation [49] and the Galileon field in Horndeski gravity. In the reduced bi-conformal sector, the fluctuation operator $\mathcal{D}(\varphi_0)$ acquires explicit $e^{\pm\varphi_0}$ dependence, so the shift symmetry of the reduced theory is preserved only at EFT precision; a fully rigorous all-loop Ward identity on the constrained sector is deferred to a separate technical analysis. For ISPG, the perturbative argument supports the masslessness of φ and the long-range nature of gravity at all orders within the EFT regime.

One-loop divergence structure.—The one-loop divergences of the coupled scalar-metric system are obtained via the Seeley–DeWitt heat-kernel expansion [41, 50]. In dimensional regularization ($d = 4 - 2\epsilon$), the divergent part of (86) takes the form

$$\Gamma^{(1)}_{\rm div} = -\frac{1}{16\pi^2\epsilon}\int d^4x\,\sqrt{-g}\,\left[\alpha\, C^2_{\mu\nu\rho\sigma} + \beta\, E_4 + \gamma\,\Box R\right], \tag{88}$$

where $C_{\mu\nu\rho\sigma}$ is the Weyl tensor, $E_4 = R^2_{\mu\nu\rho\sigma} - 4R^2_{\mu\nu} + R^2$ is the Gauss–Bonnet integrand (topological in $d = 4$), and $\Box R$ is a total derivative. The coefficients α, β, γ receive contributions from both the scalar and tensor fluctuation sectors. Generically a minimally coupled scalar contributes an additional R^2 counterterm in this scheme; we display only the basis-independent $C^2 + E_4 + \Box R$ contributions and treat any residual R^2 piece as part of the standard quadratic-curvature renormalization (it does not modify weak-field/2PN predictions at the displayed parameter hierarchy).

Two structural features are noteworthy:

1. The divergence (88) involves only *curvature-squared* terms—no R or cosmological constant renormalization appears in dimensional regularization. Newton's constant G does not run at one loop.

2. Unlike pure gravity—which is one-loop finite on shell [50]—the coupled scalar-metric system retains an on-shell divergence proportional to $R^2_{\mu\nu\rho\sigma}$. This divergence is absorbed into an $R^2_{\mu\nu\rho\sigma}$ (equivalently C^2) counterterm whose physical effect scales as $(\text{curvature}/M_{\rm Pl}^2)^2$. On the bi-conformal background $\varphi_0 = -r_{\rm s}/r$, the Kretschmann scalar scales as $R^2_{\mu\nu\rho\sigma} \sim r_{\rm s}^2/r^6$ (leading weak-field term), and is negligible in the weak-field regime ($r \gg r_{\rm s}$), so the one-loop correction to any observable is suppressed by $r_{\rm s}^3/(M_{\rm Pl}^2\, r^5) \sim 10^{-78}$ at astrophysical scales.

EFT power counting at all orders.—Beyond one loop, the effective action admits the systematic expansion

$$\Gamma[\varphi_0] = S[\varphi_0] + \sum_{n=1}^{\infty} \hbar^n\,\Gamma^{(n)}[\varphi_0], \tag{89}$$

where each n-loop contribution scales as

$$\Gamma^{(n)} \sim \frac{1}{(16\pi^2)^n} \frac{E^{2(n+1)}}{M_{\rm Pl}^{2n}} \int d^4x \sqrt{-g}\, \mathcal{O}_n, \tag{90}$$

with $\mathcal{O}_n$ a dimension-$2(n+1)$ local operator built from curvature tensors and $\partial\varphi$. The EFT expansion parameter is $(E/M_{\rm Pl})^2 \sim 10^{-76}$ at astrophysical scales (Eq. 83), so the n-loop correction is suppressed by $[10^{-76}/(16\pi^2)]^n$ relative to the tree-level action.

At two loops, Goroff and Sagnotti [51] showed that pure gravity develops a genuine divergence proportional to $R_{\mu\nu}{}^{\rho\sigma} R_{\rho\sigma}{}^{\alpha\beta} R_{\alpha\beta}{}^{\mu\nu}$. In the ISPG EFT, this and all higher-loop divergences are absorbed into higher-dimensional operators in the Wilsonian action, with Wilson coefficients of order $M_{\rm Pl}^{-2n}$. Since these operators contribute to physical amplitudes only at energies $E \sim M_{\rm Pl}$, they are irrelevant for all predictions of the present paper.

Running of the gravitational coupling.—Although Newton's constant does not run at one loop in dimensional regularization, it receives logarithmic corrections from the curvature-squared counterterms when evaluated at a renormalization scale μ:

$$G(\mu) = G_0 \left[1 + \mathcal{O}\left(\frac{\mu^2}{M_{\rm Pl}^2} \ln \frac{\mu}{\mu_0}\right)\right]. \tag{91}$$

For all scales below $M_{\rm Pl}$, this running is negligible: at $\mu = 1$ TeV (the highest accessible collider energy), $\mu^2/M_{\rm Pl}^2 \sim 10^{-32}$. The classical value G_0 used throughout this paper is therefore exact to extraordinary precision.

Summary.—The quantum effective action of ISPG has the following properties:

(i) At the unreduced action level the shift symmetry $\varphi \to \varphi + \text{const}$ forbids perturbative radiative generation of a scalar mass or potential at all loop orders (reduced biconformal sector preserves the symmetry to EFT precision), supporting the long-range nature of gravity.

(ii) At one loop, the divergences are purely curvature-squared; Newton's constant does not run, and the on-shell corrections are suppressed by $(\text{curvature}/M_{\rm Pl}^2)^2 \sim 10^{-80}$ at astrophysical scales.

(iii) At all loops, the EFT expansion parameter $(E/M_{\rm Pl})^2 \sim 10^{-76}$ ensures that quantum corrections are negligible at every astrophysically relevant scale.

(iv) The one-loop calculations performed elsewhere in this paper (gravitational birefringence, Sec. XI; Higgs mass correction, Sec. VIII C; fine-structure variation, Sec. VI E) are the leading quantum effects and receive no significant higher-order corrections.

I. Quantum persistence of the scalar sensitivity

A central prediction of the classical theory [1] is that the scalar sensitivity of self-gravitating bodies takes the universal value $s = 1/2$ for vacuum compact objects (exact) and at leading order in $|\varphi|$ for matter-filled bodies, so that the excess sensitivity $s_{\rm excess} \equiv s - 1/2$ vanishes in the vacuum case and at leading order in matter (the exact non-perturbative cancellation for matter-filled bodies of arbitrary compactness remains an open task per MAIN/16). This supports suppressed scalar dipole radiation in binary systems and a perturbatively controlled strong-equivalence statement in the scalar sector. We now present the structural argument for the all-orders status of this result within that scope.

Shift-symmetry Ward identity.—The classical proof rests on a single structural property: the ISPG field equations depend on φ only through $\partial_\mu\varphi$, endowing the theory with an exact shift symmetry $\varphi \to \varphi + c$ (c = const). The action (2) contains the metric factors $e^{\pm\varphi}$ explicitly, but the φ-dependent terms combine into total derivatives when the Euler–Lagrange variation is performed; the bulk Lagrangian can therefore be written as $\tilde{\mathcal{L}}(\partial_\mu\varphi)$ modulo boundary terms. This symmetry is preserved by renormalization: the counterterms (87) are all derivative-type, and the symmetry, being Abelian and non-chiral, admits no perturbative anomaly. At the unreduced action level the symmetry is also free of the standard instanton-type breaking channels available to chiral or non-Abelian symmetries; on the reduced bi-conformal sector and in the matter-coupled theory, exact non-perturbative preservation in quantum gravity is an open question, so the protection is asserted at perturbative/EFT precision rather than as an exact non-perturbative theorem.

The preserved shift symmetry implies a Ward identity for the quantum-corrected equations of motion:

$$\frac{\delta\Gamma[\varphi_0 + c]}{\delta\varphi_0(x)} = \frac{\delta\Gamma[\varphi_0]}{\delta\varphi_0(x)} \qquad \forall\, c = \text{const}. \tag{92}$$

Equivalently, $\Gamma[\varphi_0 + c] = \Gamma[\varphi_0] + f(c)$ where $f(c)$ is independent of φ_0. At one loop, this follows from the shift symmetry of the *unreduced* action (2), in which $g_{\mu\nu}$ and φ are independent variables and the scalar enters only through $\partial_\mu\varphi$. The counterterms (87) inherit this symmetry, so the quantum-corrected field equations depend on φ_0 only through $\partial_\mu\varphi_0$. On the bi-conformal constraint surface the fluctuation operator $\mathcal{D}(\varphi_0)$ acquires an explicit φ_0-dependence through the metric factor $e^{-2\varphi_0}$ in the temporal sector; however, this dependence produces corrections suppressed by $(E/M_{\rm Pl})^2 \sim 10^{-76}$ (Eq. 83), so the Ward identity holds to EFT precision at every loop order.

Controlled-sector quantum scalar sensitivity.—The Ward identity is used here only in the sectors where the classical rigid-shift response is under control. For vacuum compact objects the exterior solution has no matter source to spoil the shift, and the classical analysis [1] shows that if $\varphi^*(x)$ satisfies $\varphi^* \to 0$ at spatial infinity, then $\varphi^*(x)+c$ is the shifted boundary-value solution and the Komar mass scales as

$$M[\varphi^* + c] = e^{-c}\, M[\varphi^*]. \tag{93}$$

Within this vacuum sector, the Ward identity (92) preserves the shifted family of quantum-corrected equations to EFT precision at every loop order. The quantum-corrected Komar mass, extracted from the asymptotic behavior of $g_{\mu\nu}[\varphi^* + c]$, then inherits the scaling (93): the metric components $g_{00} = -e^{\varphi^*+c}$ and $g_{ij} = e^{-(\varphi^*+c)}\delta_{ij}$ each acquire the factor $e^{\pm c}$, and the Komar integral transforms as $M \to e^{-c}M$. The vacuum compact-object quantum scalar sensitivity is therefore

$$s_{\text{quantum}} = -\frac{1}{2}\frac{\partial \ln M_{\text{eff}}}{\partial \varphi_{\text{ext}}} = \frac{1}{2} \qquad \text{(vacuum compact objects: exact at all loop orders)}, \tag{94}$$

and the excess sensitivity vanishes:

$$s_{\text{excess}}^{\text{quantum}} = 0 \qquad \text{(vacuum exact; matter-filled bodies leading-order in } |\varphi|\text{, exact non-perturbative case ope} \tag{95}$$

For matter-filled bodies the reduced scalar equation contains the explicit $e^{-\varphi}$ matter-source factor discussed in MAIN/16, so the rigid-shift map is not established beyond leading order. The matter statement in Eq. (95) is therefore only the leading-order controlled result; the exact non-perturbative matter-filled cancellation remains the open sensitivity-programme task.

Explicit one-loop verification.—The Ward identity (92) ensures that, within the same vacuum/leading-order controlled sectors, the one-loop corrected field equation is shift-invariant. If φ^* is the classical solution with $\varphi^* \to 0$ at infinity, and $\delta\varphi^{(1)}$ is its one-loop correction, then $\varphi^* + c + \delta\varphi^{(1)}$ solves the one-loop corrected equation with boundary value c—the quantum correction $\delta\varphi^{(1)}$ is the same regardless of the asymptotic field value. The total quantum-corrected Komar mass at boundary value c is therefore

$$M_{\text{eff}}(c) = M\big[g_{\mu\nu}\big[\varphi^* + c + \delta\varphi^{(1)}\big]\big] = e^{-c}\, M_{\text{eff}}(0), \tag{96}$$

where the factor e^{-c} arises from the metric rescaling, not from the quantum correction itself. Since the total mass—classical plus all quantum corrections—obeys the same exponential scaling, the one-loop correction to the scalar sensitivity is

$$\delta s^{(1)} \equiv s^{(\text{1-loop})} - s_{\text{classical}} = 0 \qquad \text{(within the same controlled scope)}. \tag{97}$$

Physical consequences.—Within the controlled scope above, three consequences follow:

1. *Suppressed dipole radiation in the controlled sectors.*—For vacuum compact objects, and at leading order for matter-filled bodies, $s_A = s_B = 1/2$, so the scalar dipole luminosity $\dot{E}_{\text{dipole}} \propto (s_A - s_B)^2$ vanishes in those limits. Pulsar-timing bounds on dipole radiation from neutron-star–white-dwarf binaries are therefore consistency targets for the matter-filled strong-field open task, not already closed all-body theorems.

2. *Perturbative strong-equivalence statement.*—In the same controlled sectors, bodies respond identically to a background scalar-field shift and the differential (observable) response is zero. This supports a perturbative/EFT strong-equivalence statement in the scalar sector; an exact non-perturbative matter-coupled theorem is left to MAIN/16.

3. *Nordtvedt-effect status.*—The shift symmetry suppresses body-dependent gravitational self-energy contributions to the scalar sensitivity in the vacuum and perturbative/EFT sectors. Thus $\eta_N^{\text{quantum}} \simeq 0$ is supported within that scope, while the exact matter-filled strong-field statement remains contingent on the open sensitivity programme.

IV. THE BERNOULLI MECHANISM FOR GRAVITY

The classical ISPG theory uses the exterior Coulomb profile $\varphi = -r_s/r$ fixed by the vacuum scalar equation together with the reduced-sector source normalization, but does not by itself specify the *physical mechanism* by which matter is associated with a pressure deficit in the surrounding medium. The idea that gravity may originate as a pressure force in a fluid-like medium has antecedents in Arminjon's scalar ether theory [39], where the gravitational acceleration is postulated as $\mathbf{g} = \nabla p_e/\rho_e$ for an imagined perfect fluid. The present approach differs in three respects: the medium is not postulated but emerges from the bi-conformal field equations, the Bernoulli identity is an *exact* consequence of the stress-energy tensor on the static spherically symmetric branch (and a leading zero-frequency identity for time-periodic oscillons) rather than an analogy, and the framework passes all PPN tests ($\gamma = \beta = 1$) while yielding a falsifiable decoherence prediction absent in Arminjon's model.

We show below that the mechanism is the scalar-field counterpart of Bernoulli's principle in fluid dynamics—exact on the static spherically symmetric branch, and a leading zero-frequency bookkeeping for time-periodic oscillons: the kinetic energy of particle resonances drains the static pressure of the scalar-metric medium.

A. Stress-energy decomposition of the scalar field

The scalar-field contribution to the gravitational field equations [1] is

$$\tau^{(\varphi)}_{\mu\nu} = \partial_\mu\varphi\,\partial_\nu\varphi - \frac{1}{2}\,g_{\mu\nu}\,g^{\alpha\beta}\partial_\alpha\varphi\,\partial_\beta\varphi. \tag{98}$$

For a static field $\varphi_0(\mathbf{x})$ on the bi-conformal background, the kinetic invariant is purely spatial:

$$\mathcal{K} \equiv g^{\alpha\beta}\partial_\alpha\varphi_0\,\partial_\beta\varphi_0 = e^{\varphi_0}\,|\nabla\varphi_0|^2 \geq 0, \tag{99}$$

where the inequality holds because $g^{ij} = e^{\varphi_0}\delta^{ij}$ is positive definite.

The energy density measured by a static observer (with four-velocity $u^\mu = (e^{-\varphi_0/2}/c, 0, 0, 0)$ when the time coordinate is $x^0 = t$, so that $u^\mu u_\mu = -c^2$ in the displayed metric; in the natural-unit convention $c = 1$ this reduces to $u^\mu = (e^{-\varphi_0/2}, 0, 0, 0)$, the form used in Eq. 100 below) is

$$\rho_\varphi = \frac{1}{16\pi G}\,\tau^{(\varphi)}_{\mu\nu}\,u^\mu u^\nu = \frac{e^{\varphi_0}}{32\pi G}\,|\nabla\varphi_0|^2. \tag{100}$$

This is manifestly **positive**: in the ISPG phantom-sign convention, the scalar-field energy density in a static configuration is positive-definite. This positivity refers to the fluid-Bernoulli kinetic energy density appearing in Eq. (103); in the Einstein equation of the main theory [1], $\tau^{(\varphi)}_{\mu\nu}$ enters with the phantom-convention factor $-1/2$, so the actual gravitational source from the scalar sector has the opposite sign.

For the Schwarzschild-like solution $\varphi_0 = -r_s/r$:

$$\rho_\varphi(r) = \frac{e^{-r_s/r}}{32\pi G}\frac{r_s^2}{r^4} \tag{101}$$

(geometric expression in natural units; SI restoration adds an explicit c^4 prefactor, $\rho_\varphi^{\rm SI} = c^4\rho_\varphi$, so that with $r_s = 2GM/c^2$ the SI energy density reads $\rho_\varphi^{\rm SI} = (GM^2/8\pi r^4)e^{-r_s/r}$). Differentiating the dimensionless profile $u \equiv r_s/r$ shows that ρ_φ peaks at $r = r_s/4$ (not $r \sim r_s$) and falls as r^{-4} at large distances.

B. The Bernoulli identity

In fluid dynamics, Bernoulli's equation for a steady, inviscid, incompressible flow along a streamline reads

$$P_{\rm static} + \frac{1}{2}\rho v^2 = P_{\rm total} = \text{const}, \tag{102}$$

where $P_{\rm static}$ is the hydrostatic pressure, $\frac{1}{2}\rho v^2$ is the kinetic energy density of the flow, and $P_{\rm total}$ is the total (stagnation) pressure. Regions of faster flow have lower static pressure.

We now derive the exact scalar-field identity that plays the Bernoulli role in ISPG on the static spherically symmetric branch. The fluid formula above is quoted only to display the structural parallel; Eq. (103) itself is not postulated by analogy but obtained directly from the scalar stress-energy decomposition. On this branch the local bookkeeping is

$$\boxed{\underbrace{P_{\rm static}(\varphi)}_{\text{vacuum pressure}} \quad + \quad \underbrace{\frac{e^\varphi}{32\pi G}\,|\nabla\varphi|^2}_{\text{gradient (kinetic) energy}} \quad = P_{\rm total} = \text{const},} \tag{103}$$

where $P_{\rm static}$ is the static radial pressure of the scalar-metric medium relative to the undisturbed vacuum (derived explicitly below). The total pressure on this static branch $P_{\rm total}$ is set by the boundary condition at spatial infinity ($\varphi \to 0$, $\nabla\varphi \to 0$), where $P_{\rm static}(0) = 0$. Thus $P_{\rm total} = 0$: the stagnation pressure vanishes.

Physical interpretation.— Since $P_{\rm static}(\varphi) \le 0$ for $\varphi < 0$ (near matter) and the gradient energy is non-negative, the Bernoulli identity states that wherever the field has a non-zero gradient, the static pressure is *reduced* below the undisturbed value (zero). The pressure deficit $\Delta P \equiv -P_{\rm static} = \frac{e^\varphi}{32\pi G}|\nabla\varphi|^2 \ge 0$ is the local gradient energy density—the exact scalar-field analog of the Bernoulli pressure drop in fluid dynamics.

For static spherically symmetric configurations this is not merely an interpretive analogy: comparison with Eq. (100) gives

$$\Delta P = \rho_\varphi. \tag{104}$$

Thus, on the static spherical branch, the local pressure deficit and the local scalar-field energy density are the same physical quantity written in two languages. For a time-periodic oscillon, this identity is used only after extracting the zero-frequency, spherically averaged static component that sources $\varphi_{\rm grav}$ in the Poisson reconstruction below; no equality of the full time-dependent stress with a universal fluid pressure is assumed.

Derivation.—Consider the stress-energy tensor $\tau^{(\varphi)}_{\mu\nu}$ of Eq. (98) and the normalized stress-density $\mathcal{T}^{\mu}{}_{\nu} \equiv \tau^{\mu}{}_{\nu}/(16\pi G)$. For a static, spherically symmetric field, the radial component of the covariant conservation equation $\nabla_\mu \mathcal{T}^{\mu}{}_{r} = 0$ gives the exact differential identity

$$\frac{d}{dr}\mathcal{T}^{r}{}_{r} = -\frac{2}{r}\big(\mathcal{T}^{r}{}_{r} - \mathcal{T}^{\theta}{}_{\theta}\big) - \varphi'\,\mathcal{T}^{\theta}{}_{\theta}\,. \tag{105}$$

For the bi-conformal metric with $\varphi = \varphi(r)$, the Christoffel symbols and the stress-energy components yield

$$\mathcal{T}^{r}{}_{r} = \frac{e^{\varphi}}{32\pi G}\big(\varphi'\big)^2, \qquad \mathcal{T}^{\theta}{}_{\theta} = -\frac{e^{\varphi}}{32\pi G}\big(\varphi'\big)^2, \tag{106}$$

where $\varphi' \equiv d\varphi/dr$. The angular component $\mathcal{T}^{\theta}{}_{\theta}$ is the negative of the radial stress density; this is a generic consequence of the tensor structure of a static spherically symmetric scalar stress-energy and is shared by canonical and phantom scalars alike. The phantom character enters through the overall sign of $\tau^{(\varphi)}$ in the Einstein equation of the main theory [1], not through this internal identity.

Substituting Eq. (106) into the conservation equation (105) and using the explicit Christoffel symbols for the metric (1), we obtain after simplification:

$$\frac{d}{dr}\left[\frac{e^{\varphi}}{32\pi G}\big(\varphi'\big)^2\right] = \frac{e^{\varphi}}{32\pi G}\big(\varphi'\big)^3 - \frac{e^{\varphi}}{8\pi G}\frac{\big(\varphi'\big)^2}{r}\,. \tag{107}$$

Proof of the Bernoulli integral.—The vacuum field equation $\nabla^2\varphi = 0$ in spherical symmetry admits the exterior Coulomb family $\varphi = B/r$ after the asymptotic condition $\varphi \to 0$ is imposed. Source/Newtonian normalization fixes $B = -r_{\rm s}$, giving $\varphi = -r_{\rm s}/r$. For this profile, $\varphi' = r_{\rm s}/r^2$, $\varphi'' = -2\varphi'/r$, and all r-dependence can be expressed through φ via $1/r = -\varphi/r_{\rm s}$.

On the Coulomb profile, Eq. (107) reduces to a first-order equation for the radial stress:

$$\frac{d\mathcal{T}^{r}{}_{r}}{dr} = \big(\varphi' - 4/r\big)\,\mathcal{T}^{r}{}_{r}, \tag{108}$$

which is verified by direct substitution using $(\varphi')^2 = \varphi^4/r_{\rm s}^2$ and $1/r = -\varphi/r_{\rm s}$. This integrates to

$$\mathcal{T}^{r}{}_{r}(r) = C\,\frac{e^{\varphi}}{r^4}\,, \tag{109}$$

where $C > 0$ is a constant fixed by the source mass. Evaluating at any r with $\mathcal{T}^{r}{}_{r} = e^{\varphi}(\varphi')^2/(32\pi G)$ and $(\varphi')^2 = r_{\rm s}^2/r^4$ gives $C = r_{\rm s}^2/(32\pi G)$. Expressing the result through φ alone ($r^4 = r_{\rm s}^4/\varphi^4$) yields the **Bernoulli integral**:

$$P_{\rm static} + \underbrace{\frac{e^{\varphi}}{32\pi G}\big(\varphi'\big)^2}_{\mathcal{T}^{r}{}_{r}} = 0, \qquad P_{\rm static} \equiv -\mathcal{T}^{r}{}_{r} = -\frac{e^{\varphi}\,\varphi^4}{32\pi G\, r_{\rm s}^2}\,, \tag{110}$$

with $P_{\text{static}} \to 0$ as $\varphi \to 0$ (spatial infinity). On the source-normalized exterior Coulomb branch $\varphi = -r_{\rm s}/r$, the closed-form Bernoulli integral $P_{\text{static}} = -e^{\varphi}\varphi^4/(32\pi G\, r_{\rm s}^2)$ holds on the entire exterior vacuum region. Inside matter sources the closed-form *profile* is modified, because the sourced scalar relation $\Box\varphi = -8\pi G\, T/c^4$ replaces the vacuum equation and the integration constant C no longer reduces to $r_{\rm s}^2/(32\pi G)$. The *structural* identity $P_{\text{static}} + e^{\varphi}|\nabla\varphi|^2/(32\pi G) = 0$, however, follows from the static spherical anisotropy $\mathcal{T}^t{}_t = -\mathcal{T}^r{}_r$ and the static-observer contraction alone, so it continues to hold locally in matter-sourced regions; only the explicit Coulomb form fails inside matter. That interior closed-form extension is not required for the Newtonian and weak-field analyses used in the remainder of this section.

The pressure deficit depends on φ through the universal shape function $\hat{f}(\varphi) = e^{\varphi}\varphi^4$:

$$P_{\text{static}} = -\frac{\hat{f}(\varphi)}{32\pi G\, r_{\rm s}^2}\,, \qquad \hat{f}(\varphi) \equiv e^{\varphi}\varphi^4 \geq 0, \tag{111}$$

whose depth scales as $r_{\rm s}^{-2} \propto M^{-2}$ at fixed φ. This scaling is the expected physics: at a given value of φ, lighter sources have steeper gradients ($\varphi' = \varphi^2/r_{\rm s}$), hence higher kinetic-energy density. The *total* pressure deficit integrated over all space scales as M, recovering the correct gravitational mass (Sec. IV D).

Verification.—Substituting the Coulomb profile into the right-hand side of Eq. (107) using $(\varphi')^2 = \varphi^4/r_{\rm s}^2$, $1/r = -\varphi/r_{\rm s}$, and $(\varphi')^3 = \varphi^6/r_{\rm s}^3$:

$$\text{RHS} = \frac{e^{\varphi}\,\varphi^6}{32\pi G\, r_{\rm s}^3} + \frac{e^{\varphi}\,\varphi^5}{8\pi G\, r_{\rm s}^3} = \frac{e^{\varphi}\,(\varphi^6 + 4\varphi^5)}{32\pi G\, r_{\rm s}^3}\,. \tag{112}$$

The left-hand side, computed from $d\mathcal{T}^r{}_r/dr = (d\mathcal{T}^r{}_r/d\varphi)\,\varphi'$ with $\mathcal{T}^r{}_r = e^{\varphi}\varphi^4/(32\pi G\, r_{\rm s}^2)$ and $\varphi' = \varphi^2/r_{\rm s}$, gives the same expression, confirming the integral.

Physical interpretation. From the Bernoulli integral (110), the local pressure deficit is

$$\Delta P(r) \equiv -P_{\text{static}}(r) = \frac{e^{\varphi_0}}{32\pi G}\,|\nabla\varphi_0|^2 = \rho_{\varphi}(r) \geq 0. \tag{113}$$

The pressure deficit equals the local gradient energy density: wherever the field has a non-zero gradient, the static pressure is driven below zero. Gravity thus arises as a Bernoulli-type pressure deficit in the scalar-metric medium.

Time-averaged Bernoulli for oscillating fields.—The above derivation assumes a static field configuration ($\partial_t\varphi = 0$). For particle resonances, which are time-periodic oscillations $\varphi(t, r) = \varphi_0(r) + \Phi(r)\cos(\omega t)$, the Bernoulli identity applies to the time-averaged energy densities over one oscillation period $T = 2\pi/\omega$. The kinetic term $\frac{1}{2}(\partial_t\varphi)^2$ averages to $\frac{1}{4}\omega^2\Phi^2$, while the spatial gradient $\frac{1}{2}(\nabla\varphi)^2$ contributes both a static part from φ_0 and an oscillatory part from Φ. To leading order in the oscillation amplitude, the time-averaged Bernoulli identity reads

$$\langle P_{\text{static}}(\varphi)\rangle_T + \frac{e^{\varphi_0}}{32\pi G}\left[|\nabla\varphi_0|^2 + \frac{1}{2}|\nabla\Phi|^2 + \frac{1}{2}\omega^2 e^{-2\varphi_0}\Phi^2\right] = 0, \tag{114}$$

where $\langle\cdot\rangle_T$ denotes the time average. The key structural point is that the static gravitational source is the *zero-frequency projection* of the scalar stress-energy. Writing the oscillon field as

$$\varphi(t,r) = \varphi_{\rm grav}(r) + \Phi(r)\cos(\omega t) + \delta\varphi_{\geq 2}(t,r), \tag{115}$$

where $\delta\varphi_{\geq 2}$ collects higher harmonics and slow modulation, every term linear in the fast oscillation averages to zero over one period, while the surviving static contribution begins at quadratic order in the oscillatory amplitudes. Accordingly,

$$\langle T^{00}\rangle = \langle T^{00}\rangle_{\rm lead} + \delta T^{00}_{(0)}, \tag{116}$$

where the leading zero-frequency piece is the explicit source written in Eq. (132) below, while $\delta T^{00}_{(0)}$ contains higher-harmonic and nonlinear backreaction terms. In the harmonic/metastable regime used in this paper, $\delta T^{00}_{(0)}$ is subleading: it starts at higher order in the oscillon amplitudes and acquires only adiabatic drift corrections of relative size $\mathcal{O}(\Gamma/\omega)$ over one oscillation period. The static field is therefore controlled by the zero-frequency projection of the oscillon stress-energy, not by the oscillatory harmonics themselves. The oscillatory contributions represent the kinetic and potential energy of the particle resonance, which drain the static pressure below zero through the Bernoulli mechanism.

C. Microscopic origin: resonances as the kinetic-energy source

In the absence of matter, $\varphi = 0$ everywhere and $|\nabla\varphi| = 0$: the medium is “still” and the static pressure vanishes ($P_{\rm static} = 0$).

When matter is present, each constituent particle is a resonant excitation (Sec. III) that oscillates the scalar-metric medium in its vicinity. These oscillations generate nonvanishing gradients $\nabla\varphi \neq 0$, transferring energy from the static (pressure) channel to the kinetic (gradient) channel.

For a single particle of mass m at the origin, the scalar field solution is $\varphi = -r_m/r$ with $r_m = 2Gm/c^2$. The gradient energy in a shell between r and $r + dr$ is

$$dE_{\rm grad} = \frac{r_m^2}{8G\,r^2}\,e^{-r_m/r}\,dr \approx \frac{r_m^2}{8G\,r^2}\,dr \quad (r \gg r_m). \tag{117}$$

For a macroscopic body composed of oscillons centered at $\mathbf{x}_a$, the far-zone field is fixed not by pointwise addition of the local pressure deficits, but by the integrated coarse-grained source that enters the weak-field scalar equation. In the non-overlap regime,

$$\langle T^{00}\rangle_{\rm body}(\mathbf{x}) = \sum_{a=1}^{N}\langle T^{00}_a\rangle(\mathbf{x}-\mathbf{x}_a) + \delta T^{00}_{\rm int}(\mathbf{x}), \tag{118}$$

where $\delta T^{00}_{\rm int}$ collects localized overlap and binding corrections. The total gravitational mass is therefore

$$M_{\rm body} = \frac{1}{c^2}\int_{\mathbb{R}^3}\langle T^{00}\rangle_{\rm body}\,d^3x = \sum_{a=1}^{N} m_a + \frac{E_{\rm int}}{c^2}, \qquad E_{\rm int} \equiv \int \delta T^{00}_{\rm int}\,d^3x. \tag{119}$$

Outside the body ($r \gg R_{\rm body}$), a multipole expansion of the scalar Gauss-law solution gives

$$\varphi_{\rm body}(r) = -\frac{2GM_{\rm body}}{c^2 r}\left[1 + \mathcal{O}\left(\frac{R_{\rm body}^2}{r^2}\right)\right], \tag{120}$$

after choosing the origin at the center of mass so that the dipole term vanishes. Thus the precise many-body closure needed for gravity is this: microscopic Bernoulli deficits determine a localized total source, and the exterior field depends only on its monopole mass. The interior interference terms renormalize $M_{\rm body}$ but do not alter the universal $1/r$ far-zone law.

Dynamic macroscopic closure.—The same conclusion survives when the constituent oscillons move adiabatically. Let $\mathbf{x}_a(t)$ denote their centers and let each constituent obey the single-source control statement (58)–(80). Then the coarse-grained body source can be written as

$$\langle T^{00}\rangle_{\rm body}(\mathbf{x},t) = \sum_{a=1}^{N}\langle T_a^{00}\rangle\big(\mathbf{x}-\mathbf{x}_a(t);\Phi_{c,a}(t)\big) + \delta T^{00}_{\rm int}(\mathbf{x},t) + \delta T^{00}_{\rm dyn}(\mathbf{x},t), \tag{121}$$

where $\delta T^{00}_{\rm dyn}$ collects the controlled non-rigid corrections from the remainders η_a and from nonrelativistic source motion. Writing

$$\delta T^{00}_{\rm dyn} = \mathcal{O}\big(v_{\rm body}^2\big) + \mathcal{O}\Big(E_{\rm lin,body}^{1/2}\Big), \qquad v_{\rm body} \equiv \max_a \frac{|\dot{\mathbf{x}}_a|}{c}, \tag{122}$$

with $E_{\rm lin,body}$ the coarse-grained analogue of the single-source remainder energy, and choosing the center-of-mass frame

$$\mathbf{R}_{\rm CM}(t) \equiv \frac{1}{M_{\rm body}(t)}\sum_{a=1}^{N} m_a\,\mathbf{x}_a(t), \tag{123}$$

the far-zone field obeys

$$\varphi_{\rm body}(\mathbf{x},t) = -\frac{2G\,M_{\rm body}(t)}{c^2\,|\mathbf{x}-\mathbf{R}_{\rm CM}(t)|}\left[1+\mathcal{O}\left(\frac{R_{\rm body}^2}{r^2}\right)\right] + \mathcal{O}\big(e^{-\kappa_{\rm min}d_{\rm min}}\big) + \mathcal{O}\big(v_{\rm body}^2\big) + \mathcal{O}\Big(E_{\rm lin,body}^{1/2}\Big), \tag{124}$$

where $d_{\rm min}$ is the minimum pairwise separation in the non-overlap regime. Thus the entire body inherits the same translated monopole field as a single oscillon: internal interference only renormalizes $M_{\rm body}(t)$, while the exterior field is carried by the center-of-mass collective coordinate $\mathbf{R}_{\rm CM}(t)$.

The waterfall analogy.—Consider two bodies of water: a calm lake and the base of a waterfall. The calm lake (no matter, $\nabla\varphi = 0$) has high static pressure. The turbulent pool at the waterfall's base (10^{57} atomic resonances agitating the medium) has the same total energy, but much of it has been converted to kinetic energy of the flow, drastically reducing the static pressure. In the theory itself a test body does not feel a separate hydrodynamic force; it follows the geodesic of the metric defined by the same scalar profile. The analogy is therefore interpretive: the pressure deficit explains why the profile is gravitational, while the geodesic equation supplies the motion.

D. From pressure deficit to Newton's law

We now separate two statements that must not be conflated: the Bernoulli pressure deficit gives the mechanism-level interpretation of the scalar profile, while test-particle motion is fixed by geodesics of the bi-conformal metric.

The logic has two distinct steps. First, the Bernoulli identity identifies the *physical source* of gravity: the oscillatory gradient energy of a localized resonance reduces the static pressure of the scalar-metric medium. Second, once this pressure deficit is localized, the exterior field it generates is fixed by the vacuum scalar equation. Bernoulli therefore explains *why* the scalar profile is interpreted as gravitational, while the exterior field equation and boundary normalization determine *how* that field propagates at large distances. The two steps share the same scalar profile, not an additional pressure-gradient equation of motion for matter.

The static field equation in the weak-field limit ($|\varphi| \ll 1$) is $\nabla^2\varphi = 0$ outside the source, with the boundary condition set by the integrated source. For a localized oscillon, matching the exterior solution $\varphi_{\mathrm{grav}}(r) = -r_{\mathrm{s}}/r$ to the source integral gives

$$4\pi r_{\mathrm{s}} = \frac{8\pi G}{c^4}\int_{B_R}\langle T^{00}\rangle\, d^3x = \frac{8\pi G}{c^2}\, M_{\mathrm{grav}}, \qquad M_{\mathrm{grav}} \equiv \frac{1}{c^2}\int_{\mathbb{R}^3}\langle T^{00}\rangle\, d^3x. \tag{125}$$

The fundamental sourcing relation in the main theory is $\Box\varphi = -8\pi G\,T/c^4$, where T is the *trace* of the matter stress-energy tensor [1]. For pressureless matter (dust) one has $T = -\rho c^2$ and $\langle T^{00}\rangle = \rho c^2$, so $\langle T^{00}\rangle = -T$ and Eq. (125) is equivalent to the trace-sourced form in this limit. For traceless sources ($T = 0$, e.g. electromagnetic radiation) the trace-sourced equation gives no direct scalar response: photons contribute to $\langle T^{00}\rangle$ but not to T, so they do not source φ directly, and their gravitational deflection arises through geodesic motion in the bi-conformal metric determined by matter-sourced φ. The $\langle T^{00}\rangle$ notation used throughout this section should therefore be read as the effective energy-density source appropriate to the non-relativistic, trace-dominant regime in which the oscillon bodies live. The Bernoulli pressure deficit already fixes the sign of the field: positive localized energy lowers the static pressure and therefore produces $\varphi_{\mathrm{grav}} < 0$. Gauss's theorem then fixes the magnitude of the $1/r$ coefficient, yielding

$$\varphi_{\mathrm{grav}}(r) = -\frac{r_{\mathrm{s}}}{r} = -\frac{2GM_{\mathrm{grav}}}{c^2 r}. \tag{126}$$

The geodesic acceleration experienced by a test particle in this field is [1]

$$\mathbf{a}_{\mathrm{geo}} = -\frac{c^2}{2}\,\nabla\varphi_{\mathrm{grav}} = -\frac{GM_{\mathrm{grav}}}{r^2}\,\hat{r}. \tag{127}$$

The Newtonian potential is $\Phi_N = -GM_{\mathrm{grav}}/r = \frac{1}{2}\varphi_{\mathrm{grav}}\, c^2$, and Newton's law $\mathbf{a} = -\nabla\Phi_N = -GM_{\mathrm{grav}}\hat{r}/r^2$ is exactly reproduced by the geodesic equation.

The force on the test particle is the far-zone geodesic acceleration sourced by the same scalar profile that carries the Bernoulli pressure deficit:

$$\mathbf{F} = -m\,\tfrac{c^2}{2}\,\nabla\varphi = -m\,\nabla\Phi_N. \tag{128}$$

Firewall: minimal coupling and no fifth force. Matter is minimally coupled to $g_{\mu\nu}[\varphi]$ alone, so test particles follow geodesics of the bi-conformal metric. Eq. (128) is the weak-field geodesic acceleration in disguise; it is *not* an independent pressure-gradient force law on matter. A literal substitution $\mathbf{F} = -m\nabla(\Delta P/\rho_\varphi)$ would scale as r^{-1} on the Coulomb branch (since $\Delta P \propto r^{-4}$, $\rho_\varphi \propto r^{-4}$, so $\nabla(\Delta P)/\rho_\varphi \propto r^{-1}$), whereas the Newtonian acceleration scales as r^{-2}; the two are not equivalent and the literal pressure-gradient force law is forbidden. The Bernoulli identity supplies the *physical interpretation* of why the metric is gravitational (the same scalar profile carries a static pressure deficit), but it does not enter the equation of motion for matter. The test-particle weak equivalence principle is preserved exactly because matter couples only to $g_{\mu\nu}$.

In this interpretive sense the test body moves *toward* regions of lower static pressure (near the source) along the geodesic that the same scalar profile defines: the pressure deficit is the mechanism, the geodesic is the equation of motion. Gravity is not a mysterious attraction; it is the geodesic response of matter to a metric whose scalar profile also carries a Bernoulli pressure deficit in the medium.

Two-component structure of the oscillon field.—The preceding derivation reveals a fundamental duality in the field structure of particle-like resonances. The full scalar field near an oscillon decomposes into two distinct components:

$$\varphi(r,t) = \underbrace{\varphi_{\rm grav}(r)}_{\sim -M/r} + \underbrace{\Phi_0(r)\cos(\Omega t)}_{\sim e^{-\kappa r}/r}, \tag{129}$$

where $\kappa = \sqrt{1-\Omega^2}$. This split is fixed, to leading order in the oscillon harmonic expansion, by separating the time average from the first oscillatory mode:

$$\varphi_{\rm grav}(r) \equiv \langle \varphi(r,t)\rangle_t, \qquad \Phi_0(r)\cos(\Omega t) \equiv \varphi(r,t) - \langle\varphi(r,t)\rangle_t + \mathcal{O}\big(\Phi^2\big)\,, \tag{130}$$

where $\langle\cdot\rangle_t$ denotes averaging over one period $2\pi/\Omega$. The static component is therefore the part of the field that survives coarse-graining over the fast oscillation, while the localized oscillatory component carries the bound internal resonance. The oscillating component $\Phi_0(r)\cos\Omega t$ is the particle's *body*—the localized resonant mode whose amplitude decays exponentially at the rate κ set by the bound-state condition $\Omega < 1$. The static component $\varphi_{\rm grav} = -r_{\rm s}/r$ is the particle's *gravitational field*—the $1/r$ potential sourced by the time-averaged energy density of the oscillation through the Poisson equation $\nabla^2\varphi_{\rm grav} = -(8\pi G/c^4)\langle T^{00}\rangle$. The mechanism by which the oscillatory mode $\Phi_0\cos\Omega t$ acts as an effective source for the static Poisson equation is the nonlinear self-coupling of the scalar sector: the bi-conformal metric introduces φ-dependent factors in $\sqrt{-g}\,g^{\mu\nu}\partial_\mu\varphi\partial_\nu\varphi$ that mix the fast oscillatory and slow static components, and the time-averaged residue is the effective $\langle T^{00}\rangle$ entering Eq. (132). We take that effective source as given here and verify the resulting $1/r$ far-field profile numerically below; the full derivation of the effective source from the scalar action is carried out in the classical companion paper [1]. These two components obey fundamentally different equations: the oscillating mode satisfies the massive Klein–Gordon equation (Sec. III E) and decays exponentially; the gravitational potential satisfies the massless Laplace equation $\nabla^2\varphi = 0$ outside the source and decays as $1/r$. The particle is thus an exponentially localized excitation dressed by a long-range gravitational field—a composite object whose internal structure ($e^{-\kappa r}$) is invisible at distances $r \gg 1/\kappa$, while its gravitational influence ($1/r$) extends to infinity.

Combining the time-averaged Bernoulli identity with the decomposition (129), let $\delta\varphi(r,t) \equiv \Phi_0(r)\cos(\Omega t)$. The oscillatory part of the scalar-field energy density measured by a static observer is then, to leading order in the oscillon amplitude,

$$\rho_{\rm osc}(r,t) = \frac{1}{32\pi G}\left[e^{-\varphi_{\rm grav}}(\partial_t\delta\varphi)^2 + e^{\varphi_{\rm grav}}|\nabla\delta\varphi|^2\right]. \tag{131}$$

The mixed term $2\nabla\varphi_{\rm grav}\cdot\nabla\delta\varphi \propto \cos(\Omega t)$ averages to zero over one period, so the source component entering the weak-field Poisson equation is

$$\langle T^{00}\rangle = \langle\rho_{\rm osc}\rangle_t = \frac{1}{64\pi G}\left[e^{-\varphi_{\rm grav}}\Omega^2\Phi_0^2 + e^{\varphi_{\rm grav}}(\Phi_0')^2\right]. \tag{132}$$

In the Newtonian matching regime $|\varphi_{\rm grav}| \ll 1$, this becomes the dimensionless source profile

$$\rho(r) = \tfrac{1}{2}\Omega^2\Phi_0^2 + \tfrac{1}{2}(\Phi_0')^2, \tag{133}$$

up to the overall factor $1/(32\pi G)$. Thus the same time-averaged Bernoulli deficit $\langle\Delta P\rangle_t = \langle\rho_{\rm osc}\rangle_t$ is the leading zero-frequency localized source whose Poisson reconstruction produces $\varphi_{\rm grav}$ in the harmonic/metastable regime; subleading harmonic and nonlinear backreaction corrections are of relative size $\mathcal{O}(\Gamma/\omega)$ over one oscillation period, as recorded above.

Numerical verification: the oscillon gravitational potential.—We verify this decomposition by solving the Poisson equation sourced by the density profile in Eq. (133):

$$\varphi_{\rm grav}(r) = -\frac{M_{\rm enc}(r)}{r} - \int_r^\infty 4\pi\rho(r')\,r'\,dr', \qquad M_{\rm enc}(r) = 4\pi\int_0^r \rho(r')\,r'^2\,dr'. \tag{134}$$

Consequently the far-zone coefficient is fixed by the same source integral:

$$M_{\rm num} \equiv \lim_{r\to\infty}\big[-r\,\varphi_{\rm grav}(r)\big] = 4\pi\int_0^\infty \rho(r)\,r^2\,dr. \tag{135}$$

This is the natural-unit counterpart of Eq. (125): once the localized source profile is specified, the asymptotic $1/r$ coefficient is fixed. This is a direct reconstruction of the static field from the same oscillon profile that defines the localized resonance: once $\Phi_0(r)$ is fixed, there is no additional freedom in the asymptotic potential. The numerical test therefore checks not only the $1/r$ falloff but also the correct mass normalization carried by the source. For the oscillon at $\Phi_0 = 2.35$ ($\Omega = 0.866$, $M_{\rm num} = 295.5$ in natural units), the ratio $\varphi_{\rm grav}(r)/(-M_{\rm num}/r)$ equals unity to 5×10^{-7} at $r = 15$ and to machine precision ($< 10^{-13}$) at $r = 30$, confirming the $1/r$ profile with the correct mass coefficient.

Two-body interaction: Newton's law from oscillons.—For two oscillons separated by distance d (with d much larger than the oscillon radius), the gravitational interaction energy is

$$U(d) = \int \rho_2(\mathbf{r}_2)\,\varphi_{\rm grav,1}(|\mathbf{r}_2 + d\hat{z}|)\,d^3r_2\,. \tag{136}$$

Since ρ_1 is spherically symmetric, Gauss's theorem gives $\varphi_{\mathrm{grav},1}(r) = -M_1/r$ for $r > R_{\mathrm{osc}}$, and the angular average of $1/|\mathbf{r}_2 + d\hat{z}|$ over the second oscillon's density distribution yields $U(d) = -M_1M_2/d$ at leading order in the tail overlap. This is precisely the regime relevant to the Newtonian limit: each oscillon acts as a localized source dressed by its long-range field, while short-distance core overlap is deliberately excluded from the present claim. More precisely, spherical symmetry removes all exterior multipoles of each compact core, so the compact-support part of the interaction is *exactly* Newtonian rather than merely asymptotically so. The only correction comes from the exponentially small non-compact tail of the oscillon profile. Writing

$$\rho_i(r) = \rho_i^{(<R)}(r) + \rho_i^{(>R)}(r), \qquad M_{i,\mathrm{tail}}(R) \equiv 4\pi \int_R^\infty \rho_i(r)\, r^2\, dr, \tag{137}$$

with $R < d/2$, one obtains

$$U(d) = -\frac{M_1M_2}{d} + \mathcal{O}\!\left(\frac{M_{1,\mathrm{tail}}(R)\, M_2 + M_1\, M_{2,\mathrm{tail}}(R)}{d}\right). \tag{138}$$

For a bound oscillon, $\Phi_0(r) \sim Ae^{-\kappa r}/r$, so $\rho_i(r) \sim e^{-2\kappa_i r}/r^2$ and therefore

$$M_{i,\mathrm{tail}}(R) = \mathcal{O}\big(e^{-2\kappa_i R}\big)\,. \tag{139}$$

Choosing $R = d/2$ gives the explicit non-overlap estimate

$$U(d) = -\frac{M_1M_2}{d}\left[1 + \mathcal{O}\big(e^{-\kappa_{\min} d}\big)\right], \qquad \kappa_{\min} \equiv \min(\kappa_1, \kappa_2). \tag{140}$$

Thus no algebraic finite-size correction survives in the spherically symmetric core sector; the leading deviation from Newton's law is exponentially suppressed by tail overlap. Numerically, we compute $U(d)$ by evaluating the full angle-averaged integral for five pairs of oscillons with amplitudes $\Phi_0 \in \{1.0, 1.5, 2.0, 2.35, 3.0\}$ (masses $M \in \{127, 183, 246, 296, 402\}$). No Newtonian law is inserted at this stage: the input is only the localized oscillon density profile and the static field obtained from Eq. (134). The inverse distance law then re-emerges from the full source integral in the non-overlap regime. The results confirm Newton's law at three levels:

1. *Potential:* $|U(d)| \propto 1/d^n$ with $n = 1.000011$ (power-law fit for $d \geq 15$; deviation 0.001% from unity).

2. *Force:* $|F| = |dU/dd| \propto 1/d^n$ with $n = 2.000157$ (deviation 0.008% from the inverse-square law).

3. *Mass proportionality:* $U(d)/(M_1M_2/d) = 1.000000$ for all eight tested mass pairs (six significant figures), confirming $F \propto M_1\, M_2$.

Newton's gravitational law $F = G\, M_1M_2/d^2$ is therefore not an input of the theory but an *output*: the $1/r^2$ *form* of the force follows from the field equation $\Box\varphi = 0$ applied to localized energy sources. The numerical value of G appearing as the coefficient is the same G that sets the prefactor $1/(16\pi G)$ in the ISPG action (2); it is therefore a parameter of the theory, not derived from deeper principles in this paper.

E. Strong-field regime and singularity avoidance

In the strong-field regime ($r \sim r_s$), the exponential factors become significant. The gradient energy density Eq. (101) peaks at $r = r_s/4$ and vanishes as $r \to 0$ due to the $e^{-r_s/r}$ suppression.

This has a profound consequence: the pressure deficit is *self-limiting*. As the static pressure drops (increasing $|\varphi|$), the exponential factor e^{φ} in Eq. (103) suppresses the gradient energy, slowing the pressure drop. In the limit $\varphi \to -\infty$ (approaching the classical singularity), $e^{\varphi} \to 0$ and the gradient energy vanishes entirely: the medium reaches a state of "maximal depletion" where no further energy transfer from static to kinetic channels is possible.

This is the Bernoulli-mechanism explanation for the singularity resolution established in the classical theory [1]: the gravitational potential well has an infinite proper depth ($\int_0^\infty e^{-\varphi_0/2}\, dr = \infty$) but finite coordinate-volume energy integral ($\int \rho_\varphi\, d^3x_{\text{coord}} < \infty$; the proper-volume measure $\sqrt{h}\, d^3x = e^{-3\varphi/2} d^3x$ weighs the strong-field interior differently and is treated separately), because the bi-conformal structure automatically regulates the pressure deficit. No singularity forms because the Bernoulli mechanism has a built-in saturation.

F. Comparison with general relativity

In GR, gravity is described as the curvature of spacetime, and the geodesic equation $\ddot{x}^\mu + \Gamma^\mu_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = 0$ determines particle trajectories. The question "*why* does mass curve spacetime?" is not answered within GR; it is an axiom of the Einstein field equations.

In ISPG, this question receives a physical answer:

1. Mass consists of resonant excitations of the scalar-metric medium (Sec. III).
2. These excitations generate gradient (kinetic) energy in the medium via their oscillatory tails.
3. By the Bernoulli identity (103), this kinetic energy lowers the static pressure.
4. The same scalar profile that carries this pressure deficit defines the bi-conformal metric.
5. Test particles then follow geodesics of that metric; in the weak-field limit this gives Newtonian gravity.

The causal chain

$$\text{resonance} \xrightarrow{\text{oscillation}} |\nabla\varphi|^2 \xrightarrow{\text{Bernoulli}} \Delta P \quad \text{and} \quad \varphi \xrightarrow{\text{metric}} g_{\mu\nu}[\varphi] \xrightarrow{\text{geodesic}} \mathbf{a}_{\text{geo}} \tag{141}$$

provides the mechanical underpinning of gravity that Einstein's geometric description, by design, does not address. Both descriptions are compatible: the bi-conformal metric (1) encodes the pressure distribution geometrically, and geodesic motion in this geometry is the equation of motion. The pressure deficit supplies the mechanism-level interpretation, not a separate force law.

V. INERTIA, HYSTERESIS, AND THE EQUIVALENCE PRINCIPLE

The Bernoulli mechanism (Sec. IV) explains *gravity*—the static pressure deficit around matter. We now address the complementary question: *why does matter resist acceleration?* In GR, the equivalence of inertial and gravitational mass is postulated. In ISPG, it is derived: both arise from the same resonance–medium coupling.

A. Boosted resonance: uniform motion

A resonance at rest is described by the oscillon profile $\Phi(t, \mathbf{x}) = \phi(r)\, e^{-i\omega t}$ (Sec. III E). Under a Lorentz boost with velocity $\mathbf{v}$, the locally measured speed of light remains c (Eq. 27), so the boosted profile is obtained by the standard replacement $t \to \gamma(t - \mathbf{v} \cdot \mathbf{x}/c^2)$, $\mathbf{x} \to \mathbf{x} + (\gamma - 1)(\hat{v} \cdot \mathbf{x})\hat{v} - \gamma \mathbf{v} t$:

$$\Phi_v(t, \mathbf{x}) = \phi\left(\sqrt{x_\perp^2 + \gamma^2 (x_\parallel - vt)^2}\right) \exp\left[-i\gamma\omega\left(t - \frac{v\, x_\parallel}{c^2}\right)\right], \tag{142}$$

where $x_\parallel$ and $x_\perp$ are the components parallel and perpendicular to $\mathbf{v}$, and $\gamma = (1 - v^2/c^2)^{-1/2}$.

The spatial profile is Lorentz-contracted by γ along the direction of motion. Two levels of justification apply:

1. *Free particle (self-generated field).*—For an isolated oscillon, φ_0 is generated self-consistently by the resonance itself, and the full coupled system (metric $g_{\mu\nu}$ + scalar φ + excitation Φ) transforms covariantly under Lorentz boosts of the full field+source configuration, because the asymptotic spacetime admits the global Poincaré symmetry of flat space and the ISPG action is invariant under that symmetry. The boosted oscillon *together with its boosted* φ_0 is therefore an *exact* solution.

2. *Particle in an external field.*—When $\varphi_0(\mathbf{x})$ is sourced by other masses, the background breaks global Lorentz symmetry. Nevertheless, local Lorentz invariance is guaranteed by the equivalence principle: in the freely falling frame of the oscillon, the background is locally flat over scales $\ll L_\varphi$, where $L_\varphi = |\varphi_0/\nabla\varphi_0|$ is the gradient scale of the external field. Since $R_c \ll L_\varphi$ for all known particles in astrophysical fields, the boosted profile is an *adiabatic* (WKB) solution with corrections of order $(R_c/L_\varphi)^2 \ll 1$.

In the isolated case, no net self-force acts on the resonance during uniform motion. In the external-field case, no additional velocity-proportional self-drag arises in the local freely falling frame, but ordinary external gravitational acceleration along the $\nabla\varphi_0$ gradient and tidal/WKB corrections of order $(R_c/L_\varphi)^2$ remain allowed.

Physical interpretation: motion as pattern propagation.—An oscillon is not a rigid body translating through the medium; it is a resonant pattern *of* the medium. "Motion" means that the excitation amplitude is transferred from one set of field degrees of freedom to the neighbouring set, exactly as an ocean wave propagates without net transport of water. No substance moves from A to B; only the self-sustaining resonance pattern does.

This ontological point has a direct dynamical consequence. A resonance moving at constant velocity through the medium is indistinguishable from a resonance at rest in a boosted frame. The pressure distribution adjusts self-consistently: the Lorentz-contracted profile generates a Lorentz-contracted pressure deficit, and the static Bernoulli balance (103) is preserved in the comoving frame. No drag, no friction, no energy dissipation—because the pattern does not displace the medium, it *is* the medium. This is the superfluid property of the scalar-metric medium.

The maximum propagation speed of any pattern is set by the speed at which the medium transmits information. In local tetrad units the bound is the locally measured $|v_{\rm local}| < c$, with the same v that enters the Lorentz factor $\gamma = (1-v_{\rm local}^2/c^2)^{-1/2}$. In coordinate units the corresponding bound on the coordinate velocity $|d\mathbf{x}/dt|$ is $|d\mathbf{x}/dt| < c_{\rm coord}$ with $c_{\rm coord} = c\, e^{\varphi_0}$. No oscillon can outrun its own field; the relativistic speed limit is not imposed externally but follows from the medium's finite response rate.

B. Accelerated resonance: the origin of inertia

Section V A established that oscillon motion is the propagation of a resonant pattern through the medium, limited by the medium's information speed c. Inertia is the direct consequence: when the pattern is forced to accelerate, the medium cannot redistribute its pressure instantaneously.

Consider a resonance whose velocity changes: $\mathbf{v}(t) = \mathbf{v}_0 + \mathbf{a}\,\delta t$ over a short interval δt. The oscillon profile must continuously readjust its Lorentz contraction to match the instantaneous velocity. This readjustment propagates at the local speed of light c from the center of the oscillon outward.

Finite readjustment time.—The oscillon has a spatial extent of order the Compton wavelength $R_c = \hbar/(m_0 c)$. The time required for the profile to fully readjust is

$$t_{\rm adj} \sim \frac{R_c}{c} = \frac{\hbar}{m_0 c^2} = \frac{1}{\omega_0}, \tag{143}$$

which is the oscillation period of the resonance. This $t_{\rm adj} \sim 1/\omega_0$ is the microscopic oscillon-Compton tracking scale of the localized profile. It is distinct from the exterior point-source/self-field light-crossing scale $r_{\rm s}/c$ used in the perturbative test-body calculation of the main theory, which governs how rapidly the far-zone scalar dressing re-centers on the moving point source. Both scales co-exist as different response channels: microscopic profile tracking, exterior light-crossing, and the macroscopic collective relaxation treated separately below. During acceleration, if the velocity change occurs on a timescale shorter than $t_{\rm adj}$, the profile cannot follow instantaneously.

Pressure asymmetry.—The result is a transient asymmetry in the pressure distribution around the resonance:

1. **Ahead** (in the direction of acceleration): the resonance enters a region of undisturbed medium whose pressure has not yet been reduced by the Bernoulli mechanism. The

resonance encounters its own pressure deficit *incompletely formed*, resulting in a net restoring pressure.

2. **Behind**: the pressure deficit from the previous position persists for a time $\sim t_{\rm adj}$ before the medium re-equilibrates. This trailing deficit exerts a backward pull.

We now organize the leading force calculation by expanding the retarded scalar field to first order in the acceleration.

Retarded field of an accelerated source.—Let the oscillon center follow a trajectory $\mathbf{r}(t) = \frac{1}{2}\mathbf{a}\,t^2$ with $|\mathbf{a}| \ll c\omega_0$ (non-relativistic acceleration). The scalar field satisfies the sourced wave equation $\Box\varphi = -(8\pi G/c^4)\,T$ with the trace of the oscillon stress-energy concentrated at $\mathbf{r}(t)$. The retarded Green's function solution is

$$\varphi(\mathbf{x},t) = \frac{2G}{c^4}\int d^3x'\,\frac{T\big(t_r,\,\mathbf{x}'\big)}{|\mathbf{x}-\mathbf{x}'|}, \qquad t_r = t - \frac{|\mathbf{x}-\mathbf{x}'|}{c}. \tag{144}$$

For a compact oscillon ($R_c \ll r$) the source is effectively pointlike, and for nonrelativistic rest mass in the $(-,+,+,+)$ signature the trace satisfies

$$T \simeq -Mc^2\,\delta^{(3)}(\mathbf{x}'-\mathbf{r}(t)), \tag{145}$$

i.e. positive rest energy gives a negative trace. Performing the spatial integration in Eq. (144) with this sign convention gives

$$\varphi(\mathbf{x},t) = -\frac{r_s}{|\mathbf{x}-\mathbf{r}(t_r)|}, \tag{146}$$

where $t_r = t - |\mathbf{x}-\mathbf{r}(t_r)|/c$ is the retarded time. For non-relativistic acceleration, $\mathbf{r}(t_r) = \mathbf{r}(t) - \mathbf{v}(t)\,|\mathbf{x}|/c + \mathcal{O}(a^2)$, so the displacement from the instantaneous position is

$$\mathbf{r}(t) - \mathbf{r}(t_r) = \mathbf{v}(t)\,\frac{|\mathbf{x}|}{c} = \mathbf{a}\,t\,\frac{|\mathbf{x}|}{c}. \tag{147}$$

Expanding $|\mathbf{x}-\mathbf{r}(t_r)|^{-1}$ around the instantaneous position $\boldsymbol{\xi} = \mathbf{x} - \mathbf{r}(t)$:

$$\frac{1}{|\boldsymbol{\xi} + \mathbf{a}\,t\,|\mathbf{x}|/c|} = \frac{1}{|\boldsymbol{\xi}|} - \frac{(\mathbf{a}\cdot\hat{\xi})\,t\,|\mathbf{x}|}{c\,|\boldsymbol{\xi}|^2} + \mathcal{O}\big(a^2\big)\,. \tag{148}$$

The first-order correction to the equilibrium potential $\varphi_{\rm eq}(\xi) = -r_s/\xi$ is therefore

$$\varphi(\mathbf{x},t) = \varphi_{\rm eq}(|\boldsymbol{\xi}|) + \delta\varphi(\mathbf{x},t) + \mathcal{O}\big(a^2\big)\,, \tag{149}$$

with

$$\delta\varphi(\mathbf{x},t) = \frac{r_{\rm s}}{2c^2}\,(\hat{x}\cdot\mathbf{a}), \tag{150}$$

where the factor 1/2 comes from $v(t) = at$ evaluated at the retardation-averaged time $t/2$. This is the leading retarded dipolar perturbation in the Coulomb-tail region of the oscillon and matches the corresponding expansion in Appendix 17 of the main theory [1] on the same trace-fixed source convention. This dipolar perturbation is odd under $\hat{x} \to -\hat{x}$: the field is enhanced in the direction opposite to $\mathbf{a}$ (trailing side) and depleted ahead.

Pressure perturbation.—The Bernoulli pressure is $P = P_0 - \frac{e^{\varphi}}{32\pi G}|\nabla\varphi|^2$. Expanding to first order in $\delta\varphi$:

$$\delta P = -\frac{e^{\varphi_{\rm eq}}}{16\pi G}\,\nabla\varphi_{\rm eq}\cdot\nabla(\delta\varphi) - \frac{e^{\varphi_{\rm eq}}}{32\pi G}\,|\nabla\varphi_{\rm eq}|^2\,\delta\varphi. \tag{151}$$

Force integral.—The net force on the oscillon is $\mathbf{F} = -\oint d\mathbf{S}\;\delta P$. Integration by parts converts the volume integral $\int d^3x\,\nabla(\delta P)$ into the surface form; since $\delta\varphi \propto (\mathbf{a}\cdot\hat{x})$ is purely $\ell = 1$, only the dipolar harmonic survives. The detailed radial projection depends on the regular retarded kernel of the localized oscillon tail. We denote the resulting leading projection coefficient by $C_{\rm in}$ below. The angular integration projects $(\mathbf{a}\cdot\hat{x})\,\hat{x}$ onto $\mathbf{a}$:

$$\int d\Omega\;(\mathbf{a}\cdot\hat{x})\,\hat{x} = \frac{4\pi}{3}\,\mathbf{a}. \tag{152}$$

Combining the radial terms in the leading localized projection yields

$$\mathbf{F}_{\rm inertia} = -C_{\rm in}\,\frac{\mathbf{a}}{c^2}\int_0^\infty dr\;4\pi r^2\,\frac{e^{\varphi_{\rm eq}}}{32\pi G}\left(\varphi'_{\rm eq}\right)^2 \equiv -C_{\rm in}\,\frac{E_\varphi}{c^2}\,\mathbf{a}, \tag{153}$$

where $C_{\rm in}$ is the canonical leading projection coefficient for the localized-resonance inertia integral ($C_{\rm in} = 1$ at this leading order). Here

$$E_\varphi \equiv \int d^3x\,\frac{e^{\varphi_{\rm eq}}}{32\pi G}\,|\nabla\varphi_{\rm eq}|^2 \tag{154}$$

is the total gradient (kinetic) energy of the equilibrium oscillon—precisely the quantity identified in Eq. (101) as the scalar-field energy density integrated over all space. The full coefficient closure is tied to the regular-field closure of the MAIN/17 self-force calculation and to the Bernoulli normalization of the oscillon energy E_φ identified in Sec. IV.

Identification with effective mass.—Under the Bernoulli normalization tracked in Sec. IV, with the exterior-Coulomb integral matched to the localized oscillon energy through the same source normalization used in the Newtonian-recovery section, the gradient energy E_φ is identified with $m_{\rm eff}c^2$ on the spherical equilibrium oscillon profile, giving

$$\boxed{\mathbf{F}_{\rm inertia} = -C_{\rm in}\,m_{\rm eff}\,\mathbf{a} = -m_0\,e^{\varphi/2}\,\mathbf{a}\quad(C_{\rm in} = 1 \text{ at leading order}).} \tag{155}$$

This is the Newtonian inertial response in the weak-field, slow-acceleration, spherical-equilibrium regime considered here, derived from the causal (retarded) propagation of the scalar field during acceleration. The derivation is perturbative in $a/(c\omega_0) \ll 1$; at higher orders, one expects radiation-reaction corrections of the Abraham–Lorentz type by analogy with the standard electromagnetic and gravitational self-force structure, scaling as $\mathbf{F}_{\rm rad} \sim (2e^{\varphi}\,r_s/3c^3)\,\dot{\mathbf{a}}$. *Status.*—This scaling is quoted by analogy with the canonical Abraham–Lorentz expression for a charged particle and is not derived from the scalar self-field in this paper; a controlled derivation of the third-derivative term within the ISPG retarded expansion is left to future work.

For the localized resonance in the weak-field regime considered here, the leading inertial coefficient and the gravitational mass assignment are tied to the same scalar profile and Bernoulli pressure-deficit bookkeeping, with the effective redshifted mass $m_{\rm eff} = m_0\, e^{\varphi/2}$ (Eq. 36) entering through the Bernoulli normalization stated below.

C. The test-particle equivalence principle as an identity

In GR, the equality $m_{\rm inertial} = m_{\rm gravitational}$ is the weak equivalence principle, elevated to a foundational axiom. In ISPG, the test-particle version is first secured by minimal coupling: matter follows geodesics of the single physical metric $g_{\mu\nu}[\varphi]$. Within the localized resonance bookkeeping used here, the same equality is also reflected in the unified scalar-field mechanism: under the identifications established earlier in this section—namely that the gravitational mass is identified with the Bernoulli-normalized gradient energy E_φ/c^2 via the integral Eq. (154) (under the Bernoulli normalization tracked in Sec. IV, distinct from the redshift/frequency effective mass $m_{\rm eff} = m_0 e^{\varphi/2}$ of Eq. (36)), and that the inertial force takes the form (153) with the same integrand under the leading projection $C_{\rm in} = 1$—the two mass assignments reduce to the same Bernoulli-normalized quantity at leading order:

$$m_{\rm grav} = \frac{E_\varphi}{c^2} = \frac{1}{c^2}\int d^3x\, \frac{e^\varphi}{32\pi G}\, |\nabla\varphi|^2\,, \qquad m_{\rm inertial} = C_{\rm in}\, m_{\rm grav} \quad (C_{\rm in} = 1 \text{ at leading order}). \tag{156}$$

The equality is therefore not an independent postulate within this approximation: once the gravitational and inertial identifications above are accepted, both rely on the same Bernoulli-normalized gradient-energy integral over the same equilibrium field configuration, with the inertial side carrying the leading projection coefficient $C_{\rm in}$ introduced in Eq. (153). The actual equation of motion for test matter remains the geodesic equation of the metric defined by this profile; the Bernoulli-normalized gradient energy supplies the pressure-deficit interpretation of why that profile is gravitational. Inertia is the dynamic manifestation of the same profile bookkeeping, through resistance to profile readjustment during acceleration. *Status.*—Derived here under the assumptions of weak field, slow acceleration $a \ll c\omega_0$, and spherical equilibrium oscillon profile, with $C_{\rm in} = 1$ at leading order. The full coefficient closure depends on the regular-field closure of the perturbative test-body calculation in the main theory (the $C_{\rm reg}$ residual technical input) and on the Bernoulli normalization identifying E_φ with $m_{\rm eff}c^2$ on the spherical equilibrium profile (Sec. IV). The *logical* status is therefore that of a structural identity within these approximations, conditional on those two open numerical tasks, and not of an exact theorem of the full nonlinear theory.

Connection to the MOND transition: a common dynamical mechanism.—The inertial mechanism derived above has a direct galactic-scale counterpart. In both cases, *acceleration* creates a pressure deficit that the medium cannot replenish fast enough.

- **Inertia** (local): an oscillon undergoes linear acceleration $\mathbf{a}$. The pressure deficit behind the resonance cannot be refilled instantaneously because the field propagates at finite speed $c \to$ fore–aft asymmetry $\to$ restoring force $F = -m\,a$.

- **MOND** (galactic): the galactic vortex subjects the medium to centrifugal acceleration. Archion excitation nodes are driven outward, while the peripheral nodes cannot flow inward fast enough to compensate → central rarefaction (vortex pressure deficit) [2] → gravitational pull stronger than the Newtonian $1/r^2$ extrapolation from baryonic mass alone.

The structural similarity is qualitative: in both cases an acceleration field acts on the medium, the medium's finite response rate prevents equilibrium, and the resulting pressure deficit contributes either to the metric/geodesic response or to an effective inertial reaction. The analogy should not be read as a quantitative equivalence—the inertial calculation above is a microscopic perturbative result around a single oscillon profile, whereas the MOND derivation in Ref. [2] is a macroscopic fluid argument around a galactic vortex. The qualitative parallel is in the mechanism (causal delay → pressure deficit → metric or inertial response); the quantitative difference lies in the source of acceleration (oscillon recoil vs. galactic rotation) and the spatial scale ($R_c \sim \hbar/(mc)$ vs. $\ell \sim$ kpc). The MOND acceleration scale $a_0 = cH/(2\pi)$ itself reflects the largest distance over which the field can sustain a coherent pressure gradient: $\lambda_H = 2\pi c/H$, beyond which the Hubble damping renders the response overdamped [2]. This sets the boundary between the Newtonian regime (where the medium responds fully) and the MOND regime (where the centrifugal rarefaction is no longer compensated).

Contrast with Mach's principle.—The inertial mechanism derived above is intrinsically *local*: the restoring force on an accelerating oscillon arises from the retarded readjustment of its own scalar self-field within a region of order $R_c \sim \hbar/(mc)$, not from a coupling to the rest of the matter content of the universe. In Mach's original conception, the inertia of a body is determined by the gravitational influence of all distant masses; in the present picture, distant matter enters only through the asymptotic value φ_∞ that fixes the conformal factor $e^{\varphi/2}$ in $m_{\text{eff}} = m_0\, e^{\varphi/2}$. The dynamical content of inertia—the proportionality between force and acceleration—is set entirely by the local retardation of the oscillon's own field. ISPG is therefore Machian only in the weak sense that m_{eff} inherits an environmental factor through φ_∞; it is non-Machian in the strong sense that the inertial response itself does not require knowledge of the cosmological matter distribution.

D. Hysteresis: the memory of the medium

The readjustment time $t_{\text{adj}} \sim 1/\omega_0$ (Eq. 143) implies that the medium retains a memory of the resonance's previous state. When a body accelerates and then stops, the pressure distribution does not instantaneously snap to the new equilibrium—it relaxes over a finite timescale. This is hysteresis: the delayed response of the medium to changes in the source configuration.

In the classical theory [1], this memory effect is captured by the hysteresis field χ, an effective collective field with standalone covariant dynamics in the quasi-static bound-structure sector. In that one-way effective sector its source is treated as an external quasi-static functional of the background scalar, and the covariant action template is

$$S_\chi = \int d^4x \sqrt{-g} \left[-\frac{1}{2} \nabla_\mu \chi \nabla^\mu \chi - \frac{\lambda}{2} \big(\chi - \beta Y_{\rm qs}\big)^2 \right], \tag{157}$$

where $Y_{\rm qs} \equiv \sqrt{\nabla_\mu \varphi \nabla^\mu \varphi}$ is used only in the spacelike-gradient, quasi-static bound-structure regime, and $\lambda > 0$ is the stiffness parameter. The standalone χ field equation is

$$\Box \chi - \lambda \chi = -\lambda \beta Y_{\rm qs}, \tag{158}$$

a Klein–Gordon equation with mass $m_\chi = \sqrt{\lambda}$ driven by the quasi-static gradient invariant $Y_{\rm qs}$. The signs follow the $(-,+,+,+)$ signature convention adopted throughout this paper ($\Box = g^{\mu\nu}\nabla_\mu\nabla_\nu$), reproducing MAIN [1] Eq. (chi_KG) in its effective-template form. The fully varied backreacting (φ, χ) principal-symbol analysis is not closed here.

We now provide the quantum-mechanical motivation for this effective field and for the microscopic tracking scale.

Origin of χ from resonance dynamics.—Each oscillon generates a pressure deficit that extends to $r \sim R_c$. When the oscillon moves, the pressure deficit must be re-established at the new position and dissolved at the old one. The dissolution occurs via outgoing scalar waves—the same mechanism as the resonant tails of Sec. VI A—with a characteristic timescale $t_{\rm adj}$.

The hysteresis field χ is the coarse-grained dynamical variable that tracks the dissolution process. Its static equilibrium configuration is fixed by the source term in Eq. (158): setting $\partial_t \chi = 0$ and $\nabla^2 \chi \to 0$ at long wavelengths gives $\chi_{\rm eq} = \beta Y_{\rm qs}$, which coincides with MAIN [1] Eq. (chieq). The *dynamical* content of χ lives in the deviation from this equilibrium,

$$\delta\chi(\mathbf{x}, t) \equiv \chi(\mathbf{x}, t) - \chi_{\rm eq}(\mathbf{x}, t) = \chi(\mathbf{x}, t) - \beta Y_{\rm qs}(\mathbf{x}, t), \tag{159}$$

which is the quantity that can relax to zero when the source motion stops, provided damping, outgoing radiation, or coarse-graining is included. Equivalently, when the source has just moved, $Y_{\rm qs}$ evaluated on the actual (lagging) field φ differs from $Y_{\rm qs}$ evaluated on the instantaneous equilibrium field $\varphi_{\rm eq}$; this difference is what drives $\delta\chi \neq 0$ during the transient and is dissolved on the relaxation timescale discussed below. The full field $\chi = \chi_{\rm eq} + \delta\chi$ is the object that appears in the action (157) and in all subsequent macroscopic formulas, in agreement with MAIN.

Microscopic and macroscopic stiffness scales.—The microscopic tracking stiffness and the macroscopic collective relaxation scale are distinct quantities. From Sec. III E, a quasi-bound oscillon has a complex quasi-normal frequency $\omega = \omega_R - i\Gamma/2$, where the real part $\omega_R = m_0 c^2/\hbar$ gives the mass and the imaginary part Γ gives the decay width. The decay width represents the rate at which the oscillon loses energy to outgoing scalar waves—precisely the process that re-equilibrates the pressure distribution when the source moves.

For a stable particle (proton, electron), the barrier penetration is exponentially suppressed (Eq. 53):

$$\Gamma \sim \omega_R \exp\left[-2 \int_{r_1}^{r_2} \sqrt{V_{\rm eff} - \omega_R^2 e^{-2\varphi_0}}\, dr \right] \ll \omega_R. \tag{160}$$

The relaxation timescale is $t_{\rm rel} \sim 1/\Gamma \gg t_{\rm adj}$. However, for the *coherent* readjustment of the pressure profile around a *macroscopic* body composed of $N \sim 10^{68}$ oscillons, the relaxation is dominated not by individual decays but by the collective dephasing of the oscillon ensemble.

For a *single* particle, the microscopic stiffness $\lambda_{\rm micro}$ is set by the inverse Compton wavelength:

$$m_\chi^{(\rm micro)} = \sqrt{\lambda_{\rm micro}} = \frac{m_0\, c}{\hbar}\, e^{\varphi/2}. \tag{161}$$

For a proton ($m_0 \approx 1$ GeV$/c^2$), $m_\chi^{(\rm micro)} \sim 5\times 10^{15}$ m^{-1}, corresponding to $\lambda_{\rm micro} \sim 10^{31}$ m^{-2}. This extremely large stiffness means that for individual particles, χ tracks the equilibrium $\beta Y_{\rm qs}$ almost instantaneously: the readjustment is completed in one Compton period $t_{\rm adj} \sim 10^{-24}$ s.

For a *macroscopic body* (galaxy, cluster) composed of N oscillons, the collective relaxation is governed by a different scale. Each oscillon radiates outgoing scalar waves at its individual (exponentially suppressed) decay rate $\Gamma_{\rm single}$ (Eq. 160). Two limiting regimes are available for how the N single-particle rates combine into a collective relaxation rate $\Gamma_{\rm macro}$:

- the fully incoherent regime, where independent random phases give a $\sqrt{N}$ central-limit enhancement of the radiated amplitude and hence a $\sqrt{N}$ enhancement of the rate;
- the Dicke super-radiant regime, where phase coherence across the ensemble gives a linear-in-N enhancement.

Parametrically we therefore write

$$\Gamma_{\rm macro} = \mathcal{C}(N)\,\Gamma_{\rm single}, \tag{162}$$

with $\mathcal{C}(N) \in [\sqrt{N},\, N]$ bounded above by the Dicke limit and below by the fully-incoherent limit. The effective macroscopic relaxation timescale is

$$t_{\rm rel}^{(\rm macro)} = \frac{1}{\Gamma_{\rm macro}} = \frac{1}{\mathcal{C}(N)\,\Gamma_{\rm single}}. \tag{163}$$

Status of the cosmological matching.—A closed-form first-principles determination of $\mathcal{C}(N)$ in the oscillon medium (whose wavelength-to-body-size ratio lies between the two limits above) is a self-contained technical task and the topic of a separate work. Consequently the quantitative matching of $t_{\rm rel}^{(\rm macro)}$ to the Hubble time—which, taking $\Gamma_{\rm single} \sim \omega_R e^{-2S_{\rm barrier}}$ with $\omega_R \sim m_p c^2/\hbar$ and $S_{\rm barrier} \gtrsim 76$ (the lower bound fixed by the proton lifetime $\tau_p > 10^{34}$ yr, combined with the WKB expression 53) —cannot be claimed as a derived prediction at the present level of the analysis. What is derived is the parametric structure (162) and the two regime bounds; matching the Hubble-time target requires either an independent calculation of $\mathcal{C}(N)$ or an independent determination of $S_{\rm barrier}$ beyond the proton-stability lower bound.

The macroscopic Klein–Gordon mass governing galaxy-scale χ dynamics is therefore

$$\boxed{m_\chi^{(\rm macro)} = \sqrt{\lambda_{\rm macro}} = \frac{\mathcal{C}(N)\,\Gamma_{\rm single}}{c} = \frac{\mathcal{C}(N)\, e^{-2S_{\rm barrier}}\, m_0\, c}{\hbar}\, e^{\varphi/2}.} \tag{164}$$

This mass scale is suppressed relative to the microscopic value by the factor $\mathcal{C}(N)\, e^{-2S_{\rm barrier}} \ll 1$, placing $m_\chi^{(\rm macro)}$ in the intermediate regime between particle-physics energies and the Hubble scale—precisely the window relevant for galaxy-halo phenomenology. The absolute normalisation within that window depends on $\mathcal{C}(N)$, whose determination is deferred as noted above.

Two regimes of χ.—The hysteresis field thus operates in two distinct regimes:

- *Microscopic* ($\lambda = \lambda_{\rm micro}$): χ tracks $\beta Y_{\rm qs}$ instantaneously; no observable hysteresis at the single-particle level.
- *Macroscopic* ($\lambda = \lambda_{\rm macro}$): the collective relaxation is slow enough that $\chi \neq \beta Y_{\rm qs}$ for timescales comparable to galactic dynamical times, producing the halo phenomenology modeled in [1].

Universality of a_0 under N-dependent $\lambda_{\rm macro}$.—The N-dependence of $\lambda_{\rm macro}$ through $\mathcal{C}(N)$ (Eq. 164) does not contradict the universal MOND acceleration scale a_0 [1]: the covariant static limit $\chi \to \chi_{\rm eq} = \beta Y_{\rm qs}$ makes the BTFR-type halo normalisation depend only on the closure normalization β (to be fixed by the bound-structure coarse-graining and the cosmological scale $a_0 = cH_0/2\pi$), while λ controls only the *approach* to that equilibrium. Thus the system-specific $\lambda_{\rm macro}$ governs transients and relaxation spectra, whereas the time-averaged rotation-curve phenomenology tests the closure normalization.

Ghost freedom.—The χ sector is ghost-free: the Hamiltonian density

$$\mathcal{H}_\chi = \frac{c^2}{2}\,\pi_\chi^2 + \frac{1}{2}\,(\partial_i\chi)^2 + \frac{\lambda}{2}\,(\chi - \beta Y_{\rm qs})^2 \geq 0, \tag{165}$$

is manifestly non-negative for $\lambda > 0$, ensuring that no negative-norm states propagate and the theory is stable at the quantum level.

E. Microscopic motivation of the hysteresis action template

The action S_χ (Eq. 157) was introduced in the classical theory [1] as an effective closure. We now motivate its microscopic tracking scale and KG structure from resonance dynamics. This is a coarse-graining template, not a completed first-principles derivation of the full macroscopic hysteresis sector.

Non-equilibrium decomposition.—Let $\varphi_{\rm eq}(\mathbf{x}, t)$ denote the instantaneous equilibrium field—the solution of $\Box\varphi = -(8\pi G/c^4)\,T$ with the source at its current position and all transients decayed. The actual field lags behind due to finite propagation speed:

$$\varphi(\mathbf{x}, t) = \varphi_{\rm eq}(\mathbf{x}, t) + \delta\varphi(\mathbf{x}, t), \tag{166}$$

where $\delta\varphi$ is the retardation-induced excess. This is the same decomposition used in the inertia derivation (Eq. 149), where $\delta\varphi$ was shown to be a dipolar perturbation of order $a\,r/c^2$ (Eq. 150).

Perturbation dynamics inside the oscillon.—In the far field ($r \gg R_c$), $\delta\varphi$ propagates as a free massless wave ($\Box\delta\varphi = 0$) and disperses. Inside the oscillon ($r \lesssim R_c$), however, the perturbation is confined by the effective potential $V_{\rm eff}(r)$ of Eq. (52). At late times (after initial transients have radiated away), the confined perturbation is dominated by the longest-lived quasi-normal modes. We write the late-time approximation as

$$\delta\varphi(\mathbf{x}, t) \approx \sum_n a_n(t)\,\phi_n(\mathbf{x}), \tag{167}$$

where the sum runs over the discrete quasi-normal modes ϕ_n of the confining potential, and each amplitude a_n satisfies the damped oscillator equation

$$\ddot{a}_n + \Gamma_n\,\dot{a}_n + \omega_n^2\,a_n = f_n(t). \tag{168}$$

Here $\omega_n = \omega_{n,R}$ is the real part of the n-th quasi-normal frequency, Γ_n is the decay width (Eq. 53), and $f_n(t)$ is the driving force projected onto mode n from the source motion.

Fundamental mode dominance.—For a slow perturbation (acceleration timescale $\gg 1/\omega_0$), the driving force couples predominantly to the fundamental mode ($n = 0$) with locally measured frequency $\omega_0 = (m_0 c^2/\hbar)\,e^{\varphi_0/2}$ (Eq. 161). Higher overtones ($n \geq 1$) are suppressed by the ratio of the perturbation timescale to the mode spacing. The pressure deficit deviation is therefore controlled by the fundamental amplitude a_0:

$$\chi(\mathbf{x}, t) \approx \beta\,\frac{\nabla_\mu\varphi_{\rm eq}\,\nabla^\mu[\,a_0(t)\,\phi_0(\mathbf{x})\,]}{Y_{\rm qs,eq}}, \tag{169}$$

where $Y_{\rm qs,eq} = \sqrt{\nabla_\mu\varphi_{\rm eq}\,\nabla^\mu\varphi_{\rm eq}}$ and the linearization $Y_{\rm qs} \approx Y_{\rm qs,eq} + \hat{n}_\mu\nabla^\mu\delta\varphi$ (with $\hat{n}_\mu = \nabla_\mu\varphi_{\rm eq}/Y_{\rm qs,eq}$) has been used.

Covariant effective equation.—Equation (168) governs the temporal amplitude a_0 of a single oscillon. For a macroscopic body, χ is the coarse-grained superposition of the pressure deviations from all N constituent oscillons, each located at $\mathbf{r}_a$: $\chi(\mathbf{x}, t) = \sum_a \chi_a(\mathbf{x} - \mathbf{r}_a, t)$. In the continuum limit, this sum defines a smooth field. Because each individual perturbation $\delta\varphi_a$ propagates via the scalar wave equation ($\Box\delta\varphi_a = 0$ outside the source), the coarse-grained field inherits a spatial d'Alembertian; the temporal part retains the damping and restoring force from the confining potential. The resulting equation for χ is

$$\Box\chi - \frac{\Gamma_0}{c^2}\,\partial_t\chi - \frac{\omega_0^2}{c^2}\,\chi = -\frac{\omega_0^2}{c^2}\,\beta\,Y_{\rm qs}. \tag{170}$$

For stable particles (proton, electron), the decay width is exponentially suppressed: $\Gamma_0 \sim \omega_0\,e^{-2S_{\rm barrier}} \ll \omega_0$ (Eq. 160). The damping term $(\Gamma_0/c^2)\,\partial_t\chi$ is therefore negligible compared to the mass term $(\omega_0^2/c^2)\,\chi$, and Eq. (170) reduces to

$$\Box\chi - \lambda\,\chi = -\lambda\,\beta\,Y_{\rm qs}, \tag{171}$$

with

$$\lambda = \frac{\omega_0^2}{c^2} = \frac{m_0^2\,c^2}{\hbar^2}\,e^{\varphi}, \tag{172}$$

which is precisely the microscopic stiffness $\lambda_{\rm micro}$ of Eq. (161). Equation (171) is identical to the phenomenological field equation (158).

Effective-action reconstruction.—The motivated microscopic tracking equation (171) is reproduced by the Euler–Lagrange equation of the effective action template

$$S_\chi = \int d^4x\,\sqrt{-g}\,\left[-\frac{1}{2}\,\nabla_\mu\chi\,\nabla^\mu\chi - \frac{\lambda}{2}\big(\chi - \beta\,Y_{\rm qs}\big)^2\right], \tag{173}$$

which matches S_χ (Eq. 157) in the quasi-static source sector. The kinetic term arises from the wave propagation of the confined perturbation; the microscopic stiffness $\lambda = \omega_0^2/c^2$ is the square of the inverse Compton wavelength, reflecting the confining scale of the oscillon; and the source $\beta Y_{\rm qs}$ is the equilibrium value toward which the medium relaxes. This χ is the dynamical coarse-grained hysteresis field; in the quasi-static galactic limit it is the field-level representative of the local transported-response multiplier written in the companion MOND paper [2] for the coherent vortex branch.

Parameter status.—The microscopic argument identifies some, but not all, of the previously phenomenological inputs:

(i) The microscopic stiffness is $\lambda_{\rm micro} = \omega_0^2/c^2 = (m_0c/\hbar)^2\,e^{\varphi}$, set by the oscillon's fundamental frequency and hence by the particle mass. This is the origin of the hierarchy between microscopic and macroscopic hysteresis: the microscopic relaxation is completed in one Compton period $t_{\rm adj} \sim 1/\omega_0 \sim 10^{-24}$ s (proton), while the macroscopic relaxation depends on the collective decay of N oscillons (Eq. 164).

(ii) The dissipation rate Γ_0 is exponentially suppressed (Eq. 160) for stable particles, justifying the neglect of the damping term. For unstable resonances ($\Gamma \sim \omega_0$), the full damped equation (170) applies, and the hysteresis field decays on the particle's lifetime—as expected physically.

(iii) The coupling β is motivated by the Bernoulli relation (Sec. IV A): the pressure deficit $\delta P \propto Y_{\rm qs,eq}\,\delta Y_{\rm qs}$, so β normalizes χ to the pressure deviation per unit gradient. Its numerical value remains a phenomenological closure normalization until the full bound-structure coarse-graining is completed.

Status of the construction.—The hysteresis action S_χ is therefore a microscopically motivated effective template. The quasi-normal mode structure, fundamental-mode dominance for slow perturbations, and exponential suppression of the decay width for stable particles motivate the microscopic tracking stiffness $\lambda_{\rm micro}$. They do not by themselves close the macroscopic collective relaxation scale, the full backreacting principal-symbol analysis, or the first-principles value of β. Those remain Route B/coarse-graining tasks to be tested against galaxy-halo rotation curves [1].

F. Summary: the three faces of the medium

The scalar-metric medium exhibits three distinct mechanical properties, all tied to the same resonance–medium coupling:

1. **Pressure** ($\nabla\varphi \neq 0$ around matter): the Bernoulli deficit that produces gravity (Sec. IV).

2. **Zero self-drag at constant velocity**: Lorentz invariance of the covariant field+source equation ensures that isolated uniformly moving resonances experience no velocity-proportional self-friction (Sec. V A); external gravitational and tidal forces are separate.

3. **Memory** (hysteresis): the finite readjustment time of the oscillon profile during acceleration produces inertia and, on macroscopic scales, the χ-field dynamics responsible for galaxy-halo phenomenology (Secs. V D and V E).

These three properties—pressure without friction, combined with memory—characterize a unique physical medium: a *superfluid with hysteresis*. Within the weak-field, slow-acceleration, spherical-equilibrium regime considered here and under the leading-order projection $C_{\rm in} = 1$ of Eq. (156), the equivalence principle becomes a structural consequence of the fact that gravity and inertia are the static and dynamic aspects of the same Bernoulli-normalized gradient-energy bookkeeping of a resonant excitation; the exact non-perturbative statement remains tied to the open coefficient and normalization tasks recorded above.

VI. ENTROPY, SELF-REGULATION, AND DARK ENERGY

The Bernoulli mechanism (Sec. IV) explains how matter generates a local pressure deficit. We now turn to the cosmological consequences: the irreversible spreading of resonant tails produces a global entropy increase, which is dynamically regulated by a negative-feedback loop. This mechanism provides a *structural channel* for estimating the observed dark-energy scale, but it is not yet a completed solution of the full cosmological-constant problem: the cumulative tail-energy budget is itself small (see Eq. (183) below), and the relevance for ρ_Λ enters through a perturbative-branch feedback-regulated equilibrium developed in Sec. VI C. The same dynamics predicts an evolving dark-energy equation of state $w(z) \neq -1$ in the effective fluid description. The quantitative success of these predictions hinges on the magnitude of the per-particle tail rate $\mathcal{P}_{\rm tail}$; that magnitude depends on non-perturbative oscillon stability physics whose precise evaluation is beyond the scope of this paper, as explained below.

A. Resonant tails and irreversible entropy production

A localized resonant excitation (particle) consists of a concentrated core and extended oscillatory tails that propagate outward at the local speed of light (Sec. III E). These tails carry energy away from the core and into the surrounding medium. Once emitted, a tail cannot be recalled: the process is irreversible by the same argument that makes electromagnetic radiation irreversible (retarded boundary conditions).

The energy flux carried by the tail of a single particle of mass m is determined by the radiative damping of the quasi-bound oscillon. We compute this explicitly from the outgoing energy flux through a spherical surface surrounding the oscillon core.

Energy flux for the scalar field.—The stress-energy tensor component T^{0r} gives the radial energy flux. For the bi-conformal metric with $\varphi = \varphi_0 + \delta\varphi$,

$$T^{0r} = \frac{1}{16\pi G}\, g^{00}\, g^{rr}\, \partial_0(\delta\varphi)\, \partial_r(\delta\varphi) = -\frac{e^{2\varphi_0}}{16\pi G}\, \dot{\Phi}\, \Phi', \tag{174}$$

where $\dot{\Phi} \equiv \partial_t \Phi$ and $\Phi' \equiv \partial_r \Phi$.

Outgoing wave at infinity.—Far from the oscillon core ($r \gg R_{\rm core}$), the field behaves as an outgoing spherical wave:

$$\Phi(r,t) \sim \frac{A_\infty}{r}\, \cos(\omega t - kr), \qquad k = \omega/c. \tag{175}$$

The radial derivative is $\Phi' \sim -\frac{A_\infty}{r^2}\cos(\omega t - kr) + \frac{kA_\infty}{r}\sin(\omega t - kr)$; at large r, the dominant term is $\Phi' \approx k\Phi$ (in phase quadrature with $\dot{\Phi}$).

Time-averaged power.—The time-averaged flux through a sphere of radius r is

$$\langle T^{0r} \rangle = \frac{e^{2\varphi_0}}{32\pi G}\, \omega k\, \frac{A_\infty^2}{r^2}. \tag{176}$$

The total radiated power is the flux times the area $4\pi r^2$:

$$\mathcal{P}_{\rm rad} = 4\pi r^2 \langle T^{0r} \rangle = \frac{e^{2\varphi_0}}{8G}\, \omega k\, A_\infty^2. \tag{177}$$

Amplitude matching.—The asymptotic amplitude A_∞ is determined by matching to the oscillon core. For a core of radius R with amplitude $A_{\rm core}$, continuity requires $A_\infty \sim A_{\rm core} R$. The core amplitude is fixed by the oscillon mass: the core energy is $E_{\rm core} \sim A_{\rm core}^2 \omega^2 R^3 \sim mc^2$, giving $A_{\rm core} \sim \sqrt{m}/(\omega R^{3/2})$. Substituting:

$$A_\infty^2 \sim A_{\rm core}^2 R^2 \sim \frac{m}{\omega^2 R}. \tag{178}$$

Combining with Eq. (177), using $\omega = mc^2/\hbar$ and $k = \omega/c$, and tracking the c-factors of the kinetic prefactor in the action carefully (an explicit re-derivation including the $c^4/(16\pi G)$ overall normalisation of the gradient sector is left as a technical exercise; see the discussion below), the perturbative scaling reduces to the dimensionally consistent form

$$\mathcal{P}_{\rm tail}^{\rm (pert)} \sim \frac{e^{2\varphi_0}}{8G}\, \frac{m^3 c^3}{\hbar^2\, R} \xrightarrow{R\sim\hbar/(mc)} \frac{G\, m^4\, c^3}{\hbar^2}\, e^{2\varphi_0}, \tag{179}$$

where in the last step the core radius R is set by the Compton wavelength $R \sim \hbar/(mc)$ and the leading $O(1)$ numerical coefficients have been absorbed into the "$\sim$". The boxed asymptotic scaling is therefore

$$\boxed{\mathcal{P}_{\rm tail}^{\rm (pert)} \sim \frac{G\, m^4\, c^3}{\hbar^2}\, e^{2\varphi_0}\,.} \tag{180}$$

Direct evaluation of this perturbative expression for an electron at $\varphi_0 \approx 0$ gives $\mathcal{P}_{\rm tail}^{\rm (pert)} \sim 10^{-37}$ W, and for a nucleon $\sim 10^{-24}$ W.

This derivation confirms the m^4 scaling from dimensional and scaling analysis: the oscillon radiates as an outgoing scalar wave, with amplitude set by the core mass and frequency. The exponential factor $e^{2\varphi_0}$ reflects the gravitational redshift of the radiation in the local field.

Perturbative vs. non-perturbative estimate.—The expression (180) is the perturbative scaling: it captures only the gross power-counting of the radiating-oscillon calculation and treats the core as a coherent classical mode of amplitude fixed by $E_{\rm core} \sim mc^2$. In the oscillon literature, however, long-lived quasi-bound scalar configurations decay much more slowly than the perturbative m^4 scaling suggests: their lifetime is controlled by an exponentially small overlap integral between the bound profile and the free continuum, of the schematic form $\mathcal{F}_{\rm nonpert} \sim e^{-S_{\rm osc}/\hbar}$ with $S_{\rm osc}$ the oscillon stability action. An estimate of the resulting suppression for ISPG-resonance parameters is

$$\mathcal{P}_{\rm tail}^{\rm (np)} \equiv \mathcal{F}_{\rm nonpert} \cdot \mathcal{P}_{\rm tail}^{\rm (pert)} \sim 10^{-107}\,{\rm W} \times e^{2\varphi_0} \quad \text{(schematic suppressed per-particle benchmark),} \tag{181}$$

where the numerical value is not yet a derived species-specific prediction. The detailed evaluation of $\mathcal{F}_{\rm nonpert}$ from the full nonlinear ISPG oscillon profile—and in particular the verification that it is consistent both with the per-particle energy budget here *and* with the cosmological-constant match of Sec. VI C, which uses the perturbative formula at face value in Planck units—is an open technical task. The remainder of this subsection uses the schematic value $\mathcal{P}_{\rm tail}^{\rm (np)} \sim 10^{-107}$ W $\times e^{2\varphi_0}$ only as a suppressed-branch per-particle energy-budget benchmark; it is not the normalization used for the perturbative cosmological scale estimate below.

The total outgoing *power* of the observable universe into the background scalar field, for $N_b \sim 10^{80}$ baryons, is

$$\dot{E}_{\rm bg,\,tot} \sim N_b \cdot \mathcal{P}_{\rm tail}^{\rm (np)} \sim 10^{80} \cdot 10^{-107} \sim 10^{-27}\ {\rm W}, \tag{182}$$

an exceedingly small but strictly positive outgoing power in the suppressed branch. The calculation establishes positive outgoing energy flux under retarded boundary conditions. Interpreting this flux as entropy production requires a coarse-grained scalar bath and an effective temperature; those thermodynamic variables are not derived here, so we quote energy power rather than a closed entropy-production law. Over the age of the universe ($t_0 \sim 4 \times 10^{17}$ s), the cumulative energy deposited into the background scalar field is

$$E_{\rm bg} \sim \dot{E}_{\rm bg,\,tot} \cdot t_0 \sim 10^{-10}\ {\rm J}, \tag{183}$$

which is negligible compared to the total energy budget. This confirms that the deposition is a slow, secular process—consistent with the near-constancy of fundamental parameters over cosmic history.

Physical picture.—Every particle, every moment, broadcasts a faint resonant "echo" into the medium. These echoes superpose to form a stochastic background vibration that fills all of space—including regions devoid of matter (Sec. IV C). This background vibration is energy distributed in the medium that does not constitute localized particles (it has no resonant structure) and does not produce a localized Newtonian attraction: a spatially uniform $\langle(\nabla\varphi)^2\rangle$ has vanishing gradient of the average, so no local gravitational pull on a test mass arises from it. It is, however, energetically real, and the corresponding uniform energy density does enter the cosmological Friedmann equation on horizon scales; this cosmological role is precisely what is exploited in the cosmological-constant analysis of Sec. VI C.

B. The negative-feedback loop

The resonant-tail mechanism might seem to predict a runaway process: more accumulated tail energy → lower pressure → more tail emission. In fact, the opposite occurs. The system is self-regulating through a negative-feedback loop:

1. Particles emit resonant tails, increasing the background vibration level $\langle\delta\varphi^2\rangle$.
2. The increased background vibration, through the Bernoulli mechanism, lowers the global static pressure P_{static}.
3. Lower pressure reduces the effective mass of each particle: $m_{\text{eff}} = m_0\, e^{\varphi/2}$, where φ becomes more negative.
4. A lighter particle has lower resonance amplitude and emits *less* energy per unit time: $\mathcal{P}_{\text{tail}} \propto m^4$.
5. The outgoing tail power, and therefore any associated coarse-grained entropy-production proxy, decreases.

This can be modeled as a first-order dynamical system. Let $\varepsilon(t)$ denote the mean background vibration energy density. The feedback loop gives

$$\dot{\varepsilon} = \lambda\, n\, m_0^4\, e^{2\varphi(\varepsilon)}, \tag{184}$$

where $\lambda = Gc^3/\hbar^2$ is the dimensionally consistent coupling constant inherited from the boxed form (180) (subject to the same non-perturbative factor $\mathcal{F}_{\text{nonpert}}$ noted there), n is the particle number density, and $\varphi(\varepsilon) < 0$ is the global scalar-field shift induced by the accumulated background energy. At leading order the Bernoulli relation between accumulated background energy density and the global pressure shift gives

$$\varphi(\varepsilon) \simeq -\,\varepsilon/\varepsilon_*, \qquad \varepsilon_* \equiv \frac{c^2\, e^{-\varphi_0}}{16\pi G}, \tag{185}$$

to leading order around the local equilibrium. Higher-order corrections to this functional form modify the late-time $\varepsilon(t)$ asymptotics quantitatively but not qualitatively: in every case the right-hand side of Eq. (184) is monotonically decreasing in ε and approaches zero asymptotically. Since $\varphi(\varepsilon)$ becomes more negative as ε grows (more background vibration $\to$ lower pressure), the exponential factor $e^{2\varphi}$ monotonically decreases, providing the negative feedback.

The fixed point $\dot{\varepsilon} = 0$ is approached only asymptotically ($\varepsilon \to \infty$, $\varphi \to -\infty$, $e^{2\varphi} \to 0$): the process never truly stops, but decelerates without bound. The solution behaves as

$$\varepsilon(t) \sim \varepsilon_0 + A \ln(t/t_0) \quad (t \gg t_0), \tag{186}$$

a logarithmic growth—cosmologically slow.

C. Perturbative feedback estimate for the dark-energy scale

The cosmological constant problem is the 10^{120} discrepancy between the QFT prediction of vacuum energy density ($\rho_{\mathrm{QFT}} \sim M_{\mathrm{Pl}}^4$) and the observed value ($\rho_\Lambda \sim 10^{-120}\, M_{\mathrm{Pl}}^4$) [6, 7].

In QFT, the vacuum energy is computed by summing zero-point energies of all field modes up to some cutoff Λ:

$$\rho_{\mathrm{QFT}} = \int_0^\Lambda \frac{4\pi k^2}{(2\pi)^3} \frac{\hbar\omega_k}{2}\, dk \propto \Lambda^4. \tag{187}$$

This integral has no built-in regulator; it grows as the fourth power of the cutoff.

In the ISPG sector considered here, the gravitating dark-energy component is the *dynamical equilibrium* value of the accumulated echo-background energy $\varepsilon(t)$—the stochastic vibration deposited by resonant tails (Sec. VI A), distinct from the primordial ν_0-mode reservoir that depletes as matter forms. This statement does not, by itself, derive a cancellation or decoupling of the conventional QFT zero-point contribution in Eq. (187); that decoupling must be treated as a renormalization assumption or as part of a future completion. Within the echo-background sector, the negative-feedback loop (Sec. VI B) provides a regulator: the system cannot accumulate arbitrarily large echo energy because doing so would suppress the very sources (particle resonances) that generate it.

Cosmological feedback channel.—On cosmological scales, the local feedback of Sec. VI B is supplemented by a second, dominant channel: the de Sitter expansion dilutes the very sources of vacuum energy. The particle number density is $n(t) = n_0/a^3(t)$, so the cosmological feedback ODE becomes

$$\dot{\varepsilon} = \frac{\mathcal{P}_{\mathrm{tail},0}^{(\mathrm{pert})}\, n_0}{a^3(t)}, \qquad \mathcal{P}_{\mathrm{tail},0}^{(\mathrm{pert})} \equiv \frac{G\, m_0^4\, c^3}{\hbar^2}, \tag{188}$$

where $\mathcal{P}_{\mathrm{tail},0}^{(\mathrm{pert})}$ is the perturbative tail power per particle (Eq. 180). It has natural-unit dimension M^2 and SI dimension W; the dimensionally consistent c^3 form is used here, in agreement with Sec. VI A. As written, $\dot{\varepsilon}$ is a derivative with respect to the cosmic/coordinate time t, so no additional process-time factor $\mathcal{C}(z)$ is applied in this source law. The single-species approximation is justified because nucleons dominate the sum $\sum_i \mathcal{P}^{(\mathrm{pert})}_{\mathrm{tail},0,i}\, n_{0,i}$ over particle species: the m^4 scaling of $\mathcal{P}^{(\mathrm{pert})}_{\mathrm{tail},0,i}$ suppresses lighter-species contributions by $\gtrsim 10^{14}$ for electrons and by $\sim 10^{40}$ for neutrinos at comparable number densities.

When ε dominates the energy budget, the Friedmann equation gives $H = \sqrt{8\pi G\varepsilon/(3c^2)}$ and $a(t) \propto e^{Ht}$. The sources are then exponentially diluted:

$$\dot{\varepsilon} = \mathcal{P}^{(\mathrm{pert})}_{\mathrm{tail},0}\, n_0\, e^{-3Ht}. \tag{189}$$

Integrating and taking $t \gg H^{-1}$:

$$\varepsilon \;\longrightarrow\; \varepsilon_{\mathrm{eq}} = \frac{\mathcal{P}^{(\mathrm{pert})}_{\mathrm{tail},0}\, n_0}{3H}. \tag{190}$$

Self-consistency (H depends on ε) yields a closed equation for the equilibrium density:

$$\boxed{\rho_{\mathrm{vac}} = \varepsilon_{\mathrm{eq}} = \left(\frac{\mathcal{P}^{(\mathrm{pert})}_{\mathrm{tail},0}\, n_0}{3}\sqrt{\frac{3c^2}{8\pi G}}\right)^{2/3}.} \tag{191}$$

Numerical estimate.—Taking m_0 to be a representative nucleon/proton mass scale and n_0 to be the present baryon number density (for example $n_b \sim 0.25\,\mathrm{m}^{-3}$), in Planck units ($G = \hbar = c = 1$), $\mathcal{P}^{(\mathrm{pert})}_{\mathrm{tail},0} = m_0^4$, $n_0 = n_0^{\mathrm{phys}}\,\ell_{\mathrm{Pl}}^3 \sim 10^{-105}$, $m_0/M_{\mathrm{Pl}} \sim 10^{-19}$, so $\mathcal{P}^{(\mathrm{pert})}_{\mathrm{tail},0} n_0 \sim 10^{-76} \times 10^{-105} = 10^{-181}$. The 2/3 power gives

$$\rho_{\mathrm{vac}} \sim (10^{-181})^{2/3} \sim 10^{-121}\, M_{\mathrm{Pl}}^4 \sim H_0^2\, M_{\mathrm{Pl}}^2, \tag{192}$$

within an order of magnitude of the observed $\rho_\Lambda \sim 10^{-120}\, M_{\mathrm{Pl}}^4$. Equivalently, a direct SI estimate with the same proton-scale perturbative power and baryon density gives the observed scale only at the broad order-of-magnitude level, not as a precision fit.

The key point within this sector is that ρ_{vac} is set by the self-consistent balance between tail production and de Sitter dilution rather than by inserting a UV cutoff into the echo-background energy. This sector statement should not be read as a derivation that ordinary zero-point contributions fail to gravitate. The suppression $(m_0/M_{\mathrm{Pl}})^{8/3}\,(n_0\,\ell_{\mathrm{Pl}}^3)^{2/3} \sim 10^{-121}$ arises from two small dimensionless ratios: the particle mass relative to the Planck mass, and the cosmological particle density measured in Planck volumes.

Status of the no-fine-tuning claim.—The estimate above is a perturbative-branch scale estimate, not a completed solution of the full cosmological-constant problem. It uses the perturbative scaling

$$\mathcal{P}^{(\mathrm{pert})}_{\mathrm{tail},0} = G m_0^4 c^3/\hbar^2$$

at face value. Explicitly: the cosmological estimate in this subsection and the per-particle budget of Sec. VI A use numerically different values of the tail power. Section VI A adopts the non-perturbative value $\mathcal{P}^{(\mathrm{np})}_{\mathrm{tail}} \sim 10^{-107}$ W as a schematic suppressed per-particle benchmark (at $\varphi_0 \approx 0$), whereas the Planck-unit combination $\mathcal{P}^{(\mathrm{pert})}_{\mathrm{tail},0} n_0 \sim 10^{-181}$ entering Eq. (191) corresponds to the perturbative $\mathcal{P}^{(\mathrm{pert})}_{\mathrm{tail}} \sim 10^{-24}$ W—a factor $\mathcal{F}^{-1}_{\mathrm{nonpert}} \sim 10^{83}$ apart. If

$$\mathcal{P}^{(\mathrm{np})}_{\mathrm{tail}} = \mathcal{F}_{\mathrm{nonpert}}\, \mathcal{P}^{(\mathrm{pert})}_{\mathrm{tail}},$$

then Eq. (191) rescales as

$$\rho_{\mathrm{vac}} \to \mathcal{F}^{2/3}_{\mathrm{nonpert}} \rho^{(\mathrm{pert})}_{\mathrm{vac}}.$$

The structural mechanism—feedback-regulated balance between source and de Sitter dilution—does *not* depend on the precise value of $\mathcal{F}_{\mathrm{nonpert}}$; what does is the quantitative coincidence with the observed $10^{-120}\, M^4_{\mathrm{Pl}}$. In addition, the treatment of conventional zero-point energy requires a renormalization/decoupling assumption not derived in this paper. A controlled non-perturbative calculation that simultaneously satisfies the per-particle energy budget of Sec. VI A, the perturbative dark-energy scale estimate above, and the zero-point decoupling condition is an open task.

D. Dark-energy equation of state: $w(z) \neq -1$

A true cosmological constant has equation-of-state parameter $w \equiv P/\rho = -1$ exactly, for all time. In ISPG, the observable dark-energy density is a dynamical quantity—the accumulated resonant-tail echo background $\varepsilon(t)$—that evolves according to Eq. (184). Its effective equation of state is therefore time-dependent.

To derive $w(z)$, note that the echo-background sector is not separately conserved: it receives a continuous source $Q = \mathcal{P}^{(\mathrm{pert})}_{\mathrm{tail},0} n_0/a^3 > 0$ from resonant-tail power while having intrinsic pressure $p = -\varepsilon$ (the quasi-static phantom condensate satisfies $\rho + p \approx 0$). An observer who models the dark-energy sector as a conserved effective fluid applies the standard continuity equation $\dot{\varepsilon} + 3H(1 + w)\varepsilon = 0$, obtaining

$$w_{\mathrm{eff}}(z) = -1 - \frac{\dot{\varepsilon}(z)}{3H(z)\, \varepsilon(z)}\,. \tag{193}$$

Since $\dot{\varepsilon} > 0$ at all epochs (Eq. 188), the echo-background density *grows*, which in conservation language means $w < -1$: a growing energy density dilutes less than Λ, placing it on the phantom side. In the present *effective fluid* description this is not a kinetic-ghost instability but a *source-driven* effect: matter continuously feeds the vacuum through irreversible tail emission, and any observer unaware of the source term would infer phantom behaviour. The underlying scalar action of ISPG separately carries phantom kinetic sign ($\kappa = -1/2$ in Eq. (2)); its all-orders quantum stability is established in Sec. III G via cosmological redshift damping and the higher-derivative cutoff. The effective $w < -1$ inferred above is therefore compatible with the kinetic-stability analysis of the underlying field, not in addition to it.

The correction $\delta w(z) \equiv |w + 1| = \dot{\varepsilon}/(3H\varepsilon) > 0$ is small. At early times ($z \gg 1$), when particles were more massive (m_{eff} larger due to higher pressure) and more numerous (pre-recombination), $|\delta w|$ was larger. At late times ($z \to 0$), the feedback suppression makes

$\delta w \to 0^+$: the dark energy asymptotically approaches a cosmological constant from the phantom side.

The predicted behavior is:

$$\boxed{w(z) = -1 - \delta w_0 \,\frac{(1+z)^3}{E(z)}, \quad E(z) \equiv \frac{H(z)}{H_0}, \quad \delta w_0 \sim 10^{-2}\text{–}10^{-1},} \tag{194}$$

with $w < -1$ at all epochs (source-driven phantom, no ghost instability). The sign follows from the positive echo-background source term; the magnitude of δw_0 inherits the tail-power normalization discussed above and therefore remains conditional until the non-perturbative normalization is closed.

Observational status.—In 2024, the Dark Energy Spectroscopic Instrument (DESI) reported first-year baryon acoustic oscillation data preferring time-varying dark energy over a cosmological constant at the $\sim 2.6\sigma$ level [24]. The subsequent Data Release 2 [25], based on 1.4×10^7 tracers, strengthened the preference to 3.1σ (BAO+CMB), rising to 4.2σ with DES-SN5YR supernovae. The CPL fit yields $w_0 = -0.45^{+0.34}_{-0.21}$ and $w_a = -1.79^{+0.48}_{-1.00}$, while the constant-$w$ fit gives $w = -0.99^{+0.15}_{-0.13}$.

The ISPG prediction is $w_0^{\rm ISPG} = -1 - \delta w_0$ with $\delta w_0 \sim 10^{-2}$–10^{-1} (Eq. 194); for definiteness we use the central value $\delta w_0 \simeq 0.06$, giving $w_0 \simeq -1.06$ and $w_a \simeq -0.15$. This value lies on the *phantom* side of -1 and within the DESI DR2 *constant-w* interval at 1σ. Under the CPL parametrisation, however, the DESI central value $w_0 \simeq -0.45$ sits on the *quintessence* side at $> 4\sigma$ from -1; the ISPG prediction is therefore in tension with the DR2 CPL fit by virtue of an opposite sign of $w_0 + 1$, even though the sign of w_a is reproduced (with a magnitude smaller by roughly an order of magnitude). Neither parametrisation is yet definitive: the constant-w and CPL fits make different assumptions about the time-evolution of w, and current data do not discriminate between an ISPG-type phantom-side w that approaches -1 at $z = 0$ and a CPL-type quintessence-side w with strongly negative w_a. The Euclid space telescope and the Vera Rubin Observatory, with projected $\sigma(w_0) \sim 0.02$, are expected to discriminate between these scenarios within the current decade.

a. Cumulative process-time and the process age of the universe. The local clock-rate response derived from the bi-conformal metric (Sec. III, Eq. 27, $d\tau = e^{\varphi/2}\, dt$ with the local metric scalar φ), when applied as a constitutive pressure-ontology postulate to the coarse-grained cosmological pressure-history diagnostic $\varphi_{\rm bg}(z)$ of the main theory (rather than to the homogeneous FLRW metric scalar $\varphi_0 = 0$), implies that the total process duration accumulated by pressure-controlled matter-sector processes since redshift z_f exceeds the coordinate lookback time whenever past coarse-grained pressure was higher than present coarse-grained pressure:

$$\tau_{\rm proc}(z_f) = \int_0^{z_f} \left[\frac{\langle P_{\rm stat}\rangle_V(z)}{\langle P_{\rm stat}\rangle_V(0)}\right]^{1/2} \frac{dz}{(1+z)\,H(z)}\,. \tag{195}$$

Here $\langle P_{\rm stat}\rangle_V$ is the volume-averaged static pressure of the medium at the indicated epoch, in the process-time notation of the main theory [1]. The pressure ratio is fully determined by the feedback ODE (184) and the dark-energy parametrisation (194): $\langle P_{\rm stat}\rangle_V(z)/\langle P_{\rm stat}\rangle_V(0) = [e^{2\varphi_{\rm bg}(z)}/e^{2\varphi_{\rm bg}(0)}]^{1/2}$, with the ratio supplied by the self-consistent cosmological solution. Here α parametrises the cosmological evolution of the per-particle tail power $\mathcal{P}_{\rm tail}(z) = \mathcal{P}_{\rm tail,0}(1+z)^\alpha$ entering Eq. (188), in the convention used by the companion MOND paper [2] (their Eq. 70); $\alpha = 1$ corresponds to the matter-dominated tracking of the proper-density factor $e^{-\varphi/2}$ at leading order. For the fiducial parameter set ($z_f = 10$, $\alpha = 1$, and a high-amplitude closure benchmark $\delta w_0 \simeq 0.24$ used here as a parameter-study input rather than a central dark-energy prediction $\delta w_0 \simeq 0.06$), the process-time age is $\tau_{\rm proc}(10) \simeq 21$ Gyr, roughly 1.6× the coordinate lookback of $\simeq 13.3$ Gyr. Substituting the central $\delta w_0 = 0.06$ instead reduces the enhancement to ~ 1.1–$1.2\times$. The companion MOND paper [2] exploits the upper-end enhancement to resolve the vortex-formation timeline; the present derivation shows that the enhancement is a direct consequence of the feedback mechanism rather than a separate assumption, and that its quantitative size is controlled by the chosen δw_0 benchmark; the central quantum echo-background range gives the smaller enhancement quoted above.

E. Cosmological fine-structure constant variation

At the classical level, the bi-conformal symmetry ensures that the fine-structure constant $\alpha = e^2/(4\pi\epsilon_0\hbar c)$ is exactly invariant: all factors of $e^{\varphi/2}$ cancel in the dimensionless ratio [1].

At the quantum level, virtual particle loops contribute corrections to electromagnetic vertices. Since virtual particles have effective masses $m_{\rm eff} = m_0\, e^{\varphi/2}$ that depend on the background pressure, these loop corrections can be parametrized as a candidate residual φ-dependence in the *effective* fine-structure constant. The following expression is a schematic fixed-reference-scale ansatz, not yet a renormalized ISPG one-loop prediction:

$$\alpha_{\rm eff}(\varphi) = \alpha_0 \left[1 + \frac{2\alpha_0}{3\pi}\,\varphi + \mathcal{O}\big(\alpha_0^2\varphi^2\big)\right], \tag{196}$$

where $\alpha_0 = 1/137.036$ is the present-day value and the displayed coefficient and sign are placeholders motivated by the electron vacuum-polarization threshold (Uehling) analogy with a φ-dependent electron mass. A complete prediction requires the renormalization prescription and the held-fixed comparison scale.

For the cosmological variation over a lookback time corresponding to redshift z, the scalar field shift $\Delta\varphi \sim \mathcal{O}(H_0 t_{\rm lookback})$, if supplied by a cosmological $\varphi(z)$ solution, gives the schematic target scale

$$\left.\frac{\Delta\alpha}{\alpha}\right|_{\rm schem} \sim \frac{2\alpha_0}{3\pi}\,\Delta\varphi \sim 10^{-7}\text{–}10^{-6} \quad (z \sim 1\text{–}3). \tag{197}$$

Equation (197) should be read as a target-scale loop estimate until the renormalization prescription, comparison scale, sign, and redshift profile are fixed. This caveat is essential because local dimensionless observables cancel under a uniform bi-conformal scaling; cosmological comparisons of $\alpha_{\rm eff}(\varphi)$ must specify what reference scale is held fixed across epochs. The same $\varphi(z)$ model must also suppress the present-day slope enough to satisfy atomic-clock drift bounds. Current low-systematics quasar constraints are already at ppm-level precision, so a generic 10^{-7}–10^{-6} signal is close to existing sensitivity rather than an observed effect. The sharp test is therefore a sign and redshift profile for ESPRESSO and ELT-ANDES, not only an order-of-magnitude estimate.

F. Cosmological expansion as resonant-tail propagation

The results of Secs. VI A–VI E imply a natural interpretation of cosmological expansion within the ISPG framework. The ingredients are already in place:

1. Every particle continuously emits resonant tails that propagate outward at the local speed of light (Sec. VI A).

2. The tails are irreversible and fill all of space, including regions devoid of localized matter (Sec. IV C).

3. The medium is infinite: a medium has no boundary, and the tails propagate indefinitely.

4. In a region where no resonance has ever arrived, the scalar field is homogeneous ($\nabla\varphi = 0$) and no localized excitation exists. Without a resonant oscillator—an atom, a particle, a clock—proper time has no operational definition.

Taken together, these statements yield a picture of cosmological expansion that does not invoke an inflaton field or an initial singularity:

The observable universe is the region that resonant tails have reached since the onset of the first excitations. Beyond the outermost tail front, the medium exists in its quiescent state: homogeneous, devoid of resonances, and operationally timeless. The tail front advances at c into the quiescent medium; where it arrives, the scalar field acquires a nonzero gradient, the first pressure deficit forms, and operational proper time becomes definable through localized resonant oscillators.

This provides a physical mechanism for the cosmological horizon: we observe precisely the region within which resonant information has had time to propagate. The horizon grows monotonically as tails continue to spread, and the process is irreversible (Sec. VI A).

The quiescent state is defined not by a particular value of φ (which is gauge-dependent under the global shift $\varphi \to \varphi + C$) but by the gauge-invariant conditions $\nabla\varphi = 0$ and the absence of localized resonant excitations.

We now show that this picture is quantitatively consistent with the standard FLRW expansion history.

Friedmann equations.—The ISPG action $S = (16\pi G)^{-1}\int d^4x\sqrt{-g}\,[R + \frac{1}{2}\,g^{\mu\nu}\partial_\mu\varphi\,\partial_\nu\varphi] + S_{\mathrm{m}}$ is Einstein gravity minimally coupled to a scalar field. On a homogeneous, isotropic background the scalar field decomposes as $\varphi(t) = \bar\varphi(t) + \delta\varphi_{\mathrm{bg}}(t)$, where $\bar\varphi$ is the piece sourced by the mean matter density (mediating gravity, exactly as in GR at 1PN [1]) and $\delta\varphi_{\rm bg}$ is the accumulated background vibration from resonant tails (Sec. VI A). The Friedmann equation retains its standard form:

$$H^2(z) = H_0^2\left[\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_{\rm DE}(z)\right], \tag{198}$$

where the dark-energy density $\Omega_{\rm DE}(z)$ is set by the dynamical echo-background energy $\varepsilon(t)$ of Eq. (186), rather than by a cosmological constant.

Particle horizon = tail front.—In FLRW cosmology, the proper (physical) particle-horizon distance at cosmic time t is

$$d_H(t) = a(t)\int_0^t \frac{c\,dt'}{a(t')}. \tag{199}$$

In ISPG, the locally measured speed of light is exactly c everywhere (Eq. 27). The outermost resonant tail, emitted at the onset of the first excitations and propagating at c, covers precisely the proper distance $d_H(t)$. Therefore,

$$\boxed{d_{\text{tail front}}(t) = d_H(t).} \tag{200}$$

The tail-propagation boundary and the standard particle horizon are not merely analogous; they are *mathematically identical*. This identity recasts the standard particle horizon in ISPG language; the horizon problem itself—the observed superhorizon CMB correlations across causally disconnected regions— is not solved by this identity alone, and is addressed below at the homogeneous-background level via universal independent nucleation plus the open primordial- perturbation calculation. The observable universe is exactly the causal domain of resonant-tail propagation.

Modified dark-energy density.—From Eq. (194) and the continuity equation, the dark-energy density evolves as

$$\frac{\rho_{\rm DE}(z)}{\rho_{\rm DE}(0)} = \exp\left[-3\,\delta w_0\int_0^z \frac{(1+z')^2}{E(z')}\,dz'\right]. \tag{201}$$

Since the exponent is negative, $\rho_{\rm DE}$ was *smaller* in the past—the hallmark of phantom behaviour. At low redshift:

$$\frac{\rho_{\rm DE}(z)}{\rho_{\rm DE}(0)} \approx 1 - 3\,\delta w_0\, z + \mathcal{O}\big(z^2\big) \approx 1 - 0.18\,z, \tag{202}$$

a percent-level deviation per unit redshift from ΛCDM—precisely the sensitivity range of DESI (year-3 and beyond) and Euclid.

Hubble law.—At low redshift ($z \ll 1$), the Taylor expansion $v = cz \approx H_0 d$ holds for any dark-energy model. The ISPG expansion history therefore reproduces the Hubble law exactly.

CMB compatibility.—The angular diameter distance to last scattering,

$$d_A = \int_0^{z_*} \frac{c\,dz}{H(z)}, \tag{203}$$

is sensitive to $\Omega_{\rm DE}(z)$ primarily at $z \lesssim 2$, where dark energy dominates. At $z_* \approx 1090$, the dark-energy fraction is negligible ($\Omega_{\rm DE}(z_*) \ll \Omega_m(z_*)$), so the acoustic-peak structure is preserved. The residual modification of d_A at low z shifts the inferred H_0 by $\delta H_0/H_0 \sim \delta w_0 \sim$ few %—within current Planck–SH0ES tension bounds and resolvable by next-generation CMB experiments (CMB-S4, LiteBIRD).

Summary.—The tail-propagation picture reproduces the standard expansion history to leading order (Friedmann equations, Hubble law, particle horizon, CMB peaks), with calculable percent-level deviations encoded in $w(z) \neq -1$. We now address the dynamics of the tail front at the earliest epochs.

G. Tail-front dynamics at the earliest epochs

The preceding subsection identifies the cosmological horizon with the outermost resonant tail front but leaves open the dynamics at $t \lesssim t_{\rm Pl}$. We now show that the ISPG framework provides a self-consistent picture of the onset—one that avoids the initial singularity and addresses the standard cosmological puzzles without invoking an inflaton field.

The quiescent state.—Before any resonance forms, the medium is in its quiescent configuration: $\varphi = \text{const}$ everywhere, with $\nabla\varphi = 0$ and $P_{\rm static} = 0$, $\nabla P = 0$. There are no localized excitations, no pressure deficits, and—since proper time is defined by the oscillation of a resonant mode (Sec. III)—no operational clock. The quiescent state is not a moment "before the Big Bang" but a timeless ground configuration of the infinite medium.

Quantum nucleation of the first excitations.—The Wheeler–DeWitt equation (Sec. II) describes the full quantum state of the scalar-metric system, including configurations where no classical resonance exists. The wave functional $\Psi[\varphi, h_{ij}]$ assigns a finite amplitude to configurations with $\nabla\varphi \neq 0$—that is, to states containing localized excitations. The transition from the quiescent state to a configuration containing one oscillon is a quantum tunneling event, analogous to Schwinger pair production in QED or Coleman–De Luccia bubble nucleation.

The tunneling amplitude can be estimated at the semiclassical dimensional-estimate level. A controlled Euclidean instanton for oscillon nucleation would be a localized, $O(4)$-symmetric field configuration $\varphi_{\rm inst}(\rho)$ (with $\rho = \sqrt{\tau^2 + |\mathbf{x}|^2}$) that interpolates between the quiescent state $\varphi = \text{const}$ at $\rho \to \infty$ and the oscillon core profile at $\rho = 0$. The Euclidean action of this candidate bounce is

$$S_E = \int d^4x_E \, \sqrt{g_E} \left[\frac{R_E}{16\pi G} + \frac{1}{2}(\nabla_E\varphi)^2 \right]_{\rm inst} . \tag{204}$$

Dimensional analysis gives $S_E \sim (R_c/\ell_{\rm Pl})^2$, where $R_c = \hbar/(m_0 c)$ is the oscillon radius and $\ell_{\rm Pl} = \sqrt{\hbar G/c^3}$ the Planck length. For Planck-mass oscillons ($m_0 \sim M_{\rm Pl}$, $R_c \sim \ell_{\rm Pl}$), $S_E \sim \mathcal{O}(1)$ at this estimate level and the nucleation rate per Planck four-volume is unsuppressed: $\Gamma_{\rm nucl} \sim \ell_{\rm Pl}^{-4}\, e^{-S_E} \sim \ell_{\rm Pl}^{-4}$. For lighter oscillons ($m_0 \ll M_{\rm Pl}$, $R_c \gg \ell_{\rm Pl}$), $S_E \gg 1$ and the rate is exponentially suppressed; the first excitations are therefore generically Planck-scale in this dimensional estimate. An explicit Euclidean bounce solution and a controlled treatment of the phantom-sign/conformal-factor structure of the Euclidean path integral remain open technical tasks before this estimate becomes a derived rate.

Since the quiescent medium is spatially infinite, nucleation occurs independently at every point—not as a single event but as a spatially uniform process. The resulting ensemble of Planck-mass oscillons is statistically homogeneous and isotropic by construction: every nucleation event uses the same local onset channel in the same quiescent background, with no preferred direction or location. This *universal nucleation* replaces the single-bubble picture and supplies statistically homogeneous background initial data; the observed large-scale CMB correlations require the separate primordial-perturbation calculation identified below.

Energy density at the onset.—The first oscillons form with Planck-scale energy densities:

$$\rho_{\rm onset} \sim \frac{M_{\rm Pl}\, c^2}{\ell_{\rm Pl}^3} \sim \frac{M_{\rm Pl}^4}{c^2} \sim 10^{113}\ \mathrm{J/m^3}. \tag{205}$$

This is the natural initial condition for the feedback ODE (188): $\varepsilon(0) = 0$ in a region of Planck-scale extent, with the first resonant tails carrying Planck-scale energy. The negative-feedback loop (Sec. VI B) together with the de Sitter dilution of sources (Eq. 188) drives, in the perturbative-branch estimate, the echo-background energy to the self-consistent equilibrium $\rho_{\rm vac} \sim H_0^2 M_{\rm Pl}^2$ at the present epoch (Eq. 191).

Coarse-grained onset closure.—The onset can be closed at the homogeneous-background level by a simple kinetic description. Let $n_{\rm osc}(t)$ denote the number density of newly nucleated Planck-mass oscillons and $\rho_r(t)$ the radiation bath generated when their overlapping tails exchange energy. To leading order,

$$\dot n_{\rm osc} + 3H n_{\rm osc} = \mathcal{J}_{\rm nucl} - \Gamma_{\rm th}\, n_{\rm osc}, \qquad \dot\rho_r + 4H\rho_r = M_{\rm Pl}c^2\, \Gamma_{\rm th}\, n_{\rm osc}, \tag{206}$$

where the nucleation source is

$$\mathcal{J}_{\rm nucl} \sim \Gamma_{\rm nucl} \sim \begin{cases} \ell_{\rm Pl}^{-4} e^{-S_E}, & c = 1, \\ c\, \ell_{\rm Pl}^{-4} e^{-S_E}, & \text{SI}, \end{cases} \tag{207}$$

for the unsuppressed Planck-scale instantons with $S_E = \mathcal{O}(1)$. The SI form has dimensions $1/(\mathrm{m^3\,s})$ for the number-density rate equation. The overlap interaction of neighboring Planckian cores is geometric, so the scattering cross section is

$$\sigma_{\rm sc} = \mathcal{O}\big(\ell_{\rm Pl}^2\big)\,. \tag{208}$$

Once the occupancy reaches its natural onset value $n_{\rm osc} \sim \ell_{\rm Pl}^{-3}$, the thermalization rate is

$$\Gamma_{\rm th} \sim n_{\rm osc}\sigma_{\rm sc}\, c \sim \ell_{\rm Pl}^{-3}\, \ell_{\rm Pl}^2\, c = \frac{c}{\ell_{\rm Pl}} = t_{\rm Pl}^{-1}. \tag{209}$$

Thus both the onset cascade and the conversion of localized oscillon energy into a thermal radiation bath occur within an order-one number of Planck times. At that stage,

$$\rho_r \sim M_{\rm Pl}c^2\, n_{\rm osc} \sim \frac{M_{\rm Pl}c^2}{\ell_{\rm Pl}^3} \sim \rho_{\rm onset}, \tag{210}$$

so the hot Big-Bang background follows directly from the nucleation kinetics rather than from an additional reheating postulate. From this point onward, the standard Friedmann evolution (Eq. 198) takes over, with the radiation density scaling as $(1+z)^4$ and the dark-energy density governed by the feedback mechanism.

Coarse-grained homogeneity and flatness.—Because the nucleation events are statistically identical and occur independently in Planck-size cells, the background smooths by the central-limit law. On a coarse-graining scale $L \gg \ell_{\rm Pl}$, the number of independent onset cells is

$$N_L \sim n_{\rm osc}\, L^3 \sim \left(\frac{L}{\ell_{\rm Pl}}\right)^3 . \tag{211}$$

Hence the fractional density fluctuation of the homogeneous background on that scale is

$$\frac{\delta\rho_L}{\rho_{\rm onset}} \sim N_L^{-1/2} \sim \left(\frac{\ell_{\rm Pl}}{L}\right)^{3/2} . \tag{212}$$

This central-limit suppression makes the homogeneous background extraordinarily smooth on cosmological scales, far below the observed primordial amplitude $\delta\rho/\rho \sim 10^{-5}$ and its measured near-scale-invariant red-tilted spectrum ($n_s \simeq 0.965$). The observed primordial perturbation amplitude and spectrum therefore require a separate generation mechanism not supplied by central-limit smoothing alone; this remains open numerical work. The corresponding curvature contribution obeys the same parametric suppression,

$$\left|\frac{k_{\rm eff}}{a^2H^2}\right|_L = \mathcal{O}\left(\frac{\delta\rho_L}{\rho_{\rm onset}}\right) = \mathcal{O}\left(\left(\frac{\ell_{\rm Pl}}{L}\right)^{3/2}\right), \tag{213}$$

so for every cosmological smoothing scale $L \gg \ell_{\rm Pl}$ the onset background is homogeneous, isotropic, and spatially flat to extremely high accuracy. This is the precise background-level closure used below: the homogeneous FLRW initial data are analytically controlled, while the detailed two-point perturbation spectrum is a separate problem.

Singularity avoidance.—In standard GR, the Penrose–Hawking theorems imply a curvature singularity at the "beginning." In ISPG, these theorems do not apply, for two independent reasons:

1. The phantom kinetic sign produces NEC violation in the effective gravitational source (the canonical scalar stress-energy tensor itself satisfies the standard energy conditions; the negative coupling $\kappa = -1/2$ flips the sign of the source-side NEC contraction; cf. the energy-conditions section of the main theory [1]), invalidating the focusing theorem that underpins the singularity theorems [1]. This is the same mechanism that resolves the Schwarzschild singularity in the classical theory.

2. The Wheeler–DeWitt equation (Sec. II) is *hyperbolic* in ISPG (due to the phantom sign), with the scalar field φ playing the role of an internal time variable. The wave functional propagates smoothly through the configuration $a \to 0$ (zero scale factor) without encountering a singular boundary.

The "Big Bang" in ISPG is therefore modeled as a quantum transition from the quiescent state to the excited (expanding) state, rather than as a classical singular boundary. These two structures evade the assumptions of the standard singularity theorems; a complete geodesic-completeness and regularity proof for the bi-conformal onset geometry remains open.

Standard cosmological puzzles.—The tail-front onset addresses several puzzles traditionally associated with inflation at the homogeneous-background level:

1. *Horizon problem.*—The homogeneous background is supplied by the universality of the nucleation mechanism: every point in the quiescent medium nucleates through the same local onset channel, in the same background, producing statistically identical background initial data. On a smoothing scale L, the residual background inhomogeneity is suppressed by Eq. (212). The tail-front identity $d_{\text{tail front}} = d_H$ (Eq. 200) identifies the causal domain of subsequent tail propagation. The observed superhorizon CMB correlation structure and the primordial two-point spectrum remain part of the separate perturbation-generation calculation.

2. *Flatness problem.*—The quiescent state is exactly flat ($k = 0$): a spatially infinite, homogeneous medium admits no intrinsic curvature. Nucleation is spatially uniform (the same local onset channel everywhere), so it cannot generate a preferred curvature scale. Perturbations of the spatial geometry are localized within each oscillon's Compton radius $R_c \ll d_H$ and average to zero on cosmological scales; more precisely, Eq. (213) gives $|k_{\text{eff}}|/(a^2H^2) = \mathcal{O}\big((\ell_{\text{Pl}}/L)^{3/2}\big) \ll 1$ on every macroscopic scale. The resulting Friedmann evolution therefore inherits $k = 0$ as a structural initial condition of the no-inflation scenario. A dynamical attractor or quantum-selection argument for the exact flatness of the quiescent medium remains open future work.

3. *Monopole problem.*—ISPG contains a single scalar field with an Abelian shift symmetry and no spontaneous breaking of a non-Abelian gauge group. No topological defects arise in the present single-scalar onset sector; if a later high-energy non-Abelian gauge-breaking transition is added in the full Standard-Model embedding, defect formation must be analyzed separately for that sector.

4. *Primordial perturbations.*—The quantum fluctuations of the nucleation epoch provide the seeds for large-scale structure. The spectrum is determined by the nonlinear dynamics of the oscillon–tail interaction at the Planck epoch; its detailed form requires numerical simulation of the coupled scalar-metric system. The analytic closure above concerns the homogeneous background only. Whether the feedback mechanism produces the observed amplitude and red tilt remains open numerical work.

Summary.—The ISPG framework provides a closed homogeneous-background picture of the earliest epoch: the quiescent medium undergoes spatially uniform quantum nucleation of Planck-mass oscillons; the overlapping tails thermalize to produce the hot Big Bang; the NEC-violating Bernoulli pressure deficit (canonical $\tau_{\mu\nu}$ satisfies the energy conditions; the

$\kappa = -1/2$ coupling flips the sign of the source-side contraction, cf. the energy-conditions section of the main theory [1]) and the hyperbolic WDW equation together evade the assumptions of the standard singularity theorems. A complete geodesic-completeness/regularity proof for the bi-conformal onset geometry remains open. The standard cosmological puzzles are addressed at the homogeneous-background level by universal nucleation and the geometric inheritance of $k = 0$; the primordial spectrum, superhorizon correlations, instanton bounce, and geodesic-completeness proof remain open numerical and analytical tasks.

VII. MASS SPECTRUM OF THREE-DIMENSIONAL RESONANCES

The identification of particles as resonant modes of the scalar-metric medium (Sec. III) raises a quantitative question: *why do the charged leptons—electron, muon, and tau—have the specific mass ratios they do?* This section separates three roles: the three-dimensional cavity supplies structural mode labels, the companion MASS/Mathieu ladder supplies the quantitative charged-lepton mass ratios, and the structure-preserving reduction from cavity modes to the Mathieu index N remains an open calculation. The cavity analysis then connects this structure to the empirical Koide formula as a consistency constraint.

A. Three-dimensional resonance eigenvalue problem

Unlike string theory, where particles are one-dimensional objects and the resonance spectrum follows from the harmonics of a string, ISPG treats particles as three-dimensional resonant structures embedded in the scalar-metric medium. The eigenvalue problem is therefore that of a three-dimensional cavity, not a one-dimensional string.

Consider a self-consistent resonant mode (Sec. III E) with a core of effective radius $R_{\rm core}$ and angular frequency ω. The spatial profile satisfies the Helmholtz equation (Eq. 28) which, for the fundamental mode structure, reduces to

$$\nabla^2\psi + k^2\psi = 0, \qquad k^2 = \omega^2\, e^{-2\varphi_0}, \tag{214}$$

inside the resonant core, with outgoing-wave (radiative) boundary conditions at large r.

In spherical coordinates, the angular dependence is given by spherical harmonics $Y_{lm}(\theta, \phi)$ and the radial dependence by spherical Bessel functions:

$$R_{nl}(r) = j_l(k_{nl}\, r), \tag{215}$$

where k_{nl} are determined by the boundary conditions at the effective cavity radius. For a resonant mode confined to radius R, the eigenfrequencies are

$$\omega_{nl} = \frac{\chi_{nl}}{R}\, e^{\varphi_0}, \tag{216}$$

where χ_{nl} is the n-th zero of the spherical Bessel function j_l (or a related transcendental equation for the quasi-bound case).

The first few eigenvalues for $l = 0$ are:

$$\chi_{10} = \pi, \quad \chi_{20} = 2\pi, \quad \chi_{30} = 3\pi. \tag{217}$$

These would give mass ratios 1 : 2 : 3, which clearly do not match the lepton masses ($m_e : m_\mu : m_\tau \approx 1 : 207 : 3477$). The resolution lies in the fact that the three charged leptons are *not* successive s-wave harmonics of the same cavity. They correspond to modes with different angular quantum numbers (n, l) in a three-dimensional resonator whose effective potential is not a simple hard-wall cavity but the self-consistent nonlinear profile of Sec. III E.

B. Self-consistent mass determination

The linearized eigenvalue problem (214)–(216) captures the angular structure of the modes but cannot fix the absolute mass scale, because the confining potential is created by the nonlinear self-interaction and must be determined self-consistently. We now cast the oscillon equation (45) as a dimensionless eigenvalue problem and identify the parameters that control the spectrum.

The nonlinear eigenvalue problem.—The self-consistent oscillon equation (45), derived from $\Box\varphi = 0$ on the bi-conformal background, constitutes a nonlinear eigenvalue problem for the angular frequency ω. In spherical symmetry, define $u(r) \equiv r\,\Phi(r)$ and write it as

$$u'' + \left[\omega^2 e^{-2\varphi_0} - \frac{\ell(\ell+1)}{r^2} - V_{\text{eff}}(r;\,\Phi)\right] u = 0, \tag{218}$$

where the effective potential

$$V_{\text{eff}}(r;\,\Phi) = -\frac{\varphi_0'}{r} - \tfrac{1}{2}\omega^2 e^{-2\varphi_0}\,\Phi + \mathcal{O}\big(\Phi^2\big) \tag{219}$$

collects the nonlinear backreaction terms from Eqs. (45) and (46). The problem is nonlinear because V_{eff} depends on Φ itself, and the static background $\varphi_0(r)$ is generated self-consistently by the time-averaged stress-energy of the oscillation.

The confining potential arises from the nonlinear coupling between φ and the bi-conformal metric: the metric $g_{\mu\nu} = \text{diag}(-e^{\varphi},\, e^{-\varphi}\delta_{ij})$ depends on φ, so the oscillating mode modifies its own propagation environment. Newton's constant G does not appear in the wave equation $\Box\varphi = 0$; it enters only through the gravitational self-energy of the oscillon (a negligible correction discussed below). The resonant cavity is therefore a *geometric* property of the nonlinear wave equation, not a gravitational effect.

The boundary conditions for bound states are:

$$u(0) = 0, \quad u'(0) = 1; \qquad u \to 0 \text{ as } r \to \infty. \tag{220}$$

The eigenvalue is ω; the oscillon amplitude $\Phi_0 \equiv \Phi(0)$ is the primary parameter that controls the depth of the confining potential.

Dimensionless formulation.—Introduce the Compton radius $R_c \equiv \hbar/(\tilde{m}\, c)$ as the length scale, where $\tilde{m}$ is a reference mass (to be identified with the ground-state mass a posteriori), and define

$$x \equiv \frac{r}{R_c}, \qquad \Omega \equiv \frac{\omega R_c}{c}, \qquad f(x) \equiv \frac{u(r)}{R_c}. \tag{221}$$

The mode equation becomes

$$f'' + \left[\Omega^2 e^{-2\varphi_0} - \frac{\ell(\ell+1)}{x^2} - \hat{V}_{\text{eff}}\right] f = 0, \tag{222}$$

with $\hat{V}_{\text{eff}}$ the dimensionless form of (219). Within the formal cavity ansatz, the mass of the n-th resonance is taken as

$$m_n = \frac{\hbar\,\Omega_n}{R_c\, c} = \tilde{m}\,\Omega_n. \tag{223}$$

Amplitude–frequency relation.—The one-dimensional reduction of Sec. III E provides the explicit oscillon solution $\Phi = A\,\text{sech}^2(Bx)$ with

$$A = \tfrac{3}{2}(1-\Omega^2), \qquad B = \tfrac{1}{2}\sqrt{1-\Omega^2}, \tag{224}$$

valid for $\Omega^2 < 1$ (bound-state condition). This links the oscillon amplitude to the eigenfrequency: a deeper potential well (larger A, smaller Ω) supports more tightly bound modes. The three-dimensional generalization modifies the numerical coefficients through the angular barrier $\ell(\ell+1)/x^2$ and the background gradient φ_0', but the qualitative structure persists: a one-parameter family of oscillons parametrized by Φ_0 (or equivalently Ω).

Formal cavity mass-ratio ansatz.—Within the self-consistent confining potential, different angular channels (ℓ) and radial overtones (n) yield distinct eigenfrequencies $\Omega_{n\ell}$. Under the same formal cavity ansatz, the mass ratios are

$$\frac{m_{n_1\ell_1}}{m_{n_2\ell_2}} = \frac{\Omega_{n_1\ell_1}}{\Omega_{n_2\ell_2}} \tag{225}$$

and would be geometric: they depend on the mode numbers and the cavity profile but are independent of G, $\hbar$, or the absolute mass scale. This relation is retained as a diagnostic of the direct-cavity programme; its physical status for charged leptons is fixed by the numerical audit below.

Epistemic status of Eq. (225).—Direct numerical evaluation of the self-consistent ISPG oscillon with the canonical nonlinearity $\Phi\, e^{-\alpha\Phi}$ ($\Phi_0 = 2.35$, $\alpha = 0.5$) yields exactly four bound cavity modes $(n,\ell) \in \{(0,0),(1,0),(0,1),(0,2)\}$ with dimensionless eigenfrequencies $\Omega_{n\ell} \in (0.66,\, 1.00)$; the resulting linear ratios $\Omega_{n\ell}/\Omega_{n'\ell'}$ do *not* reproduce the observed charged-lepton hierarchy $m_\tau : m_\mu : m_e \approx 3477 : 207 : 1$, nor does any monotone power of Ω alone match this hierarchy to better than $\sim 20\%$ (verified in the companion code `QUANTUM/Python/cavity_vs_mass_audit.py` and `QUANTUM/Python/cavity_quantum_oscillon.py`). The identification of the three charged leptons with the specific (n,ℓ) modes listed above is therefore a *structural* labelling of particle modes inherited from the cavity spectrum; the *quantitative* mass ratios are provided by the Mathieu mass formula $m_N = m_1\, b_N(q{=}1.853)$

of the companion MASS paper [3], with $(N_e, N_\mu, N_\tau) = (5, 72, 295)$ reproducing the observed ratios to better than 0.2%. A formal, structure-preserving reduction from the three-dimensional cavity spectrum to the one-dimensional Mathieu index N is not established here and is deferred as future work. The Koide relation (Sec. VII C) below functions as an independent algebraic consistency check on any candidate identification.

Absolute mass scale.—The physical mass $m_n = \hbar\omega_n/c^2$ is fixed once the effective cavity radius $R_{\rm core}$ is known:

$$R_{\rm core} \sim \frac{\hbar}{m_n c} = \frac{1}{\Omega_n}\frac{\hbar}{\tilde{m}\, c}, \tag{226}$$

the Compton wavelength of the resonance, as expected on dimensional grounds. The cavity radius is set by the nonlinear self-consistency condition: the oscillon amplitude Φ_0 must be large enough to create a confining potential through the Φ^2 and Φ^3 terms in (45), but small enough to remain in the quasi-bound regime ($\Omega < 1$). For the sech^2 oscillon of the 1D model, the core radius scales as $R_{\rm core} \sim 1/B = 2/\sqrt{1-\Omega^2}$ in Compton units, confirming $R_{\rm core} \sim \hbar/(mc)$ to within order-unity factors.

Gravitational self-energy correction.—Beyond the wave-equation dynamics, the oscillon's mass m generates a gravitational field $\varphi_{\rm grav} \sim -Gm/(c^2 r)$ that modifies the background at the core by a relative amount

$$\left.\frac{|\varphi_{\rm grav}|}{1}\right|_{r\sim R_{\rm core}} \sim \frac{Gm}{c^2 R_{\rm core}} = \frac{1}{2}\frac{m^2}{M_{\rm Pl}^2} \equiv \frac{\kappa}{2}. \tag{227}$$

For the electron, $\kappa = (m_e/M_{\rm Pl})^2 \approx 1.7 \times 10^{-45}$. The resulting eigenfrequency correction is

$$\frac{\delta m_n}{m_n} \sim \kappa \sim \left(\frac{m_n}{M_{\rm Pl}}\right)^2, \tag{228}$$

a relative shift of order 10^{-45} for the electron—utterly negligible. Within the formal cavity ansatz of Eqs. (223)–(225), the ratios are geometric and the gravitational self-energy correction is therefore negligible. The physical charged-lepton ratios are supplied by the Mathieu mass formula of the companion MASS paper [3], as recorded in the epistemic-status remark below Eq. (225).

Lattice formulation.—The nonlinear eigenvalue problem (222) admits a natural lattice discretization. Replace the continuum by a cubic lattice with spacing a and sites $\mathbf{n} \in \mathbb{Z}^3$. The spatial Laplacian becomes the discrete operator

$$(\nabla_a^2\, \varphi)_{\mathbf{n}} = \frac{1}{a^2}\sum_{|\mathbf{e}|=1}(\varphi_{\mathbf{n}+\mathbf{e}} - \varphi_{\mathbf{n}}), \tag{229}$$

and the nonlinear eigenvalue problem reduces to finding localized solutions of a finite-dimensional system of coupled nonlinear equations. For $N \sim 10^2$ sites per side (sufficient to resolve the Compton wavelength with $a \sim R_c/N$), the system is tractable with standard iterative eigenvalue algorithms.

Separately, the ultralocal kinetic structure of the canonical theory (Sec. II G) ensures that the *quantum* lattice formulation—discretizing the Hamiltonian constraint (13) rather than the classical mode equation—inherits the same nearest-neighbour coupling structure, enabling Monte Carlo evaluation of the constrained path integral as an independent non-perturbative check on the mass spectrum.

C. The Koide formula as a resonance constraint

In 1981, Koide discovered an empirical relation among the three charged lepton masses [28–30]:

$$\boxed{Q \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3},} \tag{230}$$

which is satisfied to better than 0.01% with current mass measurements. This precision is striking: a random triple of masses would satisfy $Q = 2/3$ with probability $\ll 10^{-4}$. No explanation for this relation exists within the Standard Model.

In the ISPG resonance framework, the Koide relation is an *algebraic consistency condition* that any candidate 3D-resonator eigenvalue spectrum must satisfy if it is to match the observed charged-lepton hierarchy. To see this, consider the general eigenvalue problem

$$\hat{H}\,\psi_n = \lambda_n\,\psi_n, \tag{231}$$

where $\hat{H}$ is the effective Hamiltonian of the resonant mode in the self-consistent potential (Sec. III E). The eigenvalue gives the rest energy directly, $\lambda_n = m_n c^2$.

The Koide ratio $Q = 2/3$ can be rewritten as a constraint on the eigenvalues:

$$\frac{\sum_{n=1}^{3} \lambda_n}{\left(\sum_{n=1}^{3} \lambda_n^{1/2}\right)^2} = \frac{2}{3}. \tag{232}$$

This is equivalent to requiring that the three eigenvalues lie on a specific curve in $(\lambda_1, \lambda_2, \lambda_3)$ space. We now prove that this curve is the locus of eigenvalues of a **symmetric rank-1 perturbation** of a degenerate system.

Proof.—Consider the unperturbed Hamiltonian $\hat{H}_0$ with threefold degenerate eigenvalue λ_0:

$$\hat{H}_0\psi_n^{(0)} = \lambda_0\psi_n^{(0)}, \qquad n = 1, 2, 3. \tag{233}$$

Add a symmetric rank-1 perturbation $\delta\hat{H} = \alpha|v\rangle\langle v|$, where $|v\rangle$ is a normalized vector in the degenerate subspace. In the basis where $|v\rangle = (1, 1, 1)/\sqrt{3}$, the perturbation matrix is

$$\delta H = \frac{\alpha}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}. \tag{234}$$

The perturbed eigenvalues are found by diagonalizing $\lambda_0 \mathbb{1} + \delta H$. The eigenvalues of δH are: α (non-degenerate, eigenvector $(1, 1, 1)$) and 0 (twofold degenerate, eigenvectors orthogonal to $(1, 1, 1)$). Thus the perturbed eigenvalues are:

$$\lambda_1 = \lambda_0 + \alpha, \qquad \lambda_2 = \lambda_3 = \lambda_0. \tag{235}$$

The Koide ratio for these eigenvalues is:

$$Q = \frac{(\lambda_0 + \alpha) + \lambda_0 + \lambda_0}{\left(\sqrt{\lambda_0 + \alpha} + \sqrt{\lambda_0} + \sqrt{\lambda_0}\right)^2}. \tag{236}$$

As shown in Appendix A 4, defining $x \equiv \alpha/\lambda_0$, the Koide ratio is the monotonically increasing function

$$Q(x) = \frac{3 + x}{\left(\sqrt{1 + x} + 2\right)^2}, \tag{237}$$

which interpolates from $Q(0) = 1/3$ (exact degeneracy) to $Q \to 1$ ($x \to \infty$). Solving $Q(x) = 2/3$ yields $x = 33 + 24\sqrt{2} \approx 66.9$; the physical value $Q = 2/3$ is therefore attained at a *specific*, non-perturbative perturbation strength determined by the nonlinear dynamics. The corrections to $Q = 2/3$ are controlled by the secondary degeneracy-lifting parameter ϵ (the splitting of λ_2 and λ_3), not by α itself.

This establishes the physical picture:

1. In the unperturbed system (free medium, no self-interaction), all three modes are degenerate at λ_0.

2. The nonlinear self-interaction of the resonant mode (metric backreaction) acts as a symmetric rank-1 perturbation $\delta\hat{H} \propto |\varphi_0\rangle\langle\varphi_0|$ that lifts the degeneracy.

3. The specific form of the bi-conformal coupling, through which the scalar field φ couples universally to all masses via $m_{\text{eff}} = m_0 e^{\varphi/2}$, ensures the perturbation is approximately rank-1 in the (e, μ, τ) subspace. Since neither α nor λ_0 is a free parameter—both are fixed by the self-consistent oscillon solution—the theory's self-consistent oscillon problem is required to produce the value $x \equiv \alpha/\lambda_0$ that satisfies $Q = 2/3$. Mathematically this happens if and only if $x = 33 + 24\sqrt{2} \approx 66.9$; deriving this value of x from the self-consistent oscillon problem is the residual numerical task.

Computing x from first principles requires solving the full nonlinear eigenvalue system of Sec. VII B. As stated there, the cavity geometry supplies the structural mode labels and the nonlinear eigenvalue problem; the quantitative charged-lepton mass ratios are currently carried by the MASS/Mathieu ladder of the companion MASS paper [3], with the formal 3D-to-1D reduction remaining the open numerical task. Gravitational self-energy corrections of order $(m/M_{\text{Pl}})^2 \sim 10^{-45}$ are negligible. Whether the self-consistent cavity spectrum by itself yields $x \approx 66.9$ is a concrete, falsifiable prediction of ISPG. At present the numerical cavity eigenfrequencies $\Omega_{n\ell}$ of Sec. VII B do not by themselves reproduce $Q = 2/3$: direct substitution $\lambda_n \mapsto \Omega_{n\ell}$ gives $Q \simeq 0.34$, off by $\sim 50\%$. What *does* hold numerically is the Koide identity evaluated on the companion MASS paper's Mathieu-ladder values [3], $(N_e, N_\mu, N_\tau) = (5, 72, 295)$:

$$\frac{N_e^2 + N_\mu^2 + N_\tau^2}{(N_e + N_\mu + N_\tau)^2} = \frac{92234}{138384} = 0.66651\ldots, \tag{238}$$

matching 2/3 to 0.024%, in direct correspondence with the experimental Koide ratio 2/3 (to 0.01%). The rank-1 perturbation argument therefore plays a *motivational* role: it shows that the Mathieu spectrum of the MASS paper is consistent with a rank-1 lifting structure and isolates $x = 33 + 24\sqrt{2} \approx 66.9$ as the value at which the cavity and Mathieu frameworks would coincide. A first-principles derivation of this value directly from the cavity eigenvalues of Sec. VII B is an open calculation (numerical audit: `QUANTUM/Python/cavity_vs_mass_audit.py`). The Koide constraint also serves as a powerful algebraic consistency check: any candidate resonance spectrum that does not satisfy $Q = 2/3$ can be excluded, even when the spectrum is generated from a different effective formulation.

Beyond rank-1.—The pure rank-1 perturbation gives $\lambda_2 = \lambda_3$; the physical mass splitting $m_\tau/m_\mu \approx 17$ requires a secondary perturbation ϵ that lifts this degeneracy. As shown in Appendix A 4 (Eq. A12), the Koide ratio receives corrections of order $\epsilon^2/(\lambda_0\alpha)$, not α/λ_0. For the charged leptons, $\epsilon/\alpha \sim m_\mu/m_\tau \approx 0.06$, giving corrections $\sim 4 \times 10^{-3}$—consistent with $Q = 2/3$ holding to 0.01%. The precise values of m_e, m_μ, m_τ are determined by the full nonlinear eigenvalue system (Sec. VII B), whose lattice formulation reduces the problem to a finite-dimensional nonlinear system tractable with iterative algorithms.

Numerical solution of the cavity spectrum.—We have performed this computation for the exponential nonlinearity $F(\Phi) = \Phi\big(1 - e^{-\alpha\Phi}\big)$ with $\alpha = 1/2$. We clarify the epistemic status of this choice. The Taylor expansion $F(\Phi) = \alpha\Phi^2 - \frac{1}{2}\alpha^2\Phi^3 + \mathcal{O}(\Phi^4)$ has a leading quadratic coefficient α that matches the $\frac{1}{2}\omega^2 e^{-2\varphi_0}\Phi^2$ term of Eq. (45) in natural units ($\omega^2 e^{-2\varphi_0} = 1$) precisely when $\alpha = 1/2$; in this sense the leading-order coefficient of the oscillon nonlinearity is *fixed by the ISPG action*, not a free parameter. The specific *exponential resummation* of higher-order terms Φ^3, $\Phi^4, \ldots$, however, is a model choice of ultraviolet completion within the Gleiser-type nonlinear-Klein–Gordon class identified in Sec. III E: many UV completions share the same quadratic leading term but differ at cubic and higher order, and ISPG's quadratic-only action does not by itself single out the exponential form. The cavity spectrum reported below is therefore specific to this nonlinearity class; whether the Koide relation $Q = 2/3$ persists under alternative UV completions compatible with the same leading quadratic term $\frac{1}{2}\Phi^2$ is a robustness question left for future work. The background oscillon at amplitude $\Phi_0 = 2.35$ (frequency $\Omega_{\rm bg} = 0.866$) creates a confining potential that supports exactly four bound states:

Particle	Mode	ω	κ	E_{1D}
τ	(n=0, ℓ=0)	0.664	0.748	1.156
μ	(n=1, ℓ=0)	0.970	0.241	0.067
e	(n=0, ℓ=2)	0.999	0.044	4.2×10^{-4}
grav.	(n=0, ℓ=1)	0.866	0.500	—

The $\ell = 1$ mode controls the source-following gravitational response.—We now prove that the $\ell = 1$ cavity mode is not an independent particle but the oscillon's translational (Goldstone) mode. Its gravitational role is to ensure that the long-range field follows the motion of the localized oscillon core.

(i) Exact identity $\psi_1(r) = \Phi_0'(r)$.—The background oscillon $\Phi_0(r)$ satisfies the radial field equation. An infinitesimal translation by δ along the z-axis produces $\Phi_0(|\mathbf{r} - \delta\hat{z}|) = \Phi_0(r) - \delta \cos\theta\, \Phi_0'(r) + \mathcal{O}(\delta^2)$. Since the translated oscillon also satisfies the field equation (by translational invariance of the Lagrangian), the perturbation $\delta\varphi = \cos\theta\, \Phi_0'(r)\cos(\Omega t)$ is an *exact* eigenmode at frequency $\omega = \Omega_{\rm bg}$. Because $\cos\theta = Y_{10}/\sqrt{4\pi/3}$, this is the $\ell = 1$ mode with radial eigenfunction $u_1(r) = r\,\Phi_0'(r)$. We verify this numerically: the Pearson correlation between the computed eigenvector and $r\,\Phi_0'(r)$ is 0.9999999999, with a maximum point-wise residual of 1.3×10^{-5} and $|\omega_{\ell=1} - \Omega_{\rm bg}| = 1.7 \times 10^{-6}$. The identity is exact.

(ii) Zero excitation energy (Goldstone theorem).—The oscillon is localized and thus breaks the continuous translational symmetry of the field equation. By Goldstone's theorem, this broken symmetry produces a zero-energy mode. Indeed, translating the oscillon does not change its total energy: $E[\Phi_0(r)] = E[\Phi_0(|\mathbf{r} - \delta\hat{z}|)]$ for all δ. Therefore the $\ell = 1$ perturbation carries *zero* excitation energy—it is not a particle. The formal energy $E_{1D}(\omega_{\ell=1})$ represents the oscillon's translational kinetic energy $\frac{1}{2}Mv^2$, not a new mass eigenstate. *(iii) Collective-coordinate gravitational response.*—Let $\mathbf{R}(t)$ denote the oscillon center in the adiabatic regime, where the translational collective coordinate varies on timescales much longer than $\Omega_{\rm bg}^{-1}$. The translated family of configurations is then, to leading order,

$$\varphi(\mathbf{r}, t; \mathbf{R}) \simeq \varphi_{\rm grav}(|\mathbf{r} - \mathbf{R}(t)|) + \Phi_0(|\mathbf{r} - \mathbf{R}(t)|)\cos(\Omega_{\rm bg} t). \tag{239}$$

Expanding to first order in $\mathbf{R}(t)$ gives

$$\delta\varphi_{\mathbf{R}} = -\mathbf{R}(t)\cdot\nabla\Big[\varphi_{\rm grav}(r) + \Phi_0(r)\cos(\Omega_{\rm bg}t)\Big] = -\big(\mathbf{R}(t)\cdot\hat{r}\big)\Big[\varphi_{\rm grav}'(r) + \Phi_0'(r)\cos(\Omega_{\rm bg}t)\Big]. \tag{240}$$

Because $\mathbf{R}\cdot\hat{r}$ is purely $\ell = 1$, the oscillatory part of this tangent vector is exactly the Goldstone profile found above, $\psi_1(r) \propto \Phi_0'(r)$. Time-averaging over the fast internal oscillation leaves

$$\langle \delta\varphi_{\mathbf{R}}\rangle_t = -\mathbf{R}(t)\cdot\nabla\varphi_{\rm grav}(r) = -\frac{r_s\,\mathbf{R}(t)\cdot\hat{r}}{r^2}, \tag{241}$$

which is precisely the dipole term in the translated gravitational field

$$-\frac{r_s}{|\mathbf{r} - \mathbf{R}(t)|} = -\frac{r_s}{r} - \frac{r_s\,\mathbf{R}(t)\cdot\hat{r}}{r^2} + \mathcal{O}\big(R^2\big)\,. \tag{242}$$

Thus the same collective coordinate that translates the localized oscillon core also translates its long-range gravitational dressing. We verify numerically that the dipole component of the displaced oscillon field equals $\delta\,\Phi_0'(r)$ to 0.04% accuracy. The $\ell = 1$ mode is therefore the channel through which the particle's gravitational field tracks its spatial position. It is the source-following gravitational mode of the oscillon, not an additional mass eigenstate.

Thus the cavity contains exactly **three physical particle modes**—matching the three charged lepton generations—plus one translational mode that carries the source-following gravitational response of the oscillon.

Uniqueness of the mode assignment.—The four cavity modes admit four possible triples. Only the (τ, μ, e) assignment gives Q near 2/3:

Triple	Q	$\lvert Q-2/3\rvert$
(τ, ℓ=1, μ)	0.414	0.25
(τ, ℓ=1, e)	0.510	0.16
(τ, μ, e)	0.6667	6.8×10^{-5}
(ℓ=1, μ, e)	0.583	0.084

The assignment is not chosen by hand: it is the unique combination consistent with $Q \approx 2/3$, and the excluded $\ell = 1$ mode is independently proven to be the translational Goldstone mode (see above).

The mass formula E_{1D}.—The one-dimensional oscillon equation $\Phi'' + (\Omega^2 - 1)\Phi + \Phi^2 = 0$ admits the exact solution $\Phi(x) = \frac{3}{2}\kappa^2 \operatorname{sech}^2(\kappa x/2)$ with $\kappa^2 = 1 - \Omega^2$. Its kinetic energy is

$$E_{\text{kin}} = \int_{-\infty}^{\infty} \left[\tfrac{1}{2}\Omega^2\Phi^2 + \tfrac{1}{2}(\Phi')^2\right] dx = \tfrac{3}{5}\,\kappa^3\,(4\Omega^2 + 1), \tag{243}$$

where the integrals $\int \Phi^2\,dx = 6\kappa^3$ and $\int (\Phi')^2\,dx = \frac{6}{5}\kappa^5$ follow from the standard identities $\int_{-\infty}^{\infty} \operatorname{sech}^4 u\,du = \frac{4}{3}$ and $\int_{-\infty}^{\infty} \operatorname{sech}^4 u\,\tanh^2 u\,du = \frac{4}{15}$. The factor $(4\Omega^2 + 1)$ is not arbitrary: it encodes the exact ratio of temporal oscillation energy $(3\Omega^2\kappa^3)$ to spatial gradient energy $(\frac{3}{5}\kappa^5)$. We define

$$E_{1D}(\Omega) \equiv \kappa^3\,(4\Omega^2 + 1) = \tfrac{5}{3}\,E_{\text{kin}}, \tag{244}$$

where the constant 5/3 cancels in the Koide ratio. The Koide ratio evaluates to

$$Q_{\text{E1D-on-cavity}} = 0.66674, \qquad |Q - \tfrac{2}{3}| = 6.8\times10^{-5}, \tag{245}$$

reproducing the Koide relation to 0.009% accuracy. Here E_{1D} denotes the effective one-dimensional mass prescription evaluated on the cavity data; the test `SM/python/test_e3d_vs_e1d_backgrou` shows that $\langle H\rangle/E_{1D}$ has coefficient of variation 0.82 over the oscillon family. Thus E_{1D} is not a constant multiple of the full 3D oscillon energy, and the full 3D-energy reduction remains open.

This formula is singled out by the data: among fourteen candidate mass prescriptions—including κ^p for $p = 0.5, 1, \ldots, 3$, $\kappa^p(4\Omega^2 + 1)$ for various p, and the full 3D energy $E_{3D} = 4\pi\int[\frac{1}{2}\Omega^2\Phi^2 + \frac{1}{2}(\Phi')^2]\,r^2\,dr$—only E_{1D} reproduces $Q = 2/3$ to better than 1%; the next-best formula deviates by 2.2×10^{-2}, two orders of magnitude worse. In a Monte Carlo test of 10^5 random eigenvalue triples, only 0.001% produce $|Q - 2/3| < 10^{-3}$ under E_{1D}, confirming that the cavity spectrum is non-generic.

The raw mass ratios $1 : 159 : 2740$ are $\sim$79% of the experimental values $1 : 207 : 3477$. A systematic scan of Φ_0 reveals a remarkable structural coincidence: the amplitude at which Q is closest to 2/3 ($\Phi_0 = 2.350$) is the *same* amplitude at which the mass-ratio correction factor equals $4/\pi$ ($\Phi_0 = 2.3501$, with $Q = 0.66670$). This is not an independent geometric adjustment; it is a consequence of the Koide condition itself.

In the standard Koide parametrisation $\sqrt{m_i} = M\big(1 + \sqrt{2}\cos(\theta_0 + 2\pi i/3)\big)$, the experimental angle is $\theta_0^{\text{exp}} = 0.2223 \approx 2/9$ (to 0.02%), while the E_{1D} cavity spectrum gives $\theta_0^{E_{1D}} = 0.2167$. The small shift $\Delta\theta_0 = 0.006$ rad, amplified by the extreme sensitivity of m_e

to θ_0 near the cosine minimum, produces the correction factor $4/\pi$ for all ratios involving the electron:

$$\left.\frac{m_\tau}{m_e}\right|_{\rm corr} = 3490 \quad (\text{expt: } 3477,\ 0.37\%), \qquad \left.\frac{m_\mu}{m_e}\right|_{\rm corr} = 202 \quad (\text{expt: } 207,\ 2.1\%). \tag{246}$$

Equivalently, for the $\ell = 2$ electron mode the energy exponent shifts from κ^3 to $\kappa^{3.076}$, and $\kappa_e^{0.076} = 0.789 \approx \pi/4$ (0.39%). The 2.1% residual in m_μ/m_e is the τ/μ error (both $\ell = 0$), reflecting a sub-leading curvature correction $\mathcal{O}(\kappa)$ from the 1D-to-3D mapping. A formal derivation of $\theta_0 = 2/9$ from the cavity spectrum—equivalent to the exact $4/\pi$ correction—remains an open problem. The first step toward its resolution is described next.

Hartree self-consistency and $\theta_0 = 2/9$.—We stress at the outset that the target $\theta_0 = 2/9$ is an *empirical benchmark* inherited from the data (the experimental value $\theta_0^{\rm exp} = 0.2223$ lies 0.02% from 2/9, see above); the Hartree analysis that follows does *not* derive 2/9 from first principles, but asks whether the bare cavity angle $\theta_0^{E_{1D}} = 0.2167$ can be shifted to the empirical value by a perturbative mechanism intrinsic to the ISPG oscillon, without introducing new sectors or couplings. The cavity modes backreact on the effective potential through the mean-field (Hartree) correction

$$\delta V(r) = -\tfrac{1}{2}\, F''[\Phi_{\rm bg}(r)]\, |\psi_\tau(r)|^2\, \eta, \tag{247}$$

where ψ_τ is the ground-state eigenfunction (normalized to unit L^2 norm), $F'' = d^2F_{\rm NL}/d\Phi^2$, and η measures the mode occupation strength. Because the electron eigenfrequency $\omega_e \approx 0.999$ is extremely close to the continuum threshold, even a small perturbation δV produces a large relative shift in m_e.

Numerical evaluation shows that at $\eta^* = 0.074$ the Koide angle shifts from $\theta_0 = 0.2167$ (bare) to $\theta_0 = 0.2222 \approx 2/9$ with 0.07% accuracy. The Koide ratio simultaneously remains close to 2/3: a joint optimization over (Φ_0, η) yields $Q = 0.6659$ (0.11% from 2/3) and $\theta_0 = 0.2221$ (0.07% from 2/9), reducing the combined cost function by a factor of 300 relative to the bare ($\eta = 0$) spectrum. The small coupling $\eta^* \approx 0.07$ confirms that the backreaction is perturbative: the first-order Hartree correction captures the dominant effect. Whether the residual 0.11% in Q closes at second order is an open expectation; the channel-separated self-consistent Hartree–Fock run described below improves θ_0 but does not improve Q, so the joint (Q, θ_0) closure beyond first order remains a numerical task.

The Hartree mechanism thus provides a candidate physical interpretation of the $4/\pi$ correction as a first-order self-energy shift of the electron eigenfrequency due to the ground-state backreaction on the confining potential. Whether this is the unique mechanism—as opposed to, for example, a cavity-geometry correction at higher order—is not established here.

Quantum determination of η.—In the QFT framework, the coupling η is constrained (rather than free) by the vacuum zero-point fluctuations of the cavity modes, under the selected mode-content and overlap-weighting prescription. For a mode with angular momentum ℓ and frequency ω_n, the contribution to the spherically symmetric (monopole)

Hartree potential is

$$\eta_n = \frac{2\ell+1}{8\pi\,\omega_n}\,, \tag{248}$$

where the factor $(2\ell+1)$ accounts for the magnetic degeneracy. For the tau mode ($\ell = 0$, $\omega_\tau = 0.664$), this gives $\eta_\tau = 0.060$. This single contribution shifts θ_0 from 0.217 to 0.221—capturing 81% of the required correction. Including the muon's zero-point contribution, weighted by its spatial overlap with the electron eigenfunction (overlap ratio 0.27), yields

$$\eta_{\text{pred}} = \eta_\tau + \frac{\langle\psi_\mu^2|\psi_e^2\rangle}{\langle\psi_\tau^2|\psi_e^2\rangle}\,\eta_\mu = 0.060 + 0.27\times 0.041 = 0.071\,, \tag{249}$$

in 4% agreement with the empirical $\eta^* = 0.074$. The predicted Koide angle is $\theta_0 = 0.2220 \approx 2/9$ (0.09%). The iterative Hartree–Fock procedure converges in a single iteration, confirming that the backreaction is genuinely perturbative. The residual 4% discrepancy between η_{pred} and η^* is attributed to second-order corrections and the angular coupling of the $\ell = 2$ electron mode, which enters the monopole Hartree potential only through its angle-averaged density. We stress that the electron mode does not contribute to its own Hartree potential (standard self-interaction exclusion in mean-field theory) and enters the $\ell = 0$ channel potential only through the angle-averaged density of its $\ell = 2$ wavefunction; had the electron's $(2\ell+1) = 5$ degeneracy entered (248) naively it would give $\eta_e \approx 0.199$ and overshoot the empirical value. A rigorous derivation of the cross-channel suppression factor from angular-momentum coupling is an open task and is deferred to future work.

Channel-separated Hartree–Fock.—The cavity modes reside in different angular-momentum channels: τ and μ in $\ell = 0$, the electron in $\ell = 2$. A self-consistent Hartree–Fock treatment must respect this structure. In the full channel-separated scheme, each channel receives backreaction only from modes in other channels (excluding self-interaction):

$$\delta V_{\ell=0}(r) = -\tfrac{1}{2}\,F''[\Phi_{\text{bg}}]\,|\psi_e|^2\,\eta_e\,, \tag{250}$$

$$\delta V_{\ell=2}(r) = -\tfrac{1}{2}\,F''[\Phi_{\text{bg}}]\,\big(|\psi_\tau|^2\,\eta_\tau + |\psi_\mu|^2\,\eta_\mu\big)\,. \tag{251}$$

However, the large degeneracy factor $(2\ell+1) = 5$ of the electron's $\ell = 2$ mode causes an excessive shift in the $\ell = 0$ channel, degrading Q.

A physically motivated alternative is the *semiclassical* prescription: the heavy modes τ and μ ($\kappa \gg 0$, deeply bound) are treated as classical, receiving no quantum correction, while only the light, nearly unbound electron receives the backreaction from $\tau + \mu$:

$$\delta V^{(\text{sc})}_{\ell=2}(r) = -\tfrac{1}{2}\,F''[\Phi_{\text{bg}}]\,\big(|\psi_\tau|^2\,\eta_\tau + |\psi_\mu|^2\,\eta_\mu\big), \qquad \delta V^{(\text{sc})}_{\ell=0} = 0\,. \tag{252}$$

This is the standard semiclassical hierarchy: heavy (classical) modes source the potential for light (quantum) modes, not vice versa.

A scan over the oscillon amplitude Φ_0 with four prescriptions—bare, semiclassical (252), full channel-separated (250), and same-channel (old)—shows that the semiclassical scheme minimizes the combined cost $(\Delta Q/Q_{\text{target}})^2 + (\Delta\theta_0/\theta_{\text{target}})^2$ by a factor of 132 relative to the bare spectrum, achieving $Q = 0.6662$ (0.06%) and $\theta_0 = 0.2215$ (0.3%) simultaneously. The full channel-separated scheme improves θ_0 but overshoots Q; the semiclassical prescription provides the optimal balance, consistent with the expectation that deeply bound modes are well approximated classically. We note that this selection of the semiclassical scheme is physically motivated (classical treatment of deeply bound heavy modes is a standard hierarchy argument) but remains a discrete choice among the four candidate prescriptions. A first-principles derivation that uniquely selects the semiclassical hierarchy from the underlying QFT—for example, through an explicit $1/\omega_n$ expansion of the effective action—would eliminate this residual ambiguity and is an open task.

Structural origin of the amplitude Φ_0.—The electron mode ($\ell = 2$) first becomes bound at the threshold amplitude $\Phi_0 = 2.29$. Below this threshold, the cavity supports only three modes (n=0 and n=1 for $\ell = 0$, plus the translational $\ell = 1$), and the best Koide ratio is $Q = 0.42$—far from 2/3. Above the threshold, Q is closest to 2/3 at $\Phi_0 = 2.35$, just 2.6% above the threshold. The Koide relation thus follows from the oscillon amplitude sitting close to the electron binding threshold; the small offset $\Phi_0 - \Phi_{\text{thr}} = 2.6\%$ is a continuous parameter not fixed by any symmetry argument in the present framework, and the Hartree coupling η is determined partly by QFT ($\eta_{\text{pred}} = 0.071$, with 81% from τ and a small muon contribution, see (249)) and partly by empirical optimization ($\eta^* = 0.074$, 4% residual). The mechanism is therefore best characterized as a physically motivated self-consistency construction rather than a strict parameter fit, but it does involve tunable choices that must be acknowledged. The ratio is extremely sensitive to the electron eigenfrequency: a 1% shift in ω_e degrades Q from 6.8×10^{-5} to 8.5×10^{-2}, which supports the interpretation of the threshold proximity as the structural origin of the Koide relation, but does not by itself determine why Nature selects exactly 2.6% above threshold.

Parameter count and epistemic status.—For transparency we summarize the effective parametric content of the joint (Q, θ_0) reproduction: two continuous parameters (Φ_0, η) plus one discrete prescription choice (semiclassical among four candidates), against two observables. The 0.07–0.11% residuals reported above reflect the numerical optimization tolerance on a finite (Φ_0, η) grid rather than an independent theoretical prediction. A zero-parameter determination of Φ_0 from first principles, together with a full QFT derivation of η that includes the electron cross-channel coupling and uniquely selects the semiclassical prescription, remains an open programme of the theory.

D. Particle stability and the evolving spectrum

The resonance eigenvalues ω_{nl} depend on the background pressure through Eq. (216). As the background pressure decreases over cosmic time (Sec. VI), the resonance spectrum evolves:

Higher modes become unstable.—A mode with eigenfrequency ω_{nl} is stable if it lies below the threshold set by the background pressure. As the pressure drops, the threshold descends, and modes that were formerly stable become quasi-bound: their lifetime decreases from ∞ to a finite value, and eventually they decay into lower modes.

This provides a natural explanation for the evolving particle content of the universe:

- **Early universe** (high pressure): the quark-gluon plasma phase, with free quarks and gluons, was stable. The experimental recreation of that phase at RHIC and LHC [21] is compatible with the ISPG claim that high-energy resonant modes which are unstable today were once stable constituents of matter; it does not by itself confirm the pressure-stability mechanism.

- **Present epoch** (intermediate pressure): protons, neutrons, electrons, and photons are stable. Heavier particles (muon, tau, W/Z, top quark) are unstable and decay on timescales from 10^{-25} s to $2.2\,\mu$s.

- **Far future** (low pressure): if the pressure continues to decrease, eventually even the proton may become unstable. Current experimental bounds ($\tau_p > 10^{34}$ yr from Super-Kamiokande [31]) are consistent with an extremely slow destabilization driven by the cosmological pressure decline.

The rate of change of the stability threshold is set by the entropy production rate (Sec. VI A), which is logarithmically slow. This ensures that the proton lifetime, if finite, is vastly longer than the current age of the universe—consistent with observations.

VIII. THE HIGGS MECHANISM AS CONFORMAL SYMMETRY BREAKING

In the Standard Model, particles acquire mass through the Higgs mechanism: the electroweak vacuum spontaneously breaks $SU(2)_L \times U(1)_Y \to U(1)_{\rm EM}$, and the vacuum expectation value (VEV) $v = 246$ GeV sets the mass scale. At first glance, this appears to be an independent mechanism from ISPG's pressure-dependent mass. In fact, the two are deeply connected.

A. The conformal Higgs potential

The standard Higgs potential is

$$V(H) = -\mu^2\,|H|^2 + \lambda\,|H|^4, \tag{253}$$

with the VEV $v = \mu/\sqrt{\lambda}$. The φ-dependence of the VEV is not an extra postulate: it follows from the ISPG dimensional rule applied to the parameters of Eq. (253). The derivation below uses this Standard-Model Higgs sector—the $SU(2)$ doublet H with its broken-phase potential—as an effective electroweak bookkeeping device, not as a fundamental ISPG scalar. The fundamental ISPG action contains no SM-type Higgs field; what is shown here is that the bi-conformal dimensional rule forces the broken-phase VEV to scale as $v(\varphi) = v_0 e^{\varphi/2}$ once an effective SM Higgs sector is supplied with imported $v_0 = 246$ GeV.

a. Derivation from dimensional scaling. In ISPG, any quantity with mass dimension d carries a factor $e^{d\,\varphi/2}$ when translated between a local frame and the asymptotic reference frame (this is the same rule that gives $m_{\text{eff}} = m_0\, e^{\varphi/2}$ for $d = 1$). Applying it to the two parameters of the Higgs potential:

- μ^2 has mass dimension 2, so $\mu^2(\varphi) = \mu_0^2\, e^{\varphi}$;
- λ is dimensionless and, by the bi-conformal invariance of dimensionless couplings (a structural property of the theory, not an assumption introduced here), is, at fixed local renormalization convention, φ-independent.

Loop running and cross-epoch comparisons of dimensionless couplings are separate renormalization questions, addressed in the fine-structure-variation and pressure-history discussions. Therefore

$$\boxed{v^2(\varphi) \;=\; \frac{\mu^2(\varphi)}{\lambda} \;=\; v_0^2\, e^{\varphi}, \qquad v(\varphi) = v_0\, e^{\varphi/2},} \tag{254}$$

where $v_0 = 246$ GeV is the VEV measured in the local frame. No new coupling and no independent φ-dependent parameter has been added: the scaling in Eq. (254) is forced by the ISPG dimensional rule together with the bi-conformal invariance of λ.

Locally, v is undetectably constant: an observer using local mass-dimension units measures $v_0 = 246$ GeV regardless of φ. The φ-dependence in Eq. (254) is only visible when comparing the VEV at different positions through a common (asymptotic) reference frame.

This means the Higgs mechanism is not independent of gravity: it is the **electroweak projection** of the same bi-conformal symmetry that governs the scalar-metric sector.

b. Scope of this identification. What Eq. (254) establishes is the *φ-scaling* of the electroweak VEV: once the local value $v_0 = 246$ GeV is taken as input, the functional form $v(\varphi) = v_0\, e^{\varphi/2}$ is fixed by the ISPG dimensional rule. What it does *not* provide is a microscopic derivation of the numerical value v_0 itself from the ISPG condensate parameter Φ_0. Fixing v_0 in terms of Φ_0 and the underlying hysteresis sector would require a full coarse-graining of the electroweak condensate on top of the Route B microscopic closure, and is left as a topic for a dedicated paper. The present section uses only the *scaling* of v, which is what enters every observable consequence derived below.

B. Bi-conformal link between electroweak and gravitational masses

What is being linked here is the *mass-scaling* sector, not the gauge sector. The electroweak gauge group, its couplings, and the Yukawa hierarchy are taken as Standard-Model inputs; ISPG does not derive them. What ISPG fixes is that every Standard-Model mass inherits the same φ-scaling as a gravitational mass in Eq. (3), because both originate from the same bi-conformal rule applied in Eq. (254). The identification $v(\varphi) = v_0\, e^{\varphi/2}$ has three immediate consequences:

(i) Mass generation is universal.—Every Standard Model mass (fermions via Yukawa couplings $m_f = y_f v/\sqrt{2}$, and gauge bosons via $m_W = gv/2$) inherits the $e^{\varphi/2}$ factor:

$$m_f(\varphi) = \frac{y_f\, v_0}{\sqrt{2}}\, e^{\varphi/2}, \qquad m_W(\varphi) = \frac{g\, v_0}{2}\, e^{\varphi/2}, \tag{255}$$

precisely reproducing Eq. (3). The ISPG effective mass and the Higgs-generated mass are the *same* quantity viewed from different perspectives.

(ii) The Higgs boson mass is position-dependent.—The Higgs boson mass $m_H = \sqrt{2\lambda}\, v$ also scales as $e^{\varphi/2}$, but this is locally undetectable: the ratio $m_H/m_W = \sqrt{2\lambda}/g$ is dimensionless and therefore φ-independent.

(iii) Electroweak phase transition in the early universe.—In the early universe, when the background pressure was higher (in the present FLRW convention with $\varphi_0 = 0$ and the L-3 pressure-history $\varphi_{\rm bg}(z)$ channel, φ was closer to zero), the VEV was larger: $v_{\rm early} > v_{\rm today}$ (as measured by an asymptotic observer). The electroweak phase transition temperature $T_{\rm EW} \propto v$ was correspondingly higher. However, locally measured temperatures also scale by $e^{\varphi/2}$, so the dimensionless ratio $T_{\rm EW}/T_{\rm local}$ is preserved: the phase transition occurred at the same locally measured temperature as in the standard scenario.

C. Prediction: gravitational correction to the Higgs mass

The coupling between the Higgs field and the scalar-metric sector introduces a gravitational correction to the Higgs self-energy. At one-loop level, the gravitational correction to the Higgs mass parameter is

$$\delta m_H^2 \sim \frac{m_H^2}{16\pi^2}\, \frac{m_H^2}{M_{\rm Pl}^2} \sim 10^{-34}\, m_H^2, \tag{256}$$

i.e. $\delta m_H/m_H \sim 10^{-17}$. This is many orders of magnitude below the sensitivity of any currently envisaged collider (LHC: $\sim 10^{-3}$; FCC-ee/CEPC: $\sim 10^{-4}$–10^{-5}), so Eq. (256) is *not* a near-term experimental discriminant. Its role here is theoretical: it records the gravitational piece of the Higgs self-energy that the ISPG sector unavoidably contributes, and confirms that this piece is Planck-suppressed and therefore consistent with the observed Higgs mass without requiring tuning of new parameters. Any observable φ-dependent signature of the Higgs sector has to come from the universal scaling in Eq. (254), not from this loop correction.

D. Gauge boson spectrum on the bi-conformal background

Section VIII B established that every Standard Model mass inherits the universal factor $e^{\varphi/2}$. Once the electroweak gauge algebra and couplings are taken as the experimentally fixed internal data, the broken-phase gauge sector on the bi-conformal background is fully determined. We now derive the curved-space electroweak mass matrix, the resulting gauge-boson spectrum, and its observational consequences.

Broken-phase mass matrix on the bi-conformal background.—Start from the curved-space electroweak action

$$S_{\rm EW} = \int d^4x\,\sqrt{-g}\left[-\frac{1}{4}W^a_{\mu\nu}W^{a\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} + (D_\mu H)^\dagger D^\mu H - \lambda\left(H^\dagger H - \frac{v(\varphi)^2}{2}\right)^2\right], \tag{257}$$

with

$$D_\mu H = \left(\partial_\mu - \frac{ig}{2}\tau^a W^a_\mu - \frac{ig'}{2}B_\mu\right)H. \tag{258}$$

In unitary gauge,

$$H(x) = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v(\varphi) + h(x)\end{pmatrix}, \tag{259}$$

so the quadratic gauge contribution from $(D_\mu H)^\dagger D^\mu H$ is

$$\mathcal{L}_{\rm mass} = \sqrt{-g}\,\frac{v(\varphi)^2}{8}\left[g^2 W^1_\mu W^{1\mu} + g^2 W^2_\mu W^{2\mu} + \left(gW^3_\mu - g' B_\mu\right)^2\right]. \tag{260}$$

Defining the usual broken-phase combinations

$$W^\pm_\mu \equiv \frac{W^1_\mu \mp iW^2_\mu}{\sqrt{2}}, \qquad Z_\mu \equiv \frac{gW^3_\mu - g'B_\mu}{\sqrt{g^2+g'^2}}, \qquad A_\mu \equiv \frac{g'W^3_\mu + gB_\mu}{\sqrt{g^2+g'^2}}, \tag{261}$$

diagonalizes the mass matrix:

$$\mathcal{L}_{\rm mass} = \sqrt{-g}\left[m_W(\varphi)^2\, W^+_\mu W^{-\mu} + \frac{1}{2}m_Z(\varphi)^2\, Z_\mu Z^\mu\right], \tag{262}$$

with

$$m_W(\varphi) = \frac{g\,v(\varphi)}{2}, \qquad m_Z(\varphi) = \frac{\sqrt{g^2+g'^2}}{2}\,v(\varphi), \qquad m_\gamma = 0. \tag{263}$$

Thus, once the internal electroweak algebra is specified, the entire φ-dependence of the broken-phase mass matrix is fixed by the single bi-conformal scaling law $v(\varphi) = v_0 e^{\varphi/2}$.

Electroweak mass relations.—After spontaneous symmetry breaking $SU(2)_L \times U(1)_Y \to U(1)_{\rm EM}$, the four electroweak gauge bosons have masses

$$m_{W^\pm}(\varphi) = \frac{g\,v_0}{2}\,e^{\varphi/2} \approx 80.4\ \text{GeV} \times e^{\varphi/2}, \tag{264}$$

$$m_Z(\varphi) = \frac{m_W}{\cos\theta_W} \approx 91.2\ \text{GeV} \times e^{\varphi/2}, \tag{265}$$

$$m_\gamma = 0, \tag{266}$$

where $g \approx 0.653$ is the $SU(2)_L$ coupling and θ_W is the Weinberg angle with $\sin^2\theta_W \approx 0.231$. The φ-dependence enters solely through $v(\varphi) = v_0\,e^{\varphi/2}$ (Eq. 254).

The dimensionless mass ratios are φ-independent:

$$\frac{m_W}{m_Z} = \cos\theta_W, \qquad \frac{m_W}{m_H} = \frac{g}{2\sqrt{2\lambda}}, \qquad \frac{m_W}{m_f} = \frac{g}{y_f\sqrt{2}}. \tag{267}$$

These are identical to the Standard Model relations, with no tree-level corrections from the uniform scalar-metric scaling. The calculation preserves the standard tree-level electroweak mass ratios and local dispersion relations once the SM electroweak sector, g, g', and v_0 are supplied. Full electroweak precision observables—including radiative corrections, widths, asymmetries, oblique parameters, and scheme-dependent weak mixing angles—are inherited from the imported SM sector here; they are not derived from the ISPG medium in this section.

Local Lorentz invariance of the gauge sector.—On the bi-conformal background, with the MAIN convention $ds^2 = -e^{\varphi}c^2dt^2 + e^{-\varphi}d\vec{x}^2$ (so $g^{00} = -e^{-\varphi}$, $g^{ii} = +e^{\varphi}$), the mass-shell condition $g^{\mu\nu}k_\mu k_\nu + m^2 = 0$ gives the coordinate dispersion relation

$$-e^{-\varphi}\,\omega^2 + e^{\varphi}\,k^2 + m^2(\varphi) = 0. \tag{268}$$

Converting to locally measured quantities ($\omega_{\rm loc} = \omega\, e^{-\varphi/2}$ from $d\tau = e^{\varphi/2}dt$, $k_{\rm loc} = k\, e^{\varphi/2}$ from $d\ell = e^{-\varphi/2}dx$, $m_{\rm loc} \equiv m(\varphi)$, the locally measured mass):

$$\omega_{\rm loc}^2 = k_{\rm loc}^2 + m_{\rm loc}^2. \tag{269}$$

This is the standard special-relativistic dispersion relation. Although $m(\varphi)$ varies with position, every local observer measures the *same* numerical mass value (e.g., $m_W = 80.4$ GeV) because all reference scales—atomic masses, rulers, clocks—carry the same $e^{\varphi/2}$ factor. The bi-conformal metric thus preserves local Lorentz invariance of the gauge sector exactly.

Photon mass protection.—The masslessness of the photon is protected by the unbroken $U(1)_{\rm EM}$ gauge invariance, which forbids a mass term at any loop order, provided the effective/imported gauge sector and its regularization preserve the Ward identity. In ISPG, the shift symmetry $\varphi \to \varphi + \text{const}$ of the action (2) provides secondary scalar-sector hygiene: because φ enters the action only through derivatives and through the metric, it prevents explicit non-derivative scalar terms from being introduced into the gauge bookkeeping sector. This shift symmetry is not an independent photon Ward identity; it states that the scalar-metric sector does not by itself spoil the imported gauge protection. If the electromagnetic gauge field is to be derived as an emergent medium excitation, preservation of the exact $U(1)_{\rm EM}$ Ward identity remains part of the future gauge-completion programme.

Gauge couplings as input parameters.—The couplings g, g', and the strong coupling g_s are empirical dimensionless inputs at a fixed local renormalization convention and scale. The classical bi-conformal unit scaling does not change such dimensionless inputs, so the tree-level mass-ratio algebra treats them as φ-independent:

$$g \approx 0.653, \quad g' \approx 0.350, \quad g_s(m_Z) \approx 1.22. \tag{270}$$

The derivation of these couplings from the medium's internal structure—specifically, why g/g' takes its observed value—requires solving the nonlinear resonance equations for the vector perturbation modes (Sec. III F) and is the subject of future work. Running, trace-anomaly feedback, and cosmological effective comparisons of these couplings are the separate questions flagged in Sec. VIII E; they are not fixed by the tree-level mass-matrix calculation.

Status of the gauge completion.—The calculation achieved here is therefore conditional and two-tiered. What is *verified here* is the standard broken-phase electroweak mass-matrix algebra and the local dispersion/scaling on the bi-conformal background, given the effective SM electroweak sector. What is *imported here* is the gauge algebra, representations, exact gauge redundancy and Ward identity, and the numerical values of g and g'. What is *not yet derived from the medium itself* is the microscopic origin of the Yang–Mills vector fields, their exact gauge symmetry, their couplings, and the multiplicity of the electroweak generators.

Minimal medium interpretation.—Within the resonance framework of Sec. III F, the scalar-metric vector-perturbation sector is the candidate medium channel for a future derivation of the spin-1 carriers. In the calculation above, however, the standard polarization content follows from the imported Higgsed gauge sector:

$$N_{\rm pol}(W^{\pm}) = N_{\rm pol}(Z) = 3, \qquad N_{\rm pol}(\gamma) = 2. \tag{271}$$

Thus the bi-conformal background closes the conditional kinematics and mass scaling of the electroweak gauge sector once the internal gauge algebra is specified, while the deeper question of why the medium realizes Yang–Mills spin-1 carriers with precisely that algebra remains future work.

The massive gauge bosons (W, Z) are therefore described here as the standard broken-phase spin-1 modes of the effective electroweak sector. Their possible realization as localized ISPG resonances belongs to the future gauge-completion programme; their masses scale with the effective Higgs VEV through the conditional bi-conformal law of Sec. VIII A.

The massless photon is the unbroken $U(1)_{\rm EM}$ vector mode. Its identification with the surviving massless combination of the electroweak gauge fields is standard; the ISPG framework adds the prediction that the locally measured fine-structure constant $\alpha = e^2/(4\pi\hbar c)$ is position-independent (since $e = g\sin\theta_W$ and all factors are dimensionless). The cosmological variation of α discussed in Sec. VI E arises from vacuum-polarization corrections, not from a running of the bare coupling.

The strong sector.—The eight gluons of quantum chromodynamics (QCD) mediate the strong interaction between quarks. On the bi-conformal background, the QCD Lagrangian retains its standard form with φ-independent coupling g_s; all QCD predictions—asymptotic freedom, confinement, the hadron spectrum—are preserved. The origin of the $SU(3)_c$ gauge group from the medium's tensor and nonlinear scalar excitation sectors, together with the derivation of g_s from the medium dynamics, is the subject of future work.

E. Scope and deferred items of the electroweak treatment

The present section derives only the *mass-scaling* sector of the electroweak theory on the bi-conformal background. Four adjacent questions that would require new machinery are explicitly set aside here and flagged as topics for dedicated work:

- *Proton mass and the QCD contribution.*—Most of the proton mass comes from the QCD confinement scale $\Lambda_{\rm QCD}$ via dimensional transmutation, not from the Higgs

VEV. The φ-scaling of $\Lambda_{\rm QCD}$ is controlled by the running of g_s and by the QCD trace anomaly; demonstrating that the Higgs-sector scaling $v(\varphi) = v_0\, e^{\varphi/2}$ extends consistently to the full hadronic mass spectrum is a separate derivation left to future work.

- *Hierarchy problem.*—The ISPG framework does not address the electroweak hierarchy problem: the smallness of v_0 relative to $M_{\rm Pl}$ is taken here as an empirical input. Eq. (256) shows that the gravitational correction to the Higgs mass is Planck-suppressed and therefore does not generate a naturalness issue on its own, but it also does not *resolve* the Standard-Model hierarchy problem. A mechanism fixing v_0 in terms of the condensate parameter Φ_0 is left open.

- *Conformal / trace anomaly.*—The classical bi-conformal invariance of dimensionless couplings is broken at one loop by the standard gauge-theory trace anomaly. Quantifying how the resulting anomalous φ-dependence feeds back into running couplings is an open item; it is expected to contribute only at loop level and not to affect the tree-level scaling used in Eq. (254).

- *Electroweak phase transition.*—The statement in item (iii) of Sec. VIII B that the phase transition occurs at the same *locally measured* temperature relies on the classical bi-conformal symmetry of the dimensionless ratio $T_{\rm EW}/v$. Running-coupling and finite-temperature anomaly corrections to that ratio are likewise deferred.

None of these items modifies the results derived above, which depend only on the φ-scaling of v and on the φ-independence of dimensionless couplings. They identify the direction in which the electroweak chapter of ISPG must be extended.

IX. ENTANGLEMENT AND GRAVITATIONAL DECOHERENCE

Quantum entanglement—the nonlocal correlation between spatially separated particles—is one of the most precisely tested features of quantum mechanics. In ISPG, entanglement acquires a concrete physical picture as correlated resonance structure, and the theory predicts a specific, testable mechanism for gravitational decoherence.

A. Entanglement as correlated resonance structure

When two particles are produced from a single quantum event (e.g., parametric down-conversion or pair production), the resulting state is a single, spatially extended resonance pattern with two localized cores.

In the ISPG framework, the two-particle state is described by a single resonance field configuration $\delta\varphi_{12}(t, \mathbf{x})$ that cannot be factored into independent single-particle configurations:

$$\delta\varphi_{12} \neq \delta\varphi_1 \cdot \delta\varphi_2. \tag{272}$$

The cores propagate apart, but the correlations encoded in the phase structure of the resonance field are preserved as long as the field evolves unitarily (i.e., without decoherence).

Compatibility with Bell's theorem.—ISPG respects signal locality: both the background scalar φ and its resonance excitations $\delta\varphi$ obey causal wave equations derived from the action (2), with characteristic cones bounded by the speed of light on the bi-conformal metric, so no perturbation propagates faster than c. (The massless background φ propagates at c; massive resonance excitations propagate with group velocity $\leq c$ set by the standard relativistic dispersion.) ISPG therefore cannot be used to signal superluminally. The entangled correlations arise because the two cores were *never independent*: they share a common resonance structure established at creation. The "nonlocality" is not in the dynamics but in the initial conditions—the correlated state was created locally and subsequently separated.

This position is analogous to the pilot-wave (de Broglie–Bohm) interpretation, where nonlocal correlations arise from a globally defined wave function, and to quantum field theory, where entanglement is encoded in the Fock-space structure rather than in local hidden variables.

B. Gravitational phase rate and conditional decoherence

The ISPG framework predicts a specific coherent gravitational phase channel: when entangled particles are placed at different gravitational potentials (different values of φ), the bi-conformal rescaling introduces a differential phase evolution between the two local resonance clocks. At leading post-Newtonian order this phase channel reproduces the standard GR redshift/COW baseline. The stronger claim of genuine irreversible decoherence requires an additional unobserved channel, supplied in ISPG by resonant-tail emission or stochastic scalar fluctuations, and is therefore stated below as a conditional extension of the coherent phase result.

Consider two entangled particles A and B at positions with scalar-field values φ_A and φ_B, respectively. Their local frequencies are (Eq. 33):

$$\omega_A = \omega_0\, e^{-\varphi_A/2}, \qquad \omega_B = \omega_0\, e^{-\varphi_B/2}, \tag{273}$$

where ω_0 is the coordinate frequency at production.

The coherent differential phase accumulated over coordinate time t is

$$\Delta\theta(t) = (\omega_A - \omega_B)\, t = \omega_0\, t\, \left(e^{-\varphi_A/2} - e^{-\varphi_B/2}\right). \tag{274}$$

For $|\Delta\varphi| \equiv |\varphi_A - \varphi_B| \ll 1$:

$$\Delta\theta(t) \approx \frac{\omega_0\, t}{2}\, \Delta\varphi. \tag{275}$$

When the differential phase reaches $\Delta\theta \sim \pi$, the entangled state has evolved to an orthogonal (distinguishable) configuration. This defines a coherent dephasing or orthogonalization time,

$$t_\varphi \sim \frac{2\pi}{\omega_0\, |\Delta\varphi|} = \frac{h}{m_0 c^2\, |\Delta\varphi|}, \tag{276}$$

and the corresponding leading phase/dephasing rate is

$$\boxed{\Gamma_\varphi = \frac{m_0 c^2\,|\Delta\varphi|}{h} = \frac{m_0 c^2}{h}\,\frac{2|\Delta\Phi_N|}{c^2} = \frac{2m_0\,|\Delta\Phi_N|}{h},} \tag{277}$$

where $\Delta\Phi_N = -GM\Delta r/r^2$ is the Newtonian potential difference and the factor of 2 follows from $\varphi = 2\Phi_N/c^2$.

Dephasing versus decoherence.—Eq. (277) describes a differential phase evolution, which in isolation would be reversible *dephasing*: if the two particles were brought back to the same potential, the relative phase could be undone. In ISPG this coherent channel can become genuine (irreversible) *decoherence* only if one of the following unobserved channels is active and its correlation structure is derived:

1. *Resonant-tail emission.*—Each oscillon continuously emits resonant tails (Sec. VI A) that carry away which-path information into the scalar-field vacuum. In different φ-backgrounds, the tail spectra differ: the emitted radiation from particle A at φ_A is distinguishable (in principle) from that of particle B at φ_B. When these tail modes are traced out, the reduced density matrix of the two-particle system acquires off-diagonal suppression.

2. *Stochastic fluctuations.*—The scalar field φ undergoes vacuum fluctuations $\langle\delta\varphi^2\rangle \neq 0$. These fluctuations randomly modulate the differential phase $\Delta\theta(t)$, converting the deterministic dephasing into a stochastic process. For Gaussian fluctuations the ensemble average of the phase factor evaluates to $\langle e^{i\Delta\theta(t)}\rangle = \exp\big[-\frac{1}{2}\langle\Delta\theta(t)^2\rangle\big]$. In the Markovian limit, in which the correlation time τ_c of φ-fluctuations is much shorter than the observation time ($\tau_c \ll t$), this reduces to the exponential form $\langle e^{i\Delta\theta(t)}\rangle \simeq e^{-\Gamma_{\rm dec}\,t}$, with $\Gamma_{\rm dec} \simeq \Gamma_\varphi$ only after the scalar-noise or tail-emission kernel is shown to supply the required Markovian coefficient. When the Markovian condition is not satisfied, the exponential form is replaced by the Gaussian form $\exp\big[-\frac{1}{2}\sigma^2 t^2\big]$, and no universal exponential decoherence rate follows from the coherent phase calculation alone.

The closed result in this section is therefore the leading-order coherent rate Γ_φ. The identification $\Gamma_{\rm dec} \simeq \Gamma_\varphi$ is a conditional ISPG decoherence channel, not an additional completed leading-order derivation. A detailed derivation of the correlation time τ_c and the fluctuation spectrum of $\delta\varphi$, which is needed to confirm that the Markovian limit applies to any proposed experimental regime, is deferred to dedicated work. At this leading post-Newtonian order, Γ_φ coincides with the gravitational dephasing predicted by GR through the redshift–COW correspondence $\varphi \simeq 2\Phi_N/c^2$; both depend on the orientation of the pair relative to the local gravitational gradient. A genuine discriminator between ISPG and GR for this observable therefore lies beyond the 1PN level, in the 2PN response of the bi-conformal factor (Appendix 4 of the main theory [1]). The present section quantifies the leading-order rate with the understanding that it reproduces the GR baseline at that order.

Numerical estimates.—For relativistic particles such as photons, the coherent rate formula (277) admits a formal evaluation but is of limited practical use. A photon of energy $E = 1$ eV in a gradient with $|\Delta\varphi| = 2g\,\Delta h/c^2 \approx 2.2 \times 10^{-15}$ (corresponding to $\Delta h = 10$ m in Earth's field, $g = 9.8$ m/s^2) yields the formal rate

$$\Gamma_\varphi \sim \frac{E\,|\Delta\varphi|}{h} \approx 0.5 \text{ Hz}. \tag{278}$$

However, the photon spends only $t_{\text{flight}} \sim \Delta h/c \sim 33$ ns in the gradient region, so the accumulated coherent phase contribution from the bi-conformal factor is $\Delta\theta_{\text{geom}} = E\,|\Delta\varphi|\,t_{\text{flight}}/\hbar \sim 1.1 \times 10^{-7}$ rad. The formal 0.5 Hz coefficient is therefore not a clean direct test in this simple flight geometry. It is physically meaningful as an accumulation rate only for systems confined in the gradient region on timescales $t \gtrsim \Gamma_\varphi^{-1}$; in practice this shifts attention to nonrelativistic massive systems and to the still-conditional irreversible channel.

For massive particles the rate scales with the rest energy $m_0 c^2$. Taking a laboratory-scale superposition with a height difference $\Delta h \sim 1$ m in Earth's field gives $\Delta\Phi_N/c^2 \sim 10^{-16}$. For an optomechanical mass $m \sim 10^{-14}$ kg,

$$\Gamma_\varphi = \frac{mc^2 \cdot 2|\Delta\Phi_N|/c^2}{h} \sim \frac{9 \times 10^2 \text{ J} \cdot 2 \times 10^{-16}}{6.6 \times 10^{-34} \text{ J} \cdot \text{s}} \sim 3 \times 10^{20} \text{ Hz}, \tag{279}$$

i.e. the leading coherent dephasing timescale is far shorter than any accessible storage time. For nanospheres ($m \sim 10^{-20}$ kg, $\sim 10^7$ amu) at $\Delta\Phi_N/c^2 \sim 10^{-16}$ the rate drops to $\sim 3 \times 10^{14}$ Hz—still far beyond any plausible coherence time. The scaling $\Gamma_\varphi \propto m$ gives $\Gamma_\varphi \sim 3 \times 10^{12}$–$3 \times 10^{11}$ Hz for $m = 10^{-22}$–10^{-23} kg at the same meter-scale Earth-gradient benchmark. Thus this benchmark does not place MAQRO [32] or STE-QUEST masses in a kHz–MHz window. Identifying a viable low-frequency mesoscopic window requires a separate scan over mass, separation, storage time, and the still-open irreversible-channel correlator.

C. Comparison with Penrose–Diósi models

The conditional decoherence channel associated with Eq. (277) is structurally similar to the gravitational decoherence rates proposed by Penrose [8] and Diósi [9]:

$$\Gamma_{\text{P}} \sim \frac{E_G}{\hbar}, \qquad \Gamma_{\text{D}} \sim \frac{Gm^2}{\hbar\,\Delta x}, \tag{280}$$

where E_G is a gravitational self-energy and Δx is a spatial superposition size. The key differences are:

1. **Origin.** Penrose and Diósi postulate gravitational decoherence as a fundamental modification of quantum mechanics. In ISPG, the completed leading result is the φ-dependent coherent phase rate; irreversible decoherence follows only after the resonant-tail or scalar-noise channel is derived, without modifying the Schrödinger equation.

2. **Mechanism.** In Penrose's model, the decoherence comes from the "ill-definedness" of time in a superposition of different geometries. In ISPG, the calculable part is

differential phase evolution in an inhomogeneous φ-field; the irreversible part depends on the conditional tail/noise correlator.

3. **Testability.** ISPG predicts a specific *directional* dependence: the phase/decoherence rate depends on the orientation of the entangled pair relative to the gravitational gradient $\nabla\varphi$. At leading order this reproduces the GR redshift baseline; a unique ISPG signature requires either the 2PN bi-conformal response or the completed irreversible tail/noise channel.

D. Quantum tunneling in an inhomogeneous field

The resonance picture suggests a possible modification of quantum tunneling. A particle's resonant tail extends beyond the classically forbidden region (the potential barrier). If the medium on the far side of the barrier has favorable conditions (e.g., a local minimum in the effective potential), the tail "anchors" there and the resonance core transitions through the barrier. This tail-anchoring picture is retained as the candidate route to a non-Hermitian or resonant-scattering boundary condition; the controlled WKB statement below is more limited.

In an inhomogeneous φ-field, the tunneling exponent depends on the scalar profile across the forbidden region. We use x as the coordinate along a local static barrier, $x_2 - x_1 = d$ as the coordinate width, and E and $V_0(x)$ as local-tetrad energy and barrier-height data held fixed by the apparatus. With these conventions the WKB amplitude is

$$T \propto e^{-B_\varphi}, \qquad B_\varphi = \frac{2}{\hbar}\int_{x_1}^{x_2} \sqrt{2m_0\, e^{\varphi(x)/2}\,[V_0(x) - E]}\, dx. \tag{281}$$

For a locally constant φ and a rectangular barrier, this reduces to

$$T \propto \exp\left[-\frac{2d}{\hbar}\sqrt{2m_0\, e^{\varphi/2}\,(V_0 - E)}\right]. \tag{282}$$

Equation (281) is profile dependent, but it is not by itself direction-nonreciprocal. For the same real, static, Hermitian barrier, reversing the traversal direction leaves the forbidden-region action unchanged; the flux-normalized WKB transmission is reciprocal. A true directional ratio therefore requires additional structure: different boundary energies, a driven or dissipative medium, time-reversal breaking, or the ISPG-specific resonant-tail anchoring channel sketched above.

$$\boxed{\left(\frac{T_\downarrow}{T_\uparrow}\right)_{\text{tail}} = \exp\left[\mathcal{C}_{\text{tail}}\,\frac{d\,\Delta\varphi_{\text{barrier}}}{\hbar}\,\sqrt{2m_0(V_0 - E)}\right]}\,, \tag{283}$$

where $T_\downarrow$ and $T_\uparrow$ denote candidate tail-assisted tunneling rates toward and away from the gravitational source. The coefficient $\mathcal{C}_{\text{tail}}$ and its sign are not fixed by the Hermitian WKB integral; they require a dedicated resonant-scattering calculation of the tail-anchoring boundary condition. The earlier choice $\mathcal{C}_{\rm tail} = 1$ should therefore be read only as a parametrization of the candidate channel, not as a derived coefficient.

The distinctive ISPG claim is therefore not that ordinary Hermitian WKB tunneling is nonreciprocal. It is the more specific proposal that the scalar-gradient background and the resonant-tail anchoring channel can supply the extra boundary structure needed for a directional asymmetry. Standard quantum mechanics can also generate directional-looking effects when comparable extra structure is present (absorption, driving, unequal asymptotic potentials, or time-reversal breaking); the ISPG novelty is the proposed geometric origin of such structure.

The relevant quantity in the candidate directional term is the change of φ *across the barrier width* d, not across the host system's size. For nuclear barriers the barrier width is of order a femtometre, and the gravitational gradient varies over macroscopic distances, so the across-barrier shift is $\Delta\varphi_{\rm barrier} \sim (d/L)\,\Delta\varphi_{\rm system}$ where L is the gradient scale length. For a white-dwarf interior with $\Delta\varphi_{\rm system} \sim 10^{-4}$ across a radius $L \sim 10^{6}$ m, this gives $\Delta\varphi_{\rm barrier} \sim 10^{-25}$ over a fm-scale barrier—negligible for individual nuclear reactions. A detectable signal would require either very long barriers (e.g. macroscopic tunneling experiments in trapped cold atoms) or a cumulative effect integrated over many events; quantitative feasibility study of these channels is left for dedicated work. The conservative statement retained here is that the WKB exponent is profile dependent, while nonreciprocal anisotropy and its amplitude in any concrete experimental context require the resonant-tail boundary calculation and the correct barrier-scale $\Delta\varphi_{\rm barrier}$, not the system-scale value.

X. TOPOLOGICAL STRUCTURE: SPIN, ANTIMATTER, AND EXCLUSION

The resonance interpretation of particles (Sec. III) requires a mechanism to generate half-integer spin, distinguish matter from antimatter, and enforce the Pauli exclusion principle. In this section we show that all three properties emerge from the topological structure of vortex excitations in the scalar-metric medium.

A. Spin as vortex winding number

In a fluid, a vortex carries angular momentum determined by its circulation $\oint \mathbf{v}\cdot d\mathbf{l} = n\,\kappa$, where n is the (integer) winding number and κ is the circulation quantum. The angular momentum is a topological invariant: it cannot be changed by smooth deformations of the flow and can only be removed by annihilating the vortex with an anti-vortex of opposite winding.

In the scalar-metric medium, the vector perturbation δg_{0i} (the frame-dragging or gravitomagnetic sector) satisfies the linearized field equation obtained by perturbing the ISPG field equations (derived from the action (2)) around the background $g^{(0)}_{\mu\nu} = \mathrm{diag}(-e^{\varphi_0}, e^{-\varphi_0}, e^{-\varphi_0}, e^{-\varphi_0})$.

For stationary perturbations ($\partial_t = 0$) with only spatial components δg_{0i}, the linearized equation is

$$\nabla^2(e^{-\varphi_0}\delta g_{0i}) - \partial_i\partial_j(e^{-\varphi_0}\delta g_{0j}) = 0, \tag{284}$$

where ∇^2 is the flat-space Laplacian. This is the equation for a vector field in a medium with position-dependent "permeability" $e^{-\varphi_0}$. The nonlinear rotational response of this medium in the galactic regime is developed in the companion MOND paper [2]; here we focus on the topological content relevant to spin.

Vortex solution.—In cylindrical coordinates (r, ϕ, z), we seek axisymmetric solutions with only an azimuthal component. Setting $\delta g_{0r} = \delta g_{0z} = 0$ and $\delta g_{0\phi} = A(r)e^{in\phi}$ with integer n, the equation (284) reduces to a Bessel equation for the radial profile $A(r)$:

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dA}{dr}\right) - \frac{n^2}{r^2}A = 0. \tag{285}$$

The regular solution at $r = 0$ is $A(r) \propto r^{|n|}$; the solution that decays at infinity requires a cutoff or core structure (analogous to the oscillon core in Sec. III E). The phase factor $e^{in\phi}$ implies that a 2π traversal around the vortex accumulates phase $2\pi n$, defining the **winding number** $n \in \mathbb{Z}$.

Angular momentum calculation.—The angular momentum of a stationary spacetime perturbation is given by the Komar integral over a surface at infinity:

$$J = \frac{c^2}{16\pi G}\oint_\infty \nabla^\mu \xi^\nu \, dS_{\mu\nu}, \tag{286}$$

where $\xi^\mu = \partial_\phi$ is the azimuthal Killing vector. The scalar vortex with winding n sources an axially symmetric frame-dragging field $\delta g_{0\phi} = \kappa/(2\pi r)$ at large r, where κ is the circulation quantum. Evaluating the covariant derivatives on the bi-conformal background and performing the azimuthal integration, we obtain

$$J = \frac{c^2\,\kappa}{16\pi G}\, e^{\varphi_0/2}. \tag{287}$$

Derivation of the circulation quantum from topological quantization.—The circulation κ is *not* a free parameter; it is fixed by the single-valuedness of the scalar perturbation.

a. What carries the phase. The background scalar φ of MAIN is real, so a $U(1)$ winding cannot be imposed on φ itself. The phase structure used here lives on the oscillon sector: a localized time-oscillating source of the form $\Phi(t, \mathbf{x}) = \Phi_0(r)\cos[\Omega t + n\phi]$ introduced in Sec. III E naturally combines into the complex envelope $\Psi \equiv \Phi_0(r)\, e^{i(\Omega t + n\phi)}$, whose argument is what winds n times around the core. The symbol $\delta\varphi$ in Eq. (292) and below is used as a shorthand for this complex oscillon envelope, not for a naive complexification of the real background field. With this identification, the following derivation is the standard Onsager–Feynman argument for a $U(1)$-phase condensate applied to the oscillon sector.

Consider a vortex configuration in which the complex envelope $\delta\varphi = |\delta\varphi(r)|\, e^{in\phi}$ winds n times around the core. Single-valuedness requires the phase to return to its initial value after a 2π circuit:

$$\oint d\phi \, \frac{\partial}{\partial \phi}(\arg \delta\varphi) = 2\pi n, \qquad n \in \mathbb{Z}. \tag{288}$$

The circulation of the effective velocity field $\mathbf{v}_{\text{eff}} = (\hbar/m_0)\, \nabla(\arg \delta\varphi)$ is therefore quantized:

$$\oint \mathbf{v}_{\text{eff}} \cdot d\mathbf{l} = \frac{2\pi n \, \hbar}{m_0}. \tag{289}$$

This is the Onsager–Feynman quantization condition, identical to that of superfluid vortices.

Now, the gravitomagnetic sector $\delta g_{0\phi}$ is sourced by the angular momentum current of the scalar vortex. In the bi-conformal metric, the frame-dragging field is related to the circulation by $\delta g_{0\phi} = \kappa/(2\pi r)$, where

$$\kappa = \frac{4G\, m_0}{c^2\, e^{\varphi_0/2}} \cdot \frac{2\pi n\, \hbar}{m_0} = \frac{8\pi G\, n\, \hbar}{c^2\, e^{\varphi_0/2}}. \tag{290}$$

The factor $4Gm_0/(c^2 e^{\varphi_0/2})$ is the gravitomagnetic coupling of a source of mass m_{eff} in the bi-conformal geometry; the remaining factor is the quantized circulation from Eq. (289). Crucially, no assumption about the value of J has been made—only single-valuedness of $\delta\varphi$.

Result.—Substituting Eq. (290) into Eq. (287):

$$\boxed{J = n\, \frac{\hbar}{2}.} \tag{291}$$

This boxed result is conditional on the source relation $\delta g_{0\phi} = \kappa/(2\pi r)$ and on the gravitomagnetic coupling normalization in Eq. (290); a field-equation-level derivation of the scalar-phase $\to$ off-diagonal-metric channel is the bridge item listed below.

Origin of the $1/2$ *factor.*—The factor of $1/2$ is not put in by hand, but it is also not a specifically bi-conformal effect: it is the *standard* numerical ratio between the Komar prefactor $c^2/(16\pi G)$ and the gravitomagnetic coupling of a quantized circulation, namely $8\pi G/(c^2)$ per unit quantum of circulation. Inserting Eq. (290) into Eq. (287), the bi-conformal factors cancel identically: $e^{\varphi_0/2}$ from the Komar evaluation on the bi-conformal background multiplies $e^{-\varphi_0/2}$ inside κ, leaving the geometry-independent combinatorial factor $(c^2/16\pi G)\cdot(8\pi G/c^2) = 1/2$. The same calculation performed on a Minkowski background with the same quantized circulation yields the same result; a scalar vortex with an integer winding $n \in \mathbb{Z}$ and Onsager–Feynman circulation $2\pi n\hbar/m_0$ therefore *would*, under the bridge normalization above, source an axial angular momentum $J = n\hbar/2$, independently of whether the metric is Minkowski or bi-conformal. What the bi-conformal structure contributes is a non-trivial φ_0-*cancellation*, i.e. the result that J depends only on the topological winding and not on the local value of φ_0; it does not by itself explain the $1/2$. The half-integer spectrum of J therefore reflects the quantized-circulation structure of the vortex, not a metric-specific halving.

Fermions.—In this conditional route, excitations with $n = \pm 1$ carry axial spin projection $J = \hbar/2$, as computed in Eq. (291), and are identified with fermions (electrons, quarks).

b. Relation to the 4π spinor rotation. The axial Komar quantity $J = \hbar/2$ matches the spin-projection eigenvalue of a spin-1/2 state (not the spin magnitude $\sqrt{3}\hbar/2$), but it does not by itself capture the 4π rotation symmetry observed in neutron interferometry [33]. The two statements are logically distinct. Going once around the vortex of winding $n = 1$ accumulates a *vortex-phase* $2\pi n = 2\pi$ (topologically trivial), not π; it is not an active spatial rotation of the state. The 4π symmetry is instead captured by the Berry-phase argument of Sec. X C: a *half-exchange* of two $n = 1$ vortices accumulates a phase of π, which is the hallmark of a spin-1/2 state under 2π spatial rotation via the spin-statistics connection. Thus the vortex construction supplies (i) the correct half-integer angular-momentum projection and (ii), through the Berry-exchange calculation below, a proposed route to the correct 4π rotation symmetry; together these would reproduce the spin-1/2 behaviour of Rauch [33] once the Berry-exchange / spin-statistics step is closed.

Bosons.—Excitations with $n = 0$ (no vortex structure) are bosons. In the present bookkeeping the photon belongs to the imported/effective $U(1)_{\rm EM}$ gauge sector, not to the scalar vortex topology. Its two physical helicities come from the spin-1 gauge field. The transverse-traceless tensor polarizations $h_+, h_\times$ remain reserved for the spin-2 gravitational-wave sector.

c. Scope of the vortex picture for spin. Subject to the scalar-phase $\to$ gravitomagnetic-slot bridge caveat above, the vortex construction gives a candidate route to the axial spin-projection quantum $J = \hbar/2$ for an excitation with scalar winding $n = \pm 1$. It is therefore not yet a complete derivation of the full fermion spin representation. For bosonic species the angular-momentum content comes from a different part of the perturbation spectrum: photons as spin-1 modes of the imported/effective $U(1)_{\rm EM}$ gauge sector (Sec. VIII D), and massive gauge bosons from the vector sector δg_{0i} combined with the Higgs Goldstone mode. A medium derivation of the photon from vector/helicoidal/transverse channels is future work; the tensor perturbations δg_{ij}^{TT} remain the spin-2 gravitational-wave sector. The pair-exchange phases computed in Sec. X C for $n = 0, \pm 1, \pm 2$ are therefore offered as an *illustrative* pattern that the spin-statistics correspondence is compatible with this topological labeling; they should not be read as a derivation that $W^\pm$ and Z are literally $n = \pm 2$ scalar vortices. A unified topological origin covering fermions and all bosonic species on the same footing would require a simultaneous treatment of the scalar, vector, and tensor perturbation sectors and is left for future work.

B. Antimatter as opposite winding

If a particle is represented by a vortex with winding number $n = +1$, its opposite-winding structural mirror is the vortex with $n = -1$: the same spatial structure, rotating in the opposite direction. In this subsection the map $n \to -n$ is therefore an antiparticle label, not by itself the full charge-conjugation operator. The total winding number is conserved under smooth deformations, so particle–antiparticle pair creation and annihilation are represented topologically by the appearance and disappearance of $n = +1$, $n = -1$ vortex pairs with $n_{\rm total} = 0$. The energy–momentum accounting, gauge-boson final states, and annihilation amplitudes are not derived by this topological bookkeeping.

The vortex picture naturally suggests further identifications—charge quantization from the integer-valuedness of n, the electromagnetic coupling from the sign of the winding, and the detailed energy budget of annihilation—but deriving these connections from the bi-conformal field equations requires a dedicated analysis that lies beyond the scope of the present work.

a. Open structural items (deferred). Three further structural items are explicitly set aside here and flagged as topics for a dedicated continuation rather than as claims of the present work. They do not invalidate the conditional result $J = n\hbar/2$ displayed above, but they circumscribe what the vortex construction does and does not yet cover:

- *Scalar winding $\leftrightarrow$ gravitomagnetic slot bridge.*—Sec. X A uses both the scalar phase $\delta\varphi$ (whose winding is quantized by Onsager–Feynman) and the gravitomagnetic field $\delta g_{0\phi}$ (which carries the Komar angular momentum). The identification between them is made at the level of a *source relation* ($\delta g_{0\phi} \propto \kappa/r$), not derived diagram-by-diagram from the full coupled field equations that connect scalar and vector perturbations. A field-equation-level derivation of the scalar-phase $\to$ off-diagonal-metric channel is left for future work.

- *CPT and charge content of the antimatter sector.*—The identification "$n \to -n$ is the antiparticle" correctly captures the sign-flip of angular momentum and (schematically) of any $U(1)$-type charge tied to the winding. A full CPT argument built on top of this picture, and a derivation of the numerical value of the electromagnetic charge from the bi-conformal field equations, are not attempted here and remain open items for a dedicated paper.

- *Non-Abelian structure.*—An integer winding $n \in \mathbb{Z}$ labels only the abelian $U(1)$-type topological charge. The non-Abelian structures of the Standard Model ($SU(2)_L$, $SU(3)_c$) require additional internal degrees of freedom beyond the single scalar-plus-metric sector considered here and are not addressed by this vortex construction. Whether such non-Abelian structure can be realized as multi-component vortex configurations on a richer medium is a separate question left for future work.

b. Notational convention for κ. Two circulation quanta appear in Sec. X A: $\kappa_{\mathrm{sf}} \equiv 2\pi n\hbar/m_0$ (the Onsager–Feynman superfluid circulation, units [m^2/s], Eq. 289) and $\kappa_{\mathrm{gm}} \equiv \oint \delta g_{0\phi}\, d\ell$ (the quantized gravitomagnetic circulation, units [m], Eq. 290). They are linked by the gravitomagnetic coupling $\kappa_{\mathrm{gm}} = [4Gm_0/(c^2 e^{\varphi_0/2})]\, \kappa_{\mathrm{sf}}$. Where the context is unambiguous, κ alone refers to κ_{gm}; elsewhere the subscripted forms should be used to avoid dimensional confusion.

C. Pauli exclusion from vortex topology

The Pauli exclusion principle—that two identical fermions cannot occupy the same quantum state—receives a candidate geometric/topological route in the vortex picture.

Two vortices of the same winding number n cannot occupy the same spatial location, because the resulting configuration would have winding number $2n$ at that point—a topologically distinct object (not two copies of the original particle). This is the same reason two fluid vortices of the same sign cannot merge: they orbit each other but maintain their individual cores.

a. Status of this hydrodynamic remark. The merging/orbiting observation is a *heuristic* consistency check, not the Pauli principle. The quantum Pauli statement is a property of the many-body wavefunction (antisymmetry under particle exchange), not of field configurations overlapping in space. The Berry-phase exchange sketch below is the candidate quantum route; the hydrodynamic analogy only motivates why two identical vortices behave as non-coincident objects.

Within this two-dimensional transverse Berry-phase sketch, the candidate exchange phase is π for $n = \pm 1$ vortices.

Berry phase calculation.—Consider two identical vortices at positions $\mathbf{x}_1$ and $\mathbf{x}_2$ in the two-dimensional plane transverse to the vortex axes. The two-vortex scalar field configuration is

$$\delta\varphi(\mathbf{x}) = f(|\mathbf{x} - \mathbf{x}_1|)\, e^{in \arg(\mathbf{x}-\mathbf{x}_1)} + f(|\mathbf{x} - \mathbf{x}_2|)\, e^{in \arg(\mathbf{x}-\mathbf{x}_2)}, \tag{292}$$

where $f(r)$ is the radial profile. The sum ansatz in Eq. (292) should be read as a heuristic order-parameter representation; a normalized multi-vortex state, the product/phase-additive construction, and the path-, profile-, and cutoff-independence of the Berry integral remain part of the deferred Fock-space construction. Exchange of the two vortices corresponds to adiabatically transporting vortex 1 along a closed path that encircles vortex 2 and returns to the starting position of vortex 2 (a half-circuit in the relative coordinate $\mathbf{r} = \mathbf{x}_1 - \mathbf{x}_2$).

During this transport, the phase of the scalar field at any fixed observation point $\mathbf{x}$ changes by the angle subtended by the path as seen from $\mathbf{x}$. The Berry connection for the adiabatic transport of the k-th vortex is

$$\mathcal{A}_k^i = -i \int d^2x\; \delta\varphi^* \frac{\partial\, \delta\varphi}{\partial x_k^i}. \tag{293}$$

For the half-exchange path $\mathbf{x}_1(\theta) = \frac{R}{2}(\cos\theta, \sin\theta)$, $\theta : 0 \to \pi$, the Berry phase is

$$\gamma_{\text{exchange}} = \oint \mathcal{A}_1^i\, dx_1^i = n \cdot \pi. \tag{294}$$

The integral evaluates to $n\pi$ because the transported vortex of winding n accumulates a phase $n\,\Delta\phi$ for each $\Delta\phi$ of angular displacement around the other vortex, and the half-circuit contributes $\Delta\phi = \pi$. This is the Aharonov–Bohm analog for vortices: each vortex acts as a topological “flux tube” of strength $2\pi n$ for every other vortex.

For $n = \pm 1$ (fermions):

$$\gamma_{\text{exchange}} = \pm\pi \quad \Rightarrow \quad e^{i\gamma} = -1, \tag{295}$$

and therefore

$$\boxed{\psi(\mathbf{x}_2, \mathbf{x}_1) = e^{i\gamma}\,\psi(\mathbf{x}_1, \mathbf{x}_2) = -\psi(\mathbf{x}_1, \mathbf{x}_2).} \tag{296}$$

Within the candidate route above, this is the proposed exchange-sign rule for two identical $n = \pm 1$ vortex excitations; the full Hilbert/Fock-space construction that turns this sign into the Pauli theorem is deferred below.

For $n = 0$ (bosons, no vortex): $\gamma_{\text{exchange}} = 0$ and the wavefunction is symmetric—no exclusion.

For $n = \pm 2$ (even winding): $\gamma_{\text{exchange}} = \pm 2\pi \equiv 0$, symmetric. This is an illustrative consistency check that the Berry-phase formula $\gamma = n\pi$ respects Bose statistics for any even winding; it is *not* a claim that the physical $W^{\pm}, Z$ bosons are realized as $n = \pm 2$ scalar vortices (cf. the scope paragraph in Sec. X A).

This candidate topological route to the spin-statistics connection complements the standard proof via relativistic quantum field theory [34, 35] and provides a physical, geometric picture of how the fermionic exchange sign can be organized: the exchange phase is determined by the winding number, the same structural quantity that enters the conditional spin route (Sec. X A).

Dimensional caveat.—The Berry-phase calculation above is carried out in the two-dimensional plane transverse to the vortex axis. In 2+1 dimensions, the braid group permits arbitrary exchange phases (anyons); in 3+1 dimensions, only $\gamma = 0$ or π survive because exchange paths are contractible around the vortex line. The result $\gamma = n\pi$ is therefore consistent with the 3+1-dimensional requirement, but the passage from the 2D calculation to the full 3D vortex-line topology (linking, knotting, Hopf invariants) and the construction of the corresponding Hilbert/Fock-space exchange operators require a dedicated treatment and are left for future work.

In 3+1 dimensions, the rigorous origin of fermion antisymmetry is the standard QFT spin–statistics theorem, which follows from Lorentz invariance, causality and positivity of the energy [34, 35]. The vortex construction above is therefore *complementary* to, not a replacement for, that theorem: it supplies a candidate geometric picture of why the exchange phase may take the value it does, on top of the symmetry argument that guarantees the result when the QFT assumptions hold.

XI. GRAVITATIONAL BIREFRINGENCE

At the classical level, ISPG preserves local Lorentz invariance exactly: the locally measured speed of light is c for all polarizations [1], consistent with the classical light-propagation analysis in Appendix 5 of the main theory (PPN $\gamma = 1$, no birefringence at the tree level). At the quantum level, however, vacuum polarization by virtual particle–antiparticle pairs introduces a polarization-dependent correction to the photon propagation speed in the presence of a gravitational gradient. The one-loop-motivated channel estimated in this section lies beyond the tree-level regime of Appendix 5 and therefore does not contradict its conclusions. The precise comparison to the curvature-coupled GR+QED effect is deferred to §XI C.

A. Vacuum polarization in an inhomogeneous field

In QED, a photon propagating through an external electromagnetic field interacts with virtual e^+e^- pairs (the Euler–Heisenberg effective Lagrangian [36]). In ISPG, the analogous process occurs in a gravitational gradient: a photon propagating through a region with $\nabla\varphi \neq 0$ interacts with virtual pairs whose effective mass $m_{\text{eff}} = m_0\, e^{\varphi/2}$ varies across the pair's spatial extent. This effect—**gravitational birefringence**—arises from the dependence of particle masses on the scalar field and is structurally distinct from (though numerically comparable to) the one-loop GR+QED effect of Drummond–Hathrell [37]; see §XI C for a detailed comparison.

The leading EFT correction expected from a slowly varying φ-field can be parametrized as a $\nabla\varphi$-dependent contribution to photon propagation. For a photon with four-momentum k^μ and polarization ϵ^μ, the leading dispersion ansatz is

$$\omega^2 = |\mathbf{k}|^2\, c^2 \left[1 + \frac{\alpha_{\text{EM}}}{45\pi}\, \frac{(\nabla\varphi)^2}{m_e^2 c^2/\hbar^2}\, \eta(\hat{\epsilon}, \hat{\nabla}\varphi)\right], \tag{297}$$

where $\alpha_{\text{EM}} = 1/137$ is the fine-structure constant and η is a polarization-dependent factor:

$$\eta = \begin{cases} 7 & \text{if } \hat{\epsilon} \parallel \hat{\nabla}\varphi \ \ (\parallel\text{-mode}), \\ 4 & \text{if } \hat{\epsilon} \perp \hat{\nabla}\varphi \ \ (\perp\text{-mode}). \end{cases} \tag{298}$$

Note on coefficients and status.—The numerical values $\eta = 7$ and $\eta = 4$ are the Euler–Heisenberg coefficients for QED in an external electromagnetic field [36]; they are *adopted here by analogy*, not derived from ISPG first principles. The full ISPG coefficients require an explicit one-loop calculation of the φ-e^+e^- triangle diagrams with the conformal-coupling vertex $m_0\, \partial\varphi \cdot \bar{\psi}\psi$ specified by the action of the main theory [1]; this derivation is a self-contained technical task that is the topic of a separate work. The present section should be read as a *parametric* estimate: the scalar-gradient operator channel and the leading scaling $(\nabla\varphi)^2/m_e^2$ are identified here, while the existence and sign of a nonzero birefringent coefficient, its precise numerical value, and the full angular tensor require the dedicated one-loop calculation and remain open derivation targets.

Scope of the two-case parametrization.—Equation (298) records only the two *extreme* polarization geometries (electric field exactly parallel or exactly perpendicular to $\hat{\nabla}\varphi$). The parallel case is a physical photon polarization only when the propagation direction $\hat{k}$ is perpendicular to $\hat{\nabla}\varphi$, since physical photon polarizations obey $\epsilon \cdot k = 0$. For a generic propagation direction $\hat{k}$ and generic linear polarization, the one-loop effective action produces an angular dependence that interpolates between the two extremes and also involves the relative orientation between $\hat{k}$ and $\hat{\nabla}\varphi$; a general mode carries both components and experiences a weighted average. The $(7,4)$ numbers should therefore be read as bounding parallel/perpendicular values for the leading-order estimate, not as a complete angular template.

The two polarization modes propagate at slightly different phase velocities:

$$\frac{\Delta c}{c} = \frac{c_{\parallel} - c_{\perp}}{c} = \frac{3\,\alpha_{\mathrm{EM}}}{90\pi}\,\frac{(\nabla\varphi)^2}{m_e^2 c^2/\hbar^2}. \tag{299}$$

B. Order-of-magnitude estimate

For a neutron star with $M = 1.4\,M_{\odot}$ and radius $R = 10$ km:

$$|\nabla\varphi| \sim \frac{r_{\mathrm{s}}}{R^2} = \frac{2GM}{c^2 R^2} \sim 4\times10^{-5}\ \mathrm{m}^{-1}, \tag{300}$$

and

$$\frac{(\nabla\varphi)^2}{(m_e c/\hbar)^2} \sim \frac{(4\times10^{-5})^2}{(2.6\times10^{12})^2} \sim 2\times10^{-34}. \tag{301}$$

The birefringence is therefore

$$\frac{\Delta c}{c} \sim \frac{\alpha_{\mathrm{EM}}}{30\pi}\cdot 2\times10^{-34} \sim 10^{-38}. \tag{302}$$

For a photon traversing a path of length $L \sim R = 10$ km near the neutron star surface, the accumulated differential phase between the two polarizations is

$$\Delta\Phi_{\mathrm{biref}} = \frac{2\pi\nu}{c}\,\frac{\Delta c}{c}\,L \sim 2\pi\nu\cdot10^{-38}\cdot10^{4}\ \mathrm{m}/c. \tag{303}$$

For X-ray photons ($\nu \sim 10^{18}$ Hz): $\Delta\Phi_{\mathrm{biref}} \sim 10^{-24}$ rad—utterly negligible with any foreseeable technology.

For the much stronger fields near black holes ($\nabla\varphi \sim 1/r_{\mathrm{s}}$, $L \sim r_{\mathrm{s}}$):

$$\Delta\Phi_{\mathrm{biref}} \sim \frac{\alpha_{\mathrm{EM}}}{30\pi}\,\frac{(m_e c/\hbar)^{-2}}{r_{\mathrm{s}}^2}\cdot\frac{2\pi\nu\,r_{\mathrm{s}}}{c} \sim \frac{\alpha_{\mathrm{EM}}\,\nu}{30\pi\,m_e^2 c^3/\hbar^2\,r_{\mathrm{s}}}. \tag{304}$$

For a stellar-mass black hole ($r_{\mathrm{s}} \sim 10$ km) and gamma-ray photons ($\nu \sim 10^{20}$ Hz): $\Delta\Phi_{\mathrm{biref}} \sim 10^{-21}$ rad—far below any conceivable measurement sensitivity.

Regime caveat.—For the BH estimate, $|\nabla\varphi|\,r_{\mathrm{s}} \sim 1$, so the weak-field linearization of the scalar-metric background underlying Eq. (297) ceases to be rigorously controlled. The electron-Compton derivative parameter, $|\nabla\varphi|/(m_e c/\hbar) = |\nabla\varphi|\,\hbar/(m_e c)$, remains extremely small for $r_{\mathrm{s}} \sim 10$ km, of order 4×10^{-17}; the limitation is therefore not a breakdown of electron-Compton suppression, but the absence of a full curved-background one-loop self-energy including the nonlinear metric and curvature-coupled terms. In this regime the number $\Delta\Phi_{\rm biref} \sim 10^{-21}$ should therefore be read as a parametric extrapolation of the weak-field formula to its boundary of validity rather than a rigorous prediction; the non-perturbative strong-field calculation (one-loop self-energy on a full Schwarzschild-like background with $\nabla\varphi$ of order $1/r_{\rm s}$) is deferred. The quantitative smallness of the effect—already negligible in the controlled weak-field regime around neutron stars, Eq. (302)—does not depend on this caveat.

C. Distinction from other birefringence sources

Gravitational birefringence in ISPG differs from other proposed sources:

- **QED vacuum birefringence** (Euler–Heisenberg [36]): requires a strong *electromagnetic* field; scales as $B^2/B_{\rm cr}^2$. ISPG birefringence requires a *gravitational* gradient; it scales as $(\nabla\varphi)^2$. At the operator level the magnetic-field QED channel and the scalar-gradient ISPG channel are distinct. In real neutron-star or magnetar polarimetry, however, magnetic-QED, plasma, and magnetospheric propagation effects would dominate any $\sim 10^{-38}$ gravitational contribution; isolating the ISPG term would require a dedicated background model.

- **Lorentz-violation birefringence** (Standard Model Extension [38]): produces birefringence that is direction-dependent relative to a preferred frame. ISPG birefringence is direction-dependent relative to the local gravitational gradient $\nabla\varphi$ and vanishes in the absence of gravity.

- **GR + QED prediction:** At the classical level, GR predicts zero vacuum birefringence (the equivalence principle forbids polarization-dependent propagation of classical electromagnetic waves in vacuum). At one-loop in curved spacetime, however, Drummond & Hathrell [37] showed that integrating out the electron produces an effective action with curvature-coupled terms of the schematic form $R_{\mu\nu\rho\sigma}F^{\mu\nu}F^{\rho\sigma}$, yielding polarization-dependent phase velocities that scale as $R_{\mu\nu\rho\sigma}/(m_e^2c^2/\hbar^2)$. The ISPG effect (297) is structurally distinct: it scales as $(\nabla\varphi)^2$ rather than as a curvature component, and does not coincide with the Drummond–Hathrell tensorial pattern. A detection that correlates with $|\nabla\varphi|$ in regions where the Drummond–Hathrell curvature pattern is suppressed (or vice versa) would therefore discriminate ISPG from the GR+QED baseline; a generic "detection of gravitational birefringence" by itself does not falsify GR once one-loop QED corrections are included.

XII. SUMMARY OF PREDICTIONS

Table I collects the quantitative predictions of the quantum ISPG framework developed in this paper. Each prediction specifies the observable, the ISPG prediction, the standard (GR/QFT) expectation, and the experimental channel most sensitive to the effect. The gravity-core source closure derived in Sec. III E is structural rather than observational: after projection away from the neutral collective coordinates, the localized source sector has a positive reduced gap and a coercive energy bound. For that reason it is part of the logical foundation of the table below, not a separate row inside it.

TABLE I. Falsifiable predictions and prediction-status ledger of the quantum ISPG framework. "GR/SM" denotes the prediction of general relativity and/or the Standard Model.

Observable	**ISPG prediction**	**GR/SM**	**Experiment**
Dark-energy EoS $w(z)$	$w < -1$, source-driven phantom; DESI DR2 CPL comparison mixed	$w = -1$ exactly	DESI, Euclid, Rubin
Fine-structure constant $\Delta\alpha/\alpha$	$\Delta\alpha/\alpha \sim 10^{-7}$–$10^{-6}$ at $z \sim 1$–3 (future calculation; ESPRESSO ppm now, 10^{-7} ELT–ANDES target)	No spacetime variation in minimal SM/GR	ESPRESSO, ELT-ANDES
Coherent gravitational phase/dephasing	$\Gamma_\varphi = 2m\lvert\Delta\Phi_N\rvert/h$ (1PN GR/COW baseline)	Nonzero redshift/COW phase	Neutron and atom interferometry
Irreversible gravitational decoherence	Conditional $\Gamma_{\rm dec} \simeq \Gamma_\varphi$ after tail/noise correlator	0 in unitary GR+QM without environment	MAQRO, STE-QUEST parameter scan
Decoherence anisotropy	Conditional directional tail/noise channel ($\propto \nabla\varphi$)	Environment-dependent	Space quantum links
Profile-dependent tunneling / anisotropy candidate	WKB exponent depends on $\varphi(x)$; nonreciprocity conditional on tail anchoring	Static Hermitian barriers reciprocal	Long-barrier tests; cumulative modelling
Gravitational birefringence	$\Delta c/c \sim 10^{-38}$ (NS scale), $\propto (\nabla\varphi)^2/m_e^2$ (ISPG channel; coefficient future one-loop calculation)	Drummond–Hathrell: $\propto R_{\mu\nu\rho\sigma}/m_e^2$	Beyond current sensitivity
Koide relation $Q = 2/3$	$Q = 0.66673$ (E1D/rank-1 numerical support; 3D-to-1D bridge open)†	Unexplained coincidence	0.009% accuracy
Proton lifetime	Finite (very long)	Stable or GUT decay	Hyper-Kamiokande
Pair-production threshold shift	$\delta E_{\rm thr}/E \propto \varphi$	No shift	Black hole X-ray spectra
2PN Shapiro delay (ISPG−GR difference)	$\Delta B = \pi/4$ (closed; standalone absolute coefficients convention-dependent)	0 (by construction)	SKA pulsar–BH
ISCO orbital frequency	$f_{\rm ISCO} = 0.931\, f_{\rm ISCO}^{\rm GR}$	GR reference	LIGO–Virgo–KAGRA, ET
Black-hole shadow size	$b_c = e\, r_{\rm s}$ (+4.6% vs. GR)	$b_c = \frac{3\sqrt{3}}{2}\, r_{\rm s}$	EHT, ngEHT
Equivalence principle	Derived: $m_{\rm inert} \equiv m_{\rm grav}$	Postulated	MICROSCOPE, STEP
Scalar dipole radiation	$\dot{E}_{\rm dipole} \simeq 0$ (vacuum exact, leading-order in matter; matter-filled exact-cancellation open per MAIN/16)	Theory-dependent	Pulsar timing

†Requires numerical verification that the self-consistent oscillon solution yields $x = \alpha/\lambda_0 \approx 66.9$ (Sec. VII C).

The predictions span a wide range of energy scales and experimental techniques. The most immediately testable are:

1. **Dark-energy evolution** ($w \neq -1$): DESI year-3 and Euclid data, expected by 2027.

2. **Fine-structure constant variation**: ESPRESSO at the VLT is already operating at ppm precision; the 10^{-7} regime targeted here is the future ELT–ANDES-level test.

3. **Conditional gravitational decoherence**: proposed space missions (MAQRO, STE-QUEST class) can constrain the ISPG tail/noise channel once its correlation spectrum and viable parameter window are derived; the leading coherent phase rate itself is the GR/COW baseline at 1PN order.

XIII. REMARK ON THE ONTOLOGY OF THE MEDIUM

The formalism developed in the preceding sections treats space as a dynamical, pressure-carrying medium whose properties are encoded in a single scalar field φ. This ontology naturally raises a question: if space is a medium, what defines the boundaries of its existence? The remarks below do not constitute a formal derivation; they record a physical intuition that is consistent with the formalism and may guide its future development.

If space is a medium in which pressure varies—decreasing in gravitational wells, oscillating in particles, propagating as waves—then this variability logically requires two asymptotic bounds, not one: a maximum and a minimum. If the substance of space is water, it must be poured into a vessel; the vessel is the region where the substance is absent. One pole—*substance*—represents the extreme state in which pressure, mass–energy capacity, the operational clock rate of resonant structures, and spatial scale are asymptotically maximal. The other pole—*void*—represents the null limit: zero pressure, zero mass–energy content, no operational clock rate, and zero size. The void is not an independent entity; it is the logical complement of substance—the necessary consequence of something existing. Physical reality is a bounded oscillation between these two poles; a vibration that can never exceed either limit.

This motivates the ontological name *Archion*. An Archion is the primary quantum of existence: the minimal active unit of the scalar-metric medium whose measurable life is precisely this bounded oscillation between the substance pole and the void pole. The older word "oscillon" names only the dynamical profile—a localized, time-periodic resonance. *Archion* names the ontological role: the first active unit by which existence becomes measurable.

This picture becomes intuitively transparent through a single analogy: coloured soluble crystals in water. The water is space—the background medium. The crystal is an oscillon—a localized, concentrated resonant form of matter in the Archion field. As the crystal dissolves it releases colour into the surrounding water, just as an oscillon emits pressure-carrying tails into the surrounding space. The spreading of colour through the water is the spreading of the scalar profile through space; through the bi-conformal metric this profile carries gravitational influence. While dissolution is underway the water is mottled: intensely coloured near the crystals, nearly transparent far from them. This is the cosmic web—galaxy clusters where matter is concentrated and the φ-gradient is deep; filaments where colour migrates from crystal to crystal; and voids—vast stretches of water between crystals that the colour has barely reached. Gravity, in this picture, is not an additional colour-gradient force on matter; it is the geodesic response to the metric whose scalar profile is visualised by the colour distribution. No separate "dark pigment" needs to be added, just as no extra dye is required to explain the diffusion of colour in water: the dynamics of the medium suffice.

The analogy reveals one further essential property: self-regulation. A crystal dissolves from its surface: the more colour it releases, the smaller it becomes; a smaller crystal has less surface area in contact with the water and therefore dissolves more slowly. The process decelerates itself. An oscillon behaves identically: it vibrates in pressure and emits pressure tails into the surrounding medium. These tails reduce the background pressure around the oscillon. The reduction forces the oscillon to shrink in effective size and mass ($m_{\text{eff}} = m_0\, e^{\varphi/2}$). A smaller, lighter oscillon interacts less with the medium—it emits fewer tails—and consumes less pressure. This self-regulating cycle—emission, contraction, stabilisation—operates on every scale by the same principle: from oscillons to stars, from galaxy clusters to the universe as a whole. In the measurable-existence language, this is a reduction of active measurable content, not the disappearance of the underlying medium.

When all crystals have fully dissolved, the water will be uniformly tinted—an intermediate tone, neither the intensity of the crystal nor the transparency of pure water. This is maximum entropy: the vanishing of gradients, the cessation of process. In the theory this is reflected by the relation $a_0 = cH/(2\pi)$: as long as $H > 0$, structure persists. Yet the difference between maximum and minimum continuously decreases while never reaching zero. Complete equalisation would imply the disappearance of both poles—a logical contradiction, since the arena in which equalisation occurs would itself cease to exist. The process is asymptotic: the universe perpetually tends toward equilibrium but never attains it.

We do not elevate this picture to a postulate. We record it as a heuristic that is consistent with every result derived in this work and that may serve as a guide for its future development.

XIV. CONCLUSION

We have developed the quantum extension of bi-conformal scalar-tensor gravity (ISPG), building on the classical theory established in [1]. The central thesis—that elementary particles are localized resonant excitations of the scalar-metric medium—has been shown to produce a coherent, falsifiable theoretical framework. The central closed achievement of the present paper is the quantum-mechanical derivation of the gravitational mechanism itself: the oscillon source, the Bernoulli pressure deficit, the localized zero-frequency source, the exterior $1/r$ field, Newtonian recovery, the source-following collective response, and the coercive separation of neutral and non-neutral source deformations. The broader sectors below show how far the same medium picture extends into cosmology, particle structure, and quantum phenomena. They are part of the same research program, but they do not all have the same logical status in the present work: where a sector still requires a dedicated numerical or structural closure, that is stated explicitly and does not weaken the core gravity derivation. Within that scope, the principal results and extensions are:

1. **Particles as spacetime resonances** (Sec. III). The scalar field perturbation equation on the bi-conformal background admits quasi-localized oscillatory solutions whose energy reproduces the effective mass formula $m_{\text{eff}} = m_0\, e^{\varphi/2}$. Local undetectability extends to all dimensionless local quantum observables under a uniform bi-conformal background rescaling with fixed dimensionless couplings; gradient-dependent effects, loop-threshold effects, and cross-epoch comparisons require separate treatment.

2. **Bernoulli mechanism for gravity** (Sec. IV). The kinetic energy of particle resonances drains the static pressure of the medium via an exact scalar-field analog of Bernoulli's equation. This provides the first mechanical explanation for *why* matter curves spacetime. After projection onto the reduced source space, a positive spectral gap and coercive source-energy estimate keep non-neutral deformations perturbatively controlled, so the exterior field follows the same collective coordinate that moves the oscillon core.

3. **Inertia and the equivalence principle** (Sec. V). The retarded potential of an accelerated oscillon creates a dipolar pressure asymmetry whose integral yields $\mathbf{F} =$

$-m_{\text{eff}}\,\mathbf{a}$—Newton's second law derived, not postulated for the oscillon model. The identity $m_{\text{inertial}} \equiv m_{\text{gravitational}}$ is traced to the same scalar profile and coupling that set the Bernoulli energy bookkeeping; exact non-perturbative compact-body sensitivity is a separate question. The hysteresis action S_χ is an effective template motivated by the quasi-normal mode structure of the oscillon (Sec. V E): the microscopic tracking stiffness $\lambda_{\text{micro}} = \omega_0^2/c^2$ is set by the fundamental confining frequency, and the Klein–Gordon structure emerges in the limit $\Gamma \ll \omega_0$ appropriate for stable particles.

4. **Self-regulated echo-background energy** (Sec. VI). The irreversible spreading of resonant tails gives a positive outgoing energy flux, but the resulting pressure drop weakens the sources through a negative feedback loop. On the perturbative tail-power branch, the equilibrium echo-background (dark-energy) density is estimated at $\rho_\Lambda \sim H_0^2 M_{\text{Pl}}^2$; the zero-point decoupling and non-perturbative normalization required for a full cosmological-constant solution remain open.

5. **Evolving dark energy** (Sec. VI D). The dynamical nature of the echo background predicts $w(z) = -1 - \delta w_0\,(1+z)^3/E(z)$ with $w < -1$ at all epochs (source-driven phantom, no ghost instability), with a magnitude conditional on the tail-power normalization. DESI DR2 gives a mixed CPL comparison: the sign of w_a is aligned, while the sign of $w_0 + 1$ is opposite, so this remains a near-term test rather than a confirmation.

6. **Earliest-epoch dynamics without inflation** (Sec. VI G). Planck-mass oscillons nucleate uniformly from the quiescent medium in the semiclassical dimensional-estimate picture, where a candidate Planck-scale bounce has $S_E \sim \mathcal{O}(1)$; their overlapping tails thermalize to produce the hot Big Bang. The effective-source NEC violation (canonical $\tau_{\mu\nu}$ satisfies the energy conditions; the $\kappa = -1/2$ coupling flips the sign of the source-side contraction, cf. the energy-conditions section of the main theory [1]) and the hyperbolic WDW equation evade the assumptions of the standard singularity theorems. The horizon and flatness problems are addressed at the homogeneous-background level by universal nucleation and the geometric inheritance of $k = 0$, without invoking an inflaton field. The primordial spectrum, superhorizon correlations, instanton bounce, and geodesic- completeness proof remain open numerical and analytical tasks.

7. **Mass hierarchy from 3D resonances** (Sec. VII). The charged-lepton mass hierarchy is organized by the MASS/Mathieu ladder and the E1D/rank-1 cavity structure. In N-space the Koide identity holds numerically to 0.024%, while the rank-1 perturbation argument selects $x = 33 + 24\sqrt{2} \approx 66.9$ as the bridge value. A first-principles derivation of this bridge value from the self-consistent 3D oscillon eigenvalue system (Sec. VII B) is the residual numerical task, with gravitational self-energy corrections of order $(m/M_{\text{Pl}})^2 \sim 10^{-45}$.

8. **Higgs–gravity unification and electroweak spectrum** (Sec. VIII). The Higgs VEV inherits the bi-conformal factor $v(\varphi) = v_0\, e^{\varphi/2}$ for an effective electroweak sector.

Given the experimentally fixed electroweak gauge algebra, the broken-phase gauge-boson mass matrix (Sec. VIII D) preserves the standard tree-level mass ratios and local dispersion relations, with photon masslessness protected by the imported exact $U(1)_{\mathrm{EM}}$ Ward identity; the scalar shift symmetry is a compatibility condition for the scalar-metric sector, not a separate photon-mass Ward identity.

9. **Scalar sensitivity (structural argument)** (Sec. III I). The shift-symmetry Ward identity supports $s = 1/2$ at all loop orders for vacuum compact objects and at leading order for matter-filled bodies ($s_{\mathrm{excess}} = 0$ in those controlled limits; matter-coupled exact-cancellation remains an open task per MAIN/16), arguing for suppressed scalar dipole radiation and a perturbatively controlled strong-equivalence statement in the scalar sector, with the exact matter-filled Nordtvedt statement left open.

10. **Gravitational decoherence** (Sec. IX). Entangled particles in inhomogeneous φ-fields undergo differential phase evolution, producing a calculable coherent dephasing rate $\Gamma_\varphi = 2m|\Delta\Phi_N|/h$ that matches the GR/COW baseline at 1PN order. The irreversible decoherence and directional-discriminator claims are conditional on the resonant-tail or scalar-noise correlator derived in future work.

11. **Topological particle classification** (Sec. X). Fermion spin projection, antimatter labels, and the Pauli exclusion principle are proposed to be organized by the winding-number structure of vortex excitations, with the bridge and spin-statistics closures stated there.

12. **Gravitational birefringence** (Sec. XI). Vacuum polarization in a gravitational gradient identifies a polarization-dependent ISPG channel with leading scaling $\Delta c/c \propto (\nabla\varphi)^2/m_e^2$ and neutron-star scale $\Delta c/c \sim 10^{-38}$—far below current measurement sensitivity. This channel is distinct from the GR+QED Drummond–Hathrell baseline at the operator/scaling level; its numerical coefficient is the dedicated one-loop calculation target.

The framework achieves these results without introducing extra dimensions, supersymmetry, or new fundamental fields. It requires only the scalar-metric medium already present in the classical theory, extended to the quantum domain through standard techniques (canonical quantization, perturbation theory, one-loop corrections). The quantum effective action (Sec. III H) is radiatively stable due to the shift symmetry $\varphi \to \varphi+\mathrm{const}$, one-loop controlled with corrections suppressed by $(\mathrm{curvature}/M_{\mathrm{Pl}}^2)^2$ in the weak-field regime, and controlled at all loop orders by the EFT expansion parameter $(E/M_{\mathrm{Pl}})^2 \sim 10^{-76}$.

The self-consistent mass determination problem has been formulated as a dimensionless nonlinear eigenvalue system (Sec. VII B). The confining potential is a geometric property of the nonlinear wave equation, independent of G. The cavity geometry supplies the structural mode labels and the nonlinear eigenvalue problem; the quantitative charged-lepton mass ratios are currently carried by the MASS/Mathieu ladder of the companion MASS paper [3], with the formal 3D-to-1D reduction remaining the open numerical task. Gravitational self-energy corrections of order $(m/M_{\rm Pl})^2 \sim 10^{-45}$ are negligible. The lattice formulation reduces the classical eigenvalue problem to a finite-dimensional nonlinear system, while the quantum lattice (exploiting the ultralocal kinetic structure) enables Monte Carlo evaluation of the constrained path integral.

The direction for future work includes: (i) lattice simulation of the coupled scalar-metric system (to verify oscillon stability and decay properties, and to compute the primordial perturbation spectrum from the Planck-epoch nucleation dynamics of Sec. VI G); (ii) numerical solution of the self-consistent oscillon eigenvalue problem to determine whether the perturbation ratio $x = \alpha/\lambda_0$ attains the value $33 + 24\sqrt{2} \approx 66.9$ required by the Koide relation—initial numerical results (Sec. VII C) reproduce $Q = 2/3$ to 10^{-4} and the τ/e mass ratio to 0.37%, while the first-order Hartree correction, under the selected semiclassical/first-order prescription, recovers the Koide angle $\theta_0 = 2/9$ to 0.07%, with the joint (Q, θ_0) closure beyond first order remaining open; (iii) detailed confrontation with forthcoming data from DESI, Euclid, ESPRESSO, and space-based quantum experiments.

The electroweak gauge boson spectrum ($W^\pm$, Z, γ) has been treated as a conditional embedding on the bi-conformal background (Sec. VIII D): once the effective SM electroweak gauge algebra, g, g', and v_0 are supplied, the standard broken-phase mass matrix and local dispersion relations are preserved. Photon masslessness is protected by the imported exact $U(1)_{\rm EM}$ Ward identity. The derivation of the gauge couplings themselves (g, g', g_s), the exact Ward identity from the medium, and the origin of the $SU(3)_c$ colour group from the medium's internal excitation structure remain open problems for future work.

The scalar sensitivity $s = 1/2$ and the vanishing of the excess sensitivity $s_{\rm excess} = 0$ have been argued to persist at all loop orders for vacuum compact objects (Sec. III I), via the shift-symmetry Ward identity of the quantum effective action; exact non-perturbative matter-filled-body cancellation remains an open task per MAIN/16. This supports suppressed scalar dipole radiation and a perturbatively controlled strong equivalence statement in the scalar sector; the exact matter-filled Nordtvedt statement belongs to the same open sensitivity programme.

The hysteresis action S_χ (Eq. 157) is treated as a microscopically motivated effective template (Sec. V E). The microscopic tracking stiffness $\lambda_{\rm micro} = \omega_0^2/c^2$ emerges from the confining frequency of the oscillon's fundamental quasi-normal mode, and the Klein–Gordon structure follows from the damped oscillator equation in the limit $\Gamma \ll \omega_0$ appropriate for stable particles. The macroscopic collective relaxation scale and the closure normalization β remain Route B/coarse-graining tasks, with rotation curves serving as the calibration and test channel [1].

One item requires attention in future extensions:

1. *Independent numerical verification.*—The structural argument for quantum stability of the phantom sector within the pure-gravity constrained sector has been presented at all orders in the loop expansion (Sec. III G), resting on three pillars: kinematic ghost freedom from the single-DOF constraint, structurally anomaly-free Abelian constraint

algebra (full bracket-closure/anomaly proof deferred), and non-perturbative boundedness of the Euclidean action on the constrained background (matter-coupled extension via the EFT completion of Appendix 11). Independent numerical verification of these results—for example via lattice simulation of the constrained path integral—would provide a valuable cross-check.

The picture that emerges is one of radical economy: a single medium, vibrating at different frequencies and in different topological configurations, generates all the diversity of matter, the gravitational and inertial phenomena, and the arrow of time itself. Gravity is not a separate interaction added to quantum mechanics; it is the medium's response to its own internal vibrations.

Appendix A: Technical Lemmas

This appendix collects mathematical proofs and technical results that support the main text.

1. Lemma: Local Lorentz invariance of wave propagation

Lemma. For the bi-conformal metric $ds^2 = -e^{\varphi}c^2dt^2 + e^{-\varphi}(dx^2 + dy^2 + dz^2)$, the locally measured speed of scalar field perturbations is exactly c, independent of position.

Proof. The scalar field equation $\Box\varphi = 0$ on the static background $\varphi = \varphi_0(\mathbf{x})$ becomes (Eq. 25):

$$e^{-\varphi_0}\delta\ddot{\varphi} = e^{\varphi_0}\nabla^2\delta\varphi. \tag{A1}$$

The coordinate speed is $v_{\text{coord}} = e^{\varphi_0}c$. The locally measured speed uses proper time $d\tau = e^{\varphi_0/2}dt$ and proper length $d\ell = e^{-\varphi_0/2}dr$:

$$v_{\text{local}} = \frac{d\ell}{d\tau} = \frac{e^{-\varphi_0/2}dr}{e^{\varphi_0/2}dt} = e^{-\varphi_0} \cdot e^{\varphi_0}c = c. \tag{A2}$$

Thus $v_{\text{local}} = c$ for all φ_0. $\square$

2. Lemma: Positivity of scalar-field energy density

Lemma. For the ISPG phantom scalar field with static configuration $\varphi_0(\mathbf{x})$, the energy density measured by a static observer is positive-definite:

$$\rho_\varphi = \frac{e^{\varphi_0}}{32\pi G}|\nabla\varphi_0|^2 \geq 0. \tag{A3}$$

Proof. The stress-energy tensor for the scalar field is (Eq. 98):

$$\tau_{\mu\nu} = \partial_\mu\varphi\,\partial_\nu\varphi - \frac{1}{2}g_{\mu\nu}(\partial\varphi)^2. \tag{A4}$$

For a static field, the kinetic invariant is purely spatial (Eq. 99):

$$(\partial\varphi)^2 = g^{ij}\partial_i\varphi_0\,\partial_j\varphi_0 = e^{\varphi_0}|\nabla\varphi_0|^2 \geq 0. \tag{A5}$$

The static observer has $u^\mu = (e^{-\varphi_0/2}/c, 0, 0, 0)$ (equivalently $u^\mu = (e^{-\varphi_0/2}, 0, 0, 0)$ in natural units $c = 1$), giving (Eq. 100):

$$\rho_\varphi = \frac{1}{16\pi G}\tau_{\mu\nu}u^\mu u^\nu = \frac{e^{\varphi_0}}{32\pi G}|\nabla\varphi_0|^2. \tag{A6}$$

Since $e^{\varphi_0} > 0$ and $|\nabla\varphi_0|^2 \geq 0$, we have $\rho_\varphi \geq 0$. □

3. Lemma: WKB tunneling for quasi-bound states

Lemma. For the quasi-normal mode equation (Eq. 52) with effective potential $V_{\text{eff}}(r)$, the decay width is approximately:

$$\Gamma \sim \omega_R \exp\left[-2\int_{r_1}^{r_2}\sqrt{V_{\text{eff}}(r) - \omega_R^2}\,dr\right]. \tag{A7}$$

Proof. In the WKB approximation, the radial equation $\phi'' + k^2(r)\phi = 0$ with $k^2(r) = \omega^2 - V_{\text{eff}}(r)$ has turning points where $k^2(r) = 0$. In the classically forbidden region ($r_1 < r < r_2$, where $V_{\text{eff}} > \omega^2$), the solution decays exponentially: $\phi \sim \exp\big[-\int\sqrt{V-\omega^2}\,dr\big]$. The outgoing wave at infinity has amplitude reduced by the barrier penetration factor $\exp\Big[-\int_{r_1}^{r_2}\sqrt{V-\omega^2}\,dr\Big]$. The decay rate (imaginary part of ω) is proportional to the squared amplitude, giving the result. □

4. Theorem: Rank-1 perturbation and Koide ratio

Theorem (exact). Let $\hat{H} = \lambda_0 \mathbb{1} + \alpha|v\rangle\langle v|$ be a symmetric rank-1 perturbation of a threefold degenerate Hamiltonian, with $|v\rangle$ normalized. Then for any $\alpha > -\lambda_0$ (the only constraint being $\lambda_1 = \lambda_0 + \alpha > 0$), the Koide ratio is the one-parameter function

$$Q(x) = \frac{3+x}{(\sqrt{1+x}+2)^2}, \qquad x \equiv \alpha/\lambda_0, \tag{A8}$$

which increases monotonically from $Q(0) = 1/3$ to $Q(\infty) = 1$. If the remaining twofold degeneracy is lifted by a fractional amount δ, then $Q = Q(x) + \mathcal{O}(\delta^2)$.

Proof. The eigenvalues are $\lambda_1 = \lambda_0 + \alpha$ and $\lambda_2 = \lambda_3 = \lambda_0$. Define $x \equiv \alpha/\lambda_0$ (which can be arbitrarily large). The Koide ratio is

$$Q(x) = \frac{3+x}{\left(\sqrt{1+x}+2\right)^2}. \tag{A9}$$

We verify the two extreme limits:

- $x \to 0$ (degenerate): $Q = 3/9 = 1/3$. (Note: $Q = 1/3$ for exact degeneracy, since all three $\sqrt{\lambda}$ are equal.)
- $x \to \infty$ (one mass dominates): $Q \to x/x = 1$.

The function $Q(x)$ interpolates monotonically from 1/3 to 1. It equals 2/3 at a specific value of x. Setting $Q = 2/3$:

$$\frac{3+x}{(\sqrt{1+x}+2)^2} = \frac{2}{3}. \tag{A10}$$

Let $s = \sqrt{1+x}$. Then $x = s^2 - 1$ and the equation becomes

$$\frac{s^2+2}{(s+2)^2} = \frac{2}{3} \quad \Rightarrow \quad 3s^2 + 6 = 2s^2 + 8s + 8 \quad \Rightarrow \quad s^2 - 8s - 2 = 0, \tag{A11}$$

with positive root $s = 4 + 3\sqrt{2}$, giving $x = (4 + 3\sqrt{2})^2 - 1 = 33 + 24\sqrt{2} \approx 66.9$. The corresponding mass ratios are $\sqrt{\lambda_1/\lambda_2} = s = 4 + 3\sqrt{2} \approx 8.24$. For the actual leptons, $\sqrt{m_\tau/m_e} \approx 59$ and $\sqrt{m_\mu/m_e} \approx 14.4$, which means the physical masses are *not* a pure rank-1 perturbation but require lifting the $\lambda_2 = \lambda_3$ degeneracy.

Degeneracy-lifting regime.—Now consider a rank-1 perturbation plus a small degeneracy-lifting term: $\hat{H} = \lambda_0 \mathbb{1} + \alpha|v\rangle\langle v| + \epsilon\,\mathrm{diag}(0, +1, -1)$ with $\epsilon \ll \alpha$. The eigenvalues become $\lambda_1 = \lambda_0 + \alpha$, $\lambda_2 = \lambda_0 + \epsilon$, $\lambda_3 = \lambda_0 - \epsilon$. A direct calculation gives

$$Q = \frac{2}{3} + \mathcal{O}\!\left(\frac{\epsilon^2}{\lambda_0\,\alpha}\right). \tag{A12}$$

The correction is quadratic in ϵ and suppressed by $\epsilon^2/(\lambda_0\alpha)$, not by α/λ_0. This establishes the perturbative robustness of the rank-1 Koide constraint in the regime $\epsilon \ll \alpha$. For the physical charged leptons, $m_e \ll m_\mu$ implies $\epsilon \approx \lambda_0$, placing them outside the perturbative regime of this bound. The agreement of the measured Koide value with $Q = 2/3$ to 0.01% is therefore the empirical target that the rank-1/E1D consistency argument is intended to explain, not an independent confirmation of perturbative robustness in the physical hierarchy. □

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[1] G. Lotishvili, "Bi-conformal scalar-tensor gravity: exact PPN equivalence with general relativity, emergent MOND phenomenology, and falsifiable predictions," Zenodo (2026).

[2] G. Lotishvili, "Deriving MOND from Bi-Conformal Gravity: Vortex Closure, the Interpolating Function, and the Dark-Energy Connection," Zenodo (2026).

[3] G. Lotishvili, "Mathieu-Harmonic Mass Spectrum of Elementary Particles from Induced Scalar-Pressure Gravity," Zenodo (2026).

[4] R. Penrose, "Gravitational collapse and space-time singularities," Phys. Rev. Lett. **14**, 57 (1965).

[5] S. W. Hawking and R. Penrose, "The singularities of gravitational collapse and cosmology," Proc. R. Soc. Lond. A **314**, 529 (1970).

[6] S. Weinberg, "The cosmological constant problem," Rev. Mod. Phys. **61**, 1 (1989).

[7] J. Martin, "Everything you always wanted to know about the cosmological constant problem (but were afraid to ask)," C. R. Phys. **13**, 566 (2012).

[8] R. Penrose, "On gravity's role in quantum state reduction," Gen. Relativ. Gravit. **28**, 581 (1996).

[9] L. Diósi, "A universal master equation for the gravitational violation of quantum mechanics," Phys. Lett. A **120**, 377 (1987).

[10] J. Polchinski, *String Theory* (Cambridge University Press, 1998).

[11] C. Rovelli, *Quantum Gravity* (Cambridge University Press, 2004).

[12] M. Reuter and F. Saueressig, "Quantum Einstein gravity," New J. Phys. **14**, 055022 (2012).

[13] W. Thomson (Lord Kelvin), "On vortex atoms," Proc. R. Soc. Edinburgh **6**, 94 (1867).

[14] G. E. Volovik, *The Universe in a Helium Droplet* (Oxford University Press, 2003).

[15] K. G. Zloshchastiev, "Spontaneous symmetry breaking and mass generation as built-in phenomena in logarithmic nonlinear quantum theory," Acta Phys. Polon. B **42**, 261 (2011).

[16] E. Verlinde, "On the origin of gravity and the laws of Newton," JHEP **04**, 029 (2011).

[17] T. Jacobson, "Thermodynamics of spacetime: the Einstein equation of state," Phys. Rev. Lett. **75**, 1260 (1995).

[18] I. L. Bogolyubsky and V. G. Makhankov, "Lifetime of pulsating solitons in some classical models," JETP Lett. **24**, 12 (1976).

[19] M. Gleiser, "Pseudostable bubbles," Phys. Rev. D **49**, 2978 (1994).

[20] E. J. Copeland, M. Gleiser, and H.-R. Müller, "Oscillons: resonant configurations during bubble collapse," Phys. Rev. D **52**, 1920 (1995).

[21] ALICE Collaboration, "Elliptic flow of charged particles in Pb–Pb collisions at $\sqrt{s_{NN}} =$ 2.76 TeV," Phys. Rev. Lett. **105**, 252302 (2010).

[22] J. M. Bardeen, "Gauge-invariant cosmological perturbations," Phys. Rev. D **22**, 1882 (1980).

[23] H. Kodama and M. Sasaki, "Cosmological perturbation theory," Prog. Theor. Phys. Suppl. **78**, 1 (1984).

[24] DESI Collaboration, "DESI 2024 VI: Cosmological constraints from the measurements of baryon acoustic oscillations," Phys. Rev. D **110** (2024) 023507.

[25] DESI Collaboration, A. Abdul-Karim *et al.*, "DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints," Phys. Rev. D **112** (2025) 083515.

[26] J. K. Webb *et al.*, "Indications of a spatial variation of the fine structure constant," Phys. Rev. Lett. **107**, 191101 (2011).

[27] S. M. Kotus, M. T. Murphy, and R. F. Carswell, "High-precision limit on variations in the fine-structure constant from a single quasar absorption system," Mon. Not. R. Astron. Soc. **464**, 3679 (2017).

[28] Y. Koide, "New view of quark and lepton mass hierarchy," Phys. Rev. D **28**, 252 (1983).

[29] R. Foot, "A note on Koide's lepton mass relation," arXiv:hep-ph/9402242 (1994).

[30] A. Rivero and A. Gsponer, "The Koide formula and geometry," arXiv:hep-ph/0505220 (2005).

[31] Super-Kamiokande Collaboration, "Search for proton decay via $p \to e^+\pi^0$ and $p \to \mu^+\pi^0$ with

an enlarged fiducial volume," Phys. Rev. D **102**, 112011 (2020).

[32] R. Kaltenbaek *et al.*, "Macroscopic quantum resonators (MAQRO): testing quantum and gravitational physics with massive mechanical resonators," Exp. Astron. **34**, 123 (2012).

[33] H. Rauch *et al.*, "Verification of coherent spinor rotation of fermions," Phys. Lett. A **54**, 425 (1975).

[34] W. Pauli, "The connection between spin and statistics," Phys. Rev. **58**, 716 (1940).

[35] R. F. Streater and A. S. Wightman, *PCT, Spin and Statistics, and All That* (W. A. Benjamin, 1964).

[36] W. Heisenberg and H. Euler, "Consequences of Dirac's theory of positrons," Z. Phys. **98**, 714 (1936).

[37] I. T. Drummond and S. J. Hathrell, "QED vacuum polarization in a background gravitational field and its effect on the velocity of photons," Phys. Rev. D **22**, 343 (1980).

[38] D. Colladay and V. A. Kostelecký, "Lorentz-violating extension of the standard model," Phys. Rev. D **58**, 116002 (1998).

[39] M. Arminjon, "Gravity as Archimedes' thrust and a bifurcation in that theory," Found. Phys. **42**, 1341–1363 (2012).

[40] R. Arnowitt, S. Deser, and C. W. Misner, "The dynamics of general relativity," Gen. Relativ. Gravit. **40**, 1997 (2008); originally in *Gravitation: An Introduction to Current Research*, ed. L. Witten (Wiley, 1962), Ch. 7.

[41] B. S. DeWitt, "Quantum theory of gravity. I. The canonical theory," Phys. Rev. **160**, 1113 (1967).

[42] K. V. Kuchař, "Canonical quantization of gravity," in *Relativity, Astrophysics and Cosmology*, ed. W. Israel (Reidel, Dordrecht, 1973), pp. 237–288; "Canonical methods of quantization," in *Quantum Gravity 2*, eds. C. J. Isham, R. Penrose, and D. W. Sciama (Oxford Univ. Press, 1981), pp. 329–376.

[43] J. J. Halliwell, "Introductory lectures on quantum cosmology," in *Quantum Cosmology and Baby Universes*, eds. S. Coleman, J. B. Hartle, T. Piran, and S. Weinberg (World Scientific, 1991), pp. 159–243; arXiv:0909.2566 [gr-qc].

[44] M. J. Gotay, "Constraints, reduction, and quantization," J. Math. Phys. **27**, 2051 (1986).

[45] S. M. Carroll, M. Hoffman, and M. Trodden, "Can the dark energy equation-of-state parameter w be less than -1?" Phys. Rev. D **68**, 023509 (2003).

[46] J. M. Cline, S. Jeon, and G. D. Moore, "The phantom menaced: constraints on low-energy effective ghosts," Phys. Rev. D **70**, 043543 (2004).

[47] N. Afshordi, D. J. H. Chung, and G. Geshnizjani, "Cuscuton: a causal field theory with an infinite speed of sound," Phys. Rev. D **75**, 083513 (2007).

[48] A. H. Chamseddine and V. Mukhanov, "Mimetic dark matter," J. High Energy Phys. **2013**(11), 135 (2013).

[49] C. Cheung, P. Creminelli, A. L. Fitzpatrick, J. Kaplan, and L. Senatore, "The effective field theory of inflation," J. High Energy Phys. **2008**(03), 014 (2008); arXiv:0709.0293 [hep-th].

[50] G. 't Hooft and M. J. G. Veltman, "One-loop divergencies in the theory of gravitation," Ann.

Inst. Henri Poincaré A **20**, 69 (1974).

[51] M. H. Goroff and A. Sagnotti, "The ultraviolet behavior of Einstein gravity," Nucl. Phys. B **266**, 709 (1986).

[52] K. Nordtvedt, "Equivalence principle for massive bodies. II. Theory," Phys. Rev. **169**, 1017 (1968).